



BASE24-teller[®]

Transaction Processing Manual

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Applied Communications, Inc.

Preface

The ***BASE24-teller Transaction Processing Manual*** describes how BASE24-teller processes transactions. It includes basic transaction processing concepts and definitions, as well as specifics on how transactions are identified and processed by BASE24-teller. It also describes how to control and configure various aspects of BASE24-teller transaction processing.

Audience

This manual is intended for the BASE24 managers and operational staff who are responsible for configuring the BASE24-teller database for transaction processing and understanding how the transactions flow through the BASE24-teller system.

Prerequisites

Persons using this manual should be familiar with the BASE24-teller product, its basic functions, its terminology, and the BASE24 files it uses.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following BASE24 publications:

- ***BASE24 Base Files Maintenance Manual*** — Presents detailed descriptions and valid values for the configuration fields discussed in this manual.
- ***BASE24-teller Files Maintenance Manual*** — Presents detailed descriptions and valid values for the configuration fields discussed in this manual.

- ***BASE24 ANSI Host Interface Manual*** — Provides complete information on the BASE24-teller ANSI Host Interface process.
- ***BASE24 ISO Host Interface Manual*** — Provides complete information on the BASE24-teller ISO Host Interface process.
- ***BASE24 External Message Manual*** — Presents detailed descriptions and configuration issues for the ISO-based external message used with the BASE24-teller ISO Host Interface process.
- ***BASE24 Logical Network Configuration File Manual*** — Discusses in detail the assigns and params used in configuring BASE24-teller Device Handler, Authorization, and Host Interface processes.
- ***BASE24 Network Environment Source File Manual*** — Presents network-level configuration information for BASE24.
- ***BASE24 Tokens Manual*** — Presents detailed field descriptions and configuration issues for the tokens used by BASE24-teller.
- ***BASE24 Transaction Security Manual*** — Describes BASE24's support of PIN encryption, PIN verification, message authentication, card verification, and dynamic key management, along with the database settings required to implement this support.
- ***BASE24-teller Settlement and Reporting Manual*** — Presents and describes the BASE24-teller settlement and reporting tools that enable institutions to settle more efficiently and accurately with each other.

Software

This manual documents standard processing for BASE24-teller Release 5.0.

Customer-specific code modifications (i.e., RPQs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

Section 1, “Introduction,” introduces the reader to transaction processing. This section includes a discussion of card types, account types, and BASE24-teller features such as overdraft protection, passbook updating, and stop payment processing. It also includes a summary of the supported transactions.

Section 2, “BASE24-teller Transaction Path,” describes BASE24-teller messages, including their sources, destinations, and how they use tokens. It also describes the processes involved in transaction processing and the timers used along the transaction path.

Section 3, “BASE24-teller Message Flows,” presents a series of diagrams illustrating the paths taken by BASE24-teller transactions under various types of processing situations.

Section 4, “BASE24-teller Transaction Processing Controls,” describes the configuration options available for controlling BASE24-teller transaction processing.

Section 5, “BASE24-teller Transaction Routing and Authorization,” describes selected examples of the processing performed by BASE24-teller during transaction routing and authorization.

Section 6, “BASE24-teller Transaction Impacts on the Database,” illustrates the database impacts of each of the transactions that can be performed with BASE24-teller.

Appendix A, “BASE24-teller Internal Message,” describes the fields in the Teller Standard Internal Message Header (TSTMH).

Appendix B, “BASE24-teller Authorization Response Codes,” describes the response codes that can be carried in the TSTMH to identify the results of processing a transaction.

Appendix C, “Supported BASE24-teller Transactions,” describes the transaction codes that BASE24-teller supports.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (e.g., TR-AE000-07 for the ***BASE24-teller Transaction Processing Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (e.g., 04/94 for April, 1994) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions.

The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages.

Section 1

Introduction

This section provides an introduction to BASE24-teller transaction processing. It provides readers with some basic information on how BASE24-teller processes transactions.

Cards and Accounts

Transactions in the BASE24-teller system can be generally defined as specific actions carried out on accounts of some type using a card, card number, or account number as a basis for initiating the action.

BASE24 makes a distinction between cards and accounts. Activity limits and accumulators are maintained at the card level and balances are maintained at the account level. BASE24-atm and BASE24-pos can use limits, accumulators, and balances while BASE24-teller uses only balances. When cards are used to initiate transactions, BASE24-teller can use the card limit and accumulator for PIN tries.

Cards

Cards serve as evidence of an account and as a mechanism for accessing the account via electronic funds transfer (EFT) devices such as automated teller machines (ATMs) and point-of-sale (POS) terminals. These cards also can be used to access the account at a manned teller terminal; however, the account can still be accessed at the manned teller terminal when the card is unavailable.

Cards are given card numbers, which may or may not match the account numbers that they are used to access. There can be a one-to-one relationship between cards and accounts. There also can be a one-to-many relationship between cards and accounts, with one card accessing multiple accounts or one account being accessed by multiple cards, depending on card issuance procedures and BASE24 parameters.

The card types supported by BASE24 can be placed in one of two general groups based on the accounts they can access: credit and debit.

Credit Cards

Credit cards can access credit accounts (BASE24 account types 31 through 39 and 41 through 43).

Debit Cards

Debit cards can access noncredit accounts (BASE24 account types 01 through 09, 11 through 19, 21, and 50 through 55). When desired by the card issuer, debit cards can access credit accounts (BASE24 account types 31 through 39 and 41 through 43) in addition to noncredit accounts. A card that accesses credit and noncredit accounts can also be known as a dual or combination card.

Card Types

One- or two-character codes are used to identify card types in files throughout BASE24. The same codes must be used for a particular card type in all of the files. These codes are also used to identify service types in BASE24-pos. Codes used in this field are either reserved by BASE24 or user-defined.

When authorizing transactions, BASE24-teller performs the same processing for all card types except the business deposit (BD) card type. This card type identifies a card that can be used to initiate any inquiry/update transaction, as well as any financial transaction that deposits funds, but cannot be used to initiate a financial transaction that withdraws funds. Otherwise, BASE24-teller does not place any restrictions based on card type; however, BASE24 guidelines should still be used when establishing card types for BASE24-teller.

Reserved Card Types

Reserved codes are to be used only as defined, and include the following:

AD	=	Administrative (BASE24-atm only)
AX	=	American Express credit
BD	=	Business deposit (BASE24-atm and BASE24-teller only)
C*	=	Private label credit (* can be a blank, 0 through 9, A, or C through Z)
CB	=	Carte Blanche credit
D	=	Demonstration (BASE24-atm only)
DC	=	Diners Club credit
DS	=	Discover (Sears) credit
M	=	MasterCard credit
MD	=	MasterCard debit (Refer to BASE24-pos note below)
MM	=	MasterCard dual (Refer to BASE24-pos note below)
P*	=	Proprietary debit (* can be a blank, 0 through 9, or A through Z)
SC	=	Special, Check (BASE24-pos only)
SP	=	Special purpose (BASE24-atm Self-Service Banking Check Application only)

ST = Super teller (BASE24-atm Self-Service Banking Base Application only)
V = VISA credit
VD = VISA debit (Refer to BASE24-pos note below)
VV = VISA dual (Refer to BASE24-pos note below)

Codes with a first character of C, except code CB, are recommended to identify private label credit cards.

Codes with a first character of P are required to identify proprietary debit cards. BASE24 treats cards with proprietary debit codes and codes MD and VD as debit cards and treats cards with all other codes as credit cards.

Administrative (AD), Business deposit (BD), Demonstration (D), Special purpose (SP), and Super teller (ST) are special-use card types used by BASE24-atm.

Business deposit (BD) is also a special-use card type used by BASE24-teller to identify cards that can only be used to initiate deposit transactions.

MasterCard dual (MM) and VISA dual (VV) can be processed as debit or credit card types, based on the default combination card type specified in the Card Prefix File (CPF).

Special, Check (SC) is a special-use card type used to initiate BASE24-pos check guarantee and check verification transactions only.

BASE24-pos Note: BASE24-pos does not allow MasterCard debit (MD), MasterCard dual (MM), VISA debit (VD), or VISA dual (VV) card types in the POS Retailer Definition File (PRDF) and POS Terminal Data File (PTDF). BASE24-pos automatically includes the MD and MM card types with the MasterCard credit (M) card type, and automatically includes the VD and VV card types with the VISA credit (V) card type.

User-Defined Card Types

The user can add any one- or two-character code not included in the reserved code list, according to the following guidelines:

- The first character must be alphabetic (A, B, D through O, and Q through Z).
- The second character can be A through Z, 0 through 9, or a blank.
- A valid COBNAMES table entry is recommended for each user-defined code.

Accounts

BASE24 classifies all accounts as one of two types, and this classification determines which account-level limits and accumulators apply to a particular transaction. BASE24 uses codes to identify the possible account types.

Credit Accounts

Credit accounts involve funds advanced to an accountholder, by a financial institution or retailer, based on a credit agreement with the accountholder. BASE24-atm, BASE24-pos, and BASE24-teller define account types 31 through 39 as credit accounts. BASE24-teller also defines account types 41 through 43 as credit accounts.

Noncredit Accounts

Noncredit accounts involve accountholder funds on deposit with a financial institution. BASE24-atm, BASE24-pos, and BASE24-teller define account types 01 through 09 and 11 through 19 as noncredit accounts. BASE24-teller also defines account types 21 and 50 through 55 as noncredit accounts.

Number of Account Type Codes

The number of codes available to identify checking, savings, and credit accounts depends on how the transaction is entered. For manually-entered transactions, BASE24-teller recognizes a single code for each account type (i.e., 01 for checking accounts, 11 for savings accounts, and 31 for credit accounts).

For card-initiated transactions, BASE24-teller provides additional codes to identify the account types (i.e., 02 through 09 are added for checking accounts, 14 through 19 are added for savings accounts, and 33 through 39 are added for credit accounts). Institutions can use the codes within these ranges to further qualify the type of account; however, BASE24-teller treats all codes in a range just like the corresponding single code when determining how a transaction will be processed. These additional codes are not valid for manually-entered transactions.

Duplicate Account Numbers

Care must be taken when using cards to access Positive Balance File (PBF) accounts because the BASE24-teller Authorization process treats all account types in a range the same when determining whether a PBF account number is unique. When making this determination, account types 01 through 09 are in the same range, account types 11 and 14 through 19 are in the same range, and account types 31 and 33 through 39 are in the same range. As a result, the accounts shown below are considered unique when they are entered in the PBF and the Cardholder Authorization File (CAF). However, the BASE24-teller Authorization process considers the accounts to be duplicates because types 31 and 33 are in the same range.

Description	Account A	Account B
FIID	BNK0	BNK0
Account Number	123456	123456
Account Type	31	33

Account Types

The following account type codes are valid for BASE24-teller. In addition to identifying the account type, these codes determine which PBF contains the account information. BASE24 can use one, two, or three PBFs to maintain customer account information for one FIID. The first PBF (PBF 1) contains checking accounts, the second PBF (PBF 2) contains savings and NOW accounts, and the third PBF (PBF 3) contains credit card, line of credit, loan, and utility accounts.

Type	Account Description	PBF Number	Credit	Noncredit
01	Checking (DDA)	1		✓
02–09	Checking (DDA) — Same as type 01, but valid for card-initiated transactions only	1		✓
11	Savings	2		✓
12	Individual retirement account (IRA)	2		✓
13	Certificate of deposit (CD)	2		✓
14–19	Savings — Same as type 11, but valid for card-initiated transactions only	2		✓
21	Negotiable order of withdrawal (NOW)	2		✓
31	Credit	3	✓	
32	Credit line	3	✓	
33–39	Credit — Same as type 31, but valid for card-initiated transactions only	3	✓	
41	Installment loan	3	✓	
42	Mortgage loan	3	✓	
43	Commercial loan	3	✓	
50	Payment to utility	3		✓
51	Payment to utility 1	3		✓
52	Payment to utility 2	3		✓
53	Payment to utility 3	3		✓
54	Payment to utility 4	3		✓
55	Payment to utility 5	3		✓

Hosts and DPCs

Hosts can play a prominent role in the transaction processing performed by BASE24. A host is an external mainframe computer system with which BASE24-teller interacts to authorize transactions and/or update accountholder records.

Although the term host generally refers to an actual computer or system of computers, the term also refers to the institution or organization responsible for the host computer system and its processing. For example, in the phrase, “the host can opt to receive advice messages,” the term host actually refers to the BASE24-defined institution in control of the host computer, rather than the computer itself.

BASE24 also refers to hosts as data processing centers (DPCs). Generally, the terms DPC and host are synonymous. However, the term host implies either a host computer or the organization responsible for a host computer; the term DPC always characterizes the host computer or computer network itself.

A DPC can be one mainframe computer, or a DPC can represent several mainframe computers networked together at a host site. Regardless of the machine configuration, a DPC represents a single host from BASE24-teller’s point-of-view. A host site can establish multiple DPCs; however, BASE24 considers each of these DPCs to be a separate host.

DPCs are assigned identification numbers and are defined to BASE24 via the Host Configuration File (HCF). Each DPC must have a record in the HCF to be recognized by BASE24. The HCF contains options at the DPC-level that allow hosts to specify individual processing parameters for each DPC.

BASE24-teller Transaction Processing

BASE24-teller's role in transaction processing is to provide the means by which a transaction acquirer (originator) can request and receive authorization for an action on a customer card or account from an issuer (authorizer).

BASE24-teller provides the means for routing transactions, participating in or carrying out authorization for the institutions it processes, and logging the transactions for later processing and settlement.

Supported BASE24-teller Transactions

BASE24-teller supports over 300 transactions. Each of these transactions has a unique identifier, known as a transaction code, that is recognized by BASE24-teller.

Transaction Code

A BASE24-teller transaction code consists of three two-digit parts: transaction code, *from* account type, and *to* account type. The two-digit transaction code identifies the type of processing being performed. The *from* account type and *to* account type identify the type of account receiving funds from the transaction or providing the funds for the transaction. BASE24-teller uses the account type codes defined earlier in this section to identify the *from* and *to* accounts. For example, transaction code 401101, when divided into its three two-digit parts, identifies a transfer (40) from savings (11) to checking or DDA (01).

The term transaction code identifies the six-digit identifier or the first two-digit part of the six-digit identifier. References to a two-digit transaction code typically include all of the six-digit transaction codes that begin with the two-digit transaction code. For example, a reference to transaction code 11 (cash transit check) includes three six-digit transaction codes:

110000 = Cash transit check (log only)
110100 = Cash transit check (DDA)
112100 = Cash transit check (NOW)

Transaction Categories

Each BASE24-teller transaction code fits in one of three general transaction categories. The category determines which type of message BASE24-teller can use to carry the transaction information.

Financial transactions involve the movement of funds. BASE24-teller carries these transactions in 0200- and 0400-series messages.

File Inquiry/Update transactions review or update information contained in the database. BASE24-teller carries these transactions in 0300-series messages.

Administrative transactions establish the business date used when posting financial transactions. Administrative transactions also perform user-defined functions. BASE24-teller carries these transactions in 0600-series messages.

The following table lists all of the two-digit BASE24-teller transaction codes and the transaction category to which each one belongs. Appendix C provides the six-digit transaction code and a brief description for each of the BASE24-teller transactions.

Tran Code	Description	Financial	Inquiry/Update	Administrative
10	Withdrawal	✓		
11	Cash transit check	✓		
12	Cash official check	✓		
13	Cash CD	✓		
14	Cash bond	✓		
15	Cash coupon	✓		
16	Miscellaneous debit	✓		
17	Cash affiliate/correspondent check	✓		
18	Debit memo post	✓		

Tran Code	Description	Financial	Inquiry/Update	Administrative
20	Regular deposit	✓		
21	Split deposit	✓		
22	Miscellaneous credit	✓		
23	Credit memo post	✓		
30	PBF inquiry		✓	
31	PBF short inquiry		✓	
32	SPF inquiry		✓	
33	CAF inquiry		✓	
34	NBF inquiry		✓	
35	Passbook print		✓	
36	Passbook reprint		✓	
37	WHFF inquiry		✓	
40	Transfer	✓		
50	Payment	✓		
60	Purchase money order	✓		
61	Purchase cashier's check	✓		
62	Purchase traveler's checks	✓		
63	Purchase bank draft	✓		

Tran Code	Description	Financial	Inquiry/Update	Administrative
64	Purchase certified check	✓		
65	Purchase bond	✓		
66	Purchase miscellaneous	✓		
73	Change CAF card status		✓	
74	Change CAF and PBF account status		✓	
75	Verify PIN	✓		
80	Add stop payment to SPF		✓	
81	Delete stop payment from SPF		✓	
82	Change PBF account status		✓	
83	Change PBF stop pay/warning status		✓	
84	Add warning to WHFF		✓	
85	Add hold to WHFF		✓	
86	Delete hold from WHFF		✓	
87	Add float to WHFF		✓	
88	Delete float from WHFF		✓	
89	Delete warning from WHFF		✓	
90	Logon			✓
91	Logoff			✓
92	Signon			✓

Tran Code	Description	Financial	Inquiry/Update	Administrative
93	Signoff			✓
94–99	Reserved for additional administrative transactions			✓

Insufficient Funds

BASE24-teller uses overdraft limits and automatic overdraft protection to authorize checks and withdrawals for an accountholder even though he or she does not have sufficient funds in the account. A financial institution may prefer to overdraw a customer's account or transfer funds between a customer's accounts to cover the shortage of funds instead of returning checks or withdrawals unpaid, either as a service to the customer or as a way to reduce the number of items that the financial institution must handle manually.

An overdraft limit is the amount that a customer's primary account can be overdrawn and still have checks and withdrawals authorized. BASE24-teller can approve a transaction without an override even if the transaction amount exceeds the available balance, as long as the resulting negative balance does not exceed the overdraft limit amount. The value in the OVERDRAFT LIMIT field of the accountholder's PBF record controls this limit. This field can be accessed via PBF screen 1.

Automatic overdraft protection is the transfer of funds to an accountholder's primary account so that checks and withdrawals can be authorized without overdrawing the account. BASE24-teller can transfer funds from the accountholder's line of credit account or backup account as part of the processing it performs for a financial transaction. Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with automatic overdraft protection.

Processing performed by the BASE24-teller Authorization process for a subset of the BASE24-teller transaction set includes overdraft limits, automatic overdraft protection, or both. These transactions are identified in appendix C.

Preauthorized Holds

When processing transactions that would reduce the balance of an account, BASE24-teller can check for preauthorized holds when computing account balances. Preauthorized holds can be placed on an account by merchants using BASE24-pos before the customer makes a purchase to ensure that funds will be available when the purchase is actually made. As an example, a hotel clerk could place a preauthorized hold on a customer's account when he or she checks in to ensure that funds will be available in the account to pay the bill when the customer is ready to check out.

Refer to section 4 of this manual for the available options. BASE24-teller transactions that can include checking for preauthorized holds are identified in appendix C.

Passbooks

Each savings, IRA, CD, NOW, installment loan, mortgage loan, and commercial loan account maintained by BASE24-teller can be handled as a passbook or nonpassbook account. With a passbook account, a financial institution provides the customer with a passbook as evidence of the account and records each transaction in the passbook. With a nonpassbook account, a financial institution typically uses cards or coupons as evidence of the account and usually provides the customer with a receipt as a record of each transaction. A flag set in the PBF when an account is established determines whether BASE24-teller handles the account as a passbook or nonpassbook account.

When processing a financial transaction involving a passbook account, BASE24-teller processes the transaction as if the passbook is not available and writes the transaction to a No Book File (NBF) for later posting to the customer's passbook. Without the automatic passbook printing feature (described later), the teller must first post the financial transaction and then use another transaction to print the entries in the customer's passbook. Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with passbook processing.

Processing performed by the BASE24-teller Authorization process for a subset of the BASE24-teller transaction set can include writing the transaction to the NBF for later posting to the customer's passbook. The transactions in this subset are identified in appendix C.

Automatic Passbook Printing

While processing certain transactions, BASE24-teller can automatically print the transaction currently being processed and any transactions performed since the last time the passbook was updated. With the automatic passbook print feature, BASE24-teller adds the transaction to the NBF and automatically starts a passbook print transaction while processing the financial transaction. Without the automatic passbook print feature, the teller must first post the financial transaction and then use another transaction to print the entries in the customer's passbook. Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with automatic passbook printing.

Processing performed by the BASE24-teller Authorization process for a subset of the BASE24-teller transaction set can include automatically printing missing entries the customer's passbook. The transactions in this subset are identified in appendix C.

Stop Payments

As part of the processing performed for some of the check cashing transactions, BASE24-teller searches the Stop Payment File (SPF) for stop payment information before authorizing the transaction. A stop payment order can be placed on a check written by one of a financial institution's own customers to prevent the check from being cashed by anyone. Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with stop payments.

Warning, Hold, and Deposit Float Records

As part of the processing performed for financial transactions, BASE24-teller can search the Warning/Hold/Float File (WHFF) for warning information. Warnings alert tellers about possible problems involving a particular account.

The WHFF also contains hold and float information. Holds decrease the available funds balance and increase the amount on hold for a noncredit account or decrease the available credit and increase the credit balance for a credit account. Deposit floats increase the available funds balance for a noncredit account or increase the available credit for a credit account. BASE24-teller does not search the WHFF for hold and float information because balances in the Positive Balance File (PBF) already reflect all hold and float amounts.

Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with warning, hold, and deposit float records.

Section 2

BASE24-teller Transaction Path

A number of processes and processing entities form the BASE24-teller transaction path—this is, the series of BASE24-teller processes and processing entities through which a specific transaction passes in order to be authorized.

The actual path each transaction takes and the information each message contains vary based on a number of factors, including the type of transaction being processed, the transaction originator, the transaction authorizer, and the processing set up by the BASE24 processor.

This section describes BASE24-teller messages, including their sources, destinations, and how they use tokens. It also describes the processes involved in BASE24-teller transaction processing and how they are involved, as well as the timers that BASE24-teller supports along the transaction path.

Transaction Originators

BASE24-teller receives transactions only from financial institution personnel operating teller terminals. These terminals usually are connected to a controller that is responsible for communicating with BASE24-teller. Communications between the controllers and BASE24-teller are controlled by BASE24-teller Device Handlers.

BASE24-teller does not acquire transactions from interchanges, other BASE24-teller co-networks, or teller terminals controlled by hosts.

Transaction Authorizers

BASE24-teller can authorize transactions or pass them to a host for authorization. The authorizer can be specified on a transaction-by-transaction basis, but the same authorizer must process both parts of a two-part transaction (e.g., split deposit or transfer between accounts).

BASE24-teller

BASE24-teller can be set up to authorize transactions in situations where customer accounts are maintained in the Positive Balance File (PBF) on the BASE24-teller system or for transactions where BASE24-teller processing is limited to logging the transaction to the Teller Transaction Log File (TTLF).

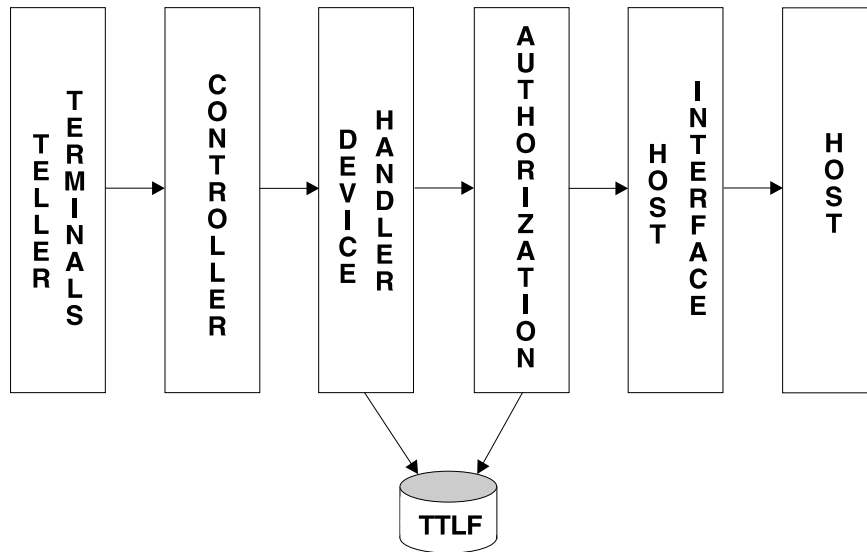
Transactions defined as log-only can be logged to the TTLF by the BASE24-teller Device Handler or the BASE24-teller Authorization process. All other transactions are logged to the TTLF by the BASE24-teller Authorization process.

Hosts

Hosts can be set up to authorize transactions in situations where customer accounts are maintained on the host. Unlike BASE24-atm and BASE24-pos, however, BASE24-teller cannot act as a stand-in authorizer in situations when the host system is unavailable. A transaction to be authorized by the host is declined when the host is unavailable. The BASE24-teller Authorization process logs responses to transactions other than advices to the TTLF and logs advice transactions to the TTLF without a response.

BASE24-teller Processes

All BASE24-teller transactions start at a teller terminal, so there is one basic path that a BASE24-teller transaction can follow. This path is illustrated below.



Along the transaction path, processing is carried out in stages by several BASE24-teller processes, each of which has its own responsibilities. Those BASE24-teller processes that can be involved in transaction processing are described below.

BASE24-teller processes all communicate with one another through the Nucleus.

BASE24-teller Device Handler

The Device Handler is responsible for communicating with the teller terminal controllers directly attached to BASE24-teller. BASE24-teller Device Handlers are written for each type of device attached to BASE24-teller, based on the customer's operating requirements. However, all Device Handlers perform essentially the same functions, which include the following:

- Recognizing native-mode messages sent from the teller terminal or controller and translating them into BASE24 internal message format.
- Writing messages defined as Device Handler log-only transactions directly to the transaction log file without sending them to any other BASE24-teller process.

- Sending messages other than those defined as Device Handler log-only transactions to a BASE24-teller Authorization process.
- Setting and stopping internal timers to control the maximum length of time the Device Handler will wait for a response from a BASE24-teller Authorization process before canceling the transaction.
- Setting flags in the Teller Standard Internal Message Header (TSTMH) and internal message tokens to control the processing performed by the BASE24-teller Authorization process or the host. The flags set depend on the transactions supported and the BASE24-teller product features used.

Note: Each BASE24-teller Device Handler is developed to meet a customer's specific requirements. Refer to the informal documentation received with the Device Handler software for information about the Device Handler's installation, configuration, and operation.

BASE24-teller Authorization Process

The Authorization process is responsible for authorizing the transactions it receives or routing them to the appropriate BASE24-teller Host Interface process. The Authorization process logs all transactions it receives to the transaction log file, and can generate completion messages to notify hosts regarding transactions approved or declined by BASE24-teller.

BASE24-teller Host Interface Process

The Host Interface process is responsible for communicating with the host system when the host is the appropriate authorizer or is to receive completion messages for transactions authorized by BASE24-teller. BASE24-teller does not accept (acquire) transactions from teller devices or controllers that are attached to a host. Because two possible message formats, ISO-based and ANSI-based, can be passed between BASE24-teller and the host, there are two types of Host Interface processes. One BASE24-teller Host Interface process is for hosts using the ANSI-based message format. The other Host Interface process is for hosts using the ISO-based message format.

The Host Interface process translates messages from the BASE24-teller internal message format to the ISO or ANSI external format recognizable to a host. The ISO-based message can contain more information than the ANSI-based message. New information can be added to the ISO-based message using tokens. When new information must be added to the ANSI-based message, the entire message must be restructured and all processes using the message are impacted.

The ISO Host Interface process supports the following BASE24-teller features that are not available with the ANSI Host Interface process:

- Interbank teller transaction routing
- Automatic passbook printing
- Credit line/backup account
- PIN processing
- Card verification
- Message authentication
- Internal administrative (06x0) messages
- Native message data from teller terminal included with the external message sent to the host
- Customer short name included in responses to the teller terminal
- Complete listing of applicable response codes provided in override situations

Each of these features is presented in greater detail in sections 4 and 5.

Note: A Host Interface process can process host messages for multiple BASE24 products. If an institution purchases multiple BASE24 products, the appropriate product modules are bound into one Host Interface process. This allows the Host Interface process to recognize and process transaction messages for different products.

For complete information on the Host Interface processes, refer to the ***BASE24 ISO Host Interface Manual*** and the ***BASE24 ANSI Host Interface Manual***.

BASE24-teller Interchange Interface Process

BASE24-teller does not have any Interchange Interface processes. Interchanges accept transactions originating at automatic teller machines (ATMs) or point of sale (POS) devices, but not teller devices; therefore, Interchange Interface support for BASE24-teller is not necessary.

BASE24-teller BIC Interface Process

BASE24 Interchange (BIC) Interface support for BASE24-teller is unavailable at the present time. A BIC Interface process allows one BASE24 network to communicate with another BASE24 or similar network. The current release of BASE24-teller routes transactions only within a single logical network; therefore, BIC Interface support for BASE24-teller is not necessary.

Process Linkages

Since multiple processes are involved in transaction processing, transactions must be sent between them for processing. The links between these processes can be configured. Each link in the transaction path is defined below.

Defining Stations Connected to BASE24

All teller terminals, controllers, and hosts are defined to the BASE24 system as stations. Stations provide the means by which communication takes place between BASE24 and a DPC or device. All messages sent to a DPC are actually sent to the stations defined for the DPC. Likewise, all messages received from a DPC are actually sent from stations.

The Nucleus controls BASE24's interaction with its stations. The Nucleus receives inbound transactions and routes them to the appropriate processes. The Nucleus determines which process to send a message to via a station table in memory, which it builds at startup from the Network Environment Source File (NEFS) station declarations.

Each station declaration in the NEFS contains a DESTINATION element, which identifies the process to which all messages from the station are to be sent. In the case of teller terminals or controllers, the DESTINATION element should specify a Device Handler. For host stations, the DESTINATION element should specify a Host Interface process. Using the DESTINATION element, the Nucleus forwards messages to the designated process.

Once the designated process receives the message from the station, that process verifies that it should be receiving messages from the station.

Inbound Messages to a Device Handler Process

A BASE24-teller Device Handler verifies that it is supposed to receive the messages from a particular device based on the value in the DH PROCESS field in the Teller Terminal Data File (TTDF), which can be accessed via TTDF screen 1. The value in this TTDF field should match the DESTINATION element entry in the NEFS.

Device Handler to Authorization Process

A BASE24-teller Device Handler determines whether a transaction must be sent to a BASE24-teller Authorization process based on the value in the TERM TRAN ALLOWED field in the Teller Transaction File (TTF), which can be accessed via TTF screen 1. One TTF record exists for each transaction code in use at each FIID. The Device Handler determines the appropriate TTF record by comparing the values in TSTMH fields to the values in corresponding TTF fields, as shown in the following table. All TTF fields are accessed via TTF screen 1.

TSTMH Fields	TTF Fields
BNK.TERM-FIID *	FIID
RQST.TRAN.CDE	TRANSACTION CODE
RQST.TRAN.FROM-ACCT-TYP	FROM ACCOUNT TYPE
RQST.TRAN.TO-ACCT-TYP	TO ACCOUNT TYPE

* The Device Handler obtains the value for the BNK.TERM-FIID field in the TSTMH from the FIID field in the Teller Terminal Data File (TTDF), which can be accessed via TTDF screen 1.

If the value in the TERM TRAN ALLOWED field of the TTF record indicates that the Device Handler should log the transaction, the Device Handler logs the transaction without passing it to an Authorization process. Otherwise, the Device Handler determines the Authorization process to which to send transaction requests using the value defined in the AUTH NAME field of the TTDF, which can be accessed via TTDF screen 1. The value in this TTDF field can indicate either a specific Authorization process or a Generic Queue process.

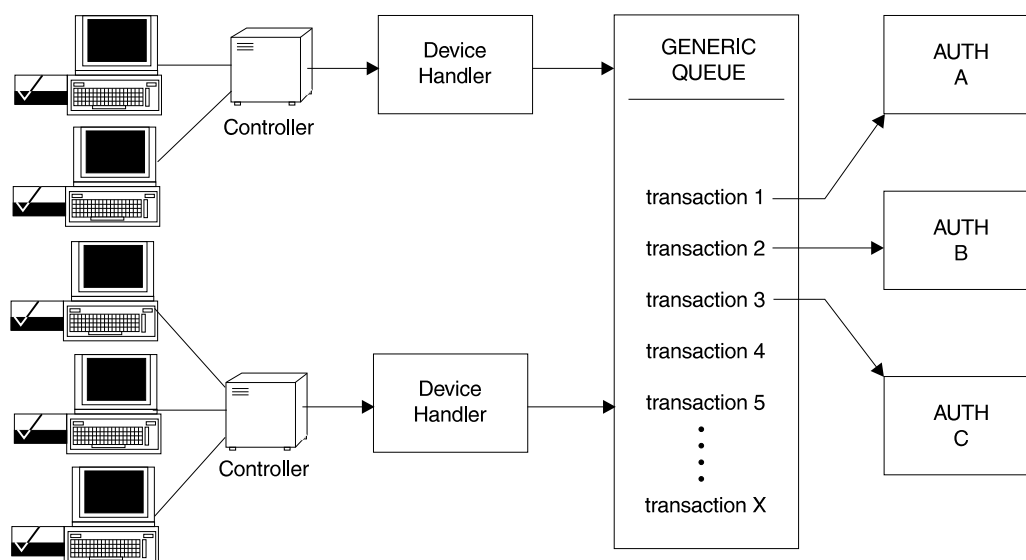
Authorization Process to Host

The BASE24-teller Authorization process determines whether a transaction must be sent to a host using the value in the AUTH LEVEL field of the TTF, which can be accessed via TTF screen 1. If the value in the AUTH LEVEL field of the TTF record specifies that a transaction must be sent to a host, the BASE24-teller Authorization process then determines the proper host using information from the PRIMARY DPC, SYMBOLIC NAME, and ACCOUNT TYPE fields in the Teller Routing Table of the Institution Definition File (IDF), which can be accessed via IDF screen 28. The BASE24-teller Authorization process determines the appropriate Host Interface process by comparing the account types in the RQST. TRAN.FROM-ACCT-TYP and RQST. TRAN.TO-ACCT-TYP fields of the TSTMH to the values in the ACCOUNT TYPE fields in the IDF Teller Routing Table.

Generic Queuing

When multiple Authorization processes are in use, BASE24-teller allows Device Handlers to send transaction requests to a Generic Queue process instead of directly to an Authorization process. This option is designed to avoid the processing inefficiencies that could develop when only one Authorization process is assigned to each Device Handler.

Without generic queuing, all of the teller terminals and controllers assigned to one Authorization process have to wait for that Authorization process to complete their transactions even if there are other Authorization processes available. With generic queuing, transactions can be distributed among multiple Authorization processes.



Under generic queuing, the name of the Generic Queue process is placed in the TTDF instead of the name of a particular Authorization process. Transaction requests are sent to the Generic Queue process, which in turn queues the requests. The Generic Queue process then sends the request to the first available Authorization process.

To use generic queuing, the name of a Generic Queue process must be placed in a PROCESS declaration in the Network Environment Source File (NEFS). The name of the Generic Queue process must also be placed in STATION declarations in the NEFS as a destination process. Then, each Authorization process involved in generic queuing must have the Generic Queue process named as the alternate

queue in its own PROCESS declaration in the NEFS. For more information about declaring a Generic Queue process in the NEFS, refer to the ***BASE24 Network Environment Source File Manual***.

Note: Generic queuing for BASE24-teller applies only between Device Handlers and Authorization processes because BASE24-teller does not acquire transactions from interchanges or host-controlled devices.

Transaction Timers

Timers enable processes to take action in situations where responses have not been received. BASE24-teller uses timers for logical network processing, device processing, and host processing. These various timers are set in the Logical Network Configuration File (LCONF) and the Host Configuration File (HCF).

General Timer

The following general timer is used by multiple BASE24 processes.

Security Device Timer Limit — BASE24-teller processes can perform transaction security activities such as PIN encryption, PIN verification, and card verification internally or by passing the transaction data to external security modules. When security modules are used, the SECURE-DEV-TIM-LMT param in the LCONF specifies the maximum amount of time, in hundredths of a second, a BASE24-teller Device Handler, Authorization, or Host Interface process waits for a response from the security module before timing out. Refer to the ***BASE24 Transaction Security Manual*** for additional information about transaction security activities.

Device Handler Timer

The following timer is used in transaction processing by BASE24-teller Device Handlers.

Device Handler Timer Limit — The TLR-DH-AUTH-TIMER param in the LCONF specifies the maximum amount of time, in seconds, that BASE24-teller Device Handlers wait for an authorization response from BASE24-teller Authorization processes before canceling a transaction.

Note: Each BASE24-teller Device Handler is developed to meet a customer's specific requirements. Refer to the informal documentation received with the Device Handler software for additional information about transaction timers and other configuration options.

Host Interface Process Timers

The following timers are used by Host Interface processes. Not all are directly related to transaction processing; some are related to network management.

Network Management Timer Limit — This timer limit can be set in the NETWORK MANAGEMENT field on HCF screen 1. It specifies the amount of time, in seconds, a BASE24 Host Interface process waits for a response after sending a network management message to a data processing center (DPC) station. If a response message is not received within the time specified, the Host Interface process marks the station as down.

Extended Network Timer Limit — This timer limit can be set in the EXTENDED NETWORK field on HCF screen 1. It specifies the amount of time, in seconds, a BASE24 Host Interface process waits, once a station has been marked down, before sending an echo-test message.

Store and Forward Timer Limit — This timer limit can be set in the STORE AND FORWARD field on HCF screen 1. It specifies the amount of time, in seconds, a BASE24 Host Interface process waits for an acknowledgment after sending a store-and-forward message to the DPC.

Wait for Traffic Timer Limit — This timer limit can be set in the WAIT FOR TRAFFIC field on HCF screen 1. It specifies the amount of time, in seconds, a BASE24 Host Interface process waits for traffic on the line to a station before initiating an echo test.

Outbound Timer Limit — This timer limit can be set in the OUTBOUND LIMIT field on HCF screen 10. It specifies the amount of time, in seconds, a BASE24 Host Interface process waits for a response message after sending a BASE24-teller transaction request message to the host.

Completion Acknowledgment Message Timer Limit — This timer limit can be set in the COMPLETION ACK field on HCF screen 10. It specifies the amount of time, in seconds, a BASE24 Host Interface process waits for an acknowledgment message after sending a BASE24-teller advice or reversal message to the DPC.

Queue Subtract Timer Limit — This timer limit can be set in the QUEUE SUBTRACT field on HCF screen 10. It specifies the amount of time, in seconds, the Nucleus is to subtract from the value in the OUTBOUND TIMER LIMIT field to compute the maximum time the Nucleus can wait before sending the BASE24-teller message to the host. If the Nucleus cannot send the message to the host within this time, it returns the message to the Host Interface process as stale. This option is designed to circumvent processing when it becomes apparent that sending a message will not give a host time to respond before the message times out.

BASE24-teller Transaction Messages

All transactions are carried out via messages exchanged between the originator and authorizer of the transaction. These messages are assigned numbers to identify their message type and function throughout a BASE24-teller system.

Basic Message Functions

Messages can be divided into four basic categories: request, response, advice/force post, and reversal.

Request — A message from a transaction originator requesting the authorizer to authorize a financial or administrative transaction, provide database information, or update database information. Automatic passbook printing can be performed for passbook accounts; multiple account selection can be performed with card-initiated transactions; and teller terminal operators can perform database inquiries and response code overrides.

The types of internal messages used with requests are as follows:

0200 = Financial transaction request
0300 = File inquiry or update request
0600 = Administrative transaction request

Response — A message sent from a transaction authorizer to the transaction originator approving or denying a financial or administrative transaction request or advice, database inquiry request, or database update request or advice.

The types of internal messages used with responses are as follows:

0210 = Financial transaction response
0230 = Financial transaction advice response
0310 = File inquiry or update response
0330 = File update advice response
0430 = Financial transaction reversal response
0610 = Administrative response
0630 = Administrative advice response

Advice/Force Post — A message sent from a transaction originator to the authorizer so the authorizer can update its files. Advices are also known as force post transactions because the authorizer must accept the transaction. Automatic passbook prints, passbook prints, passbook reprints, PIN verification, multiple account selection, database inquiries, and response code overrides cannot be performed with advice/force post messages.

The types of internal messages used with advice/force posts are as follows:

0220 = Financial transaction advice
0320 = File update advice
0620 = Administrative transaction advice

Reversal — A message indicating that any database impacts from an earlier transaction should be reversed. Reversals can be initiated by a teller to cancel the effect of a transaction that was completed incorrectly. Reversals can be initiated by BASE24 to cancel the effect of a request when BASE24 files have been updated, but a response cannot be returned to the teller.

Multiple account selection, database inquiries and updates, and response code overrides cannot be reversed because BASE24-teller does not retain the information required to perform the reversal.

Reversals of automatic passbook updates cannot be initiated by a teller; however, reversals of automatic passbook updates can be initiated by BASE24 because the information required to perform the reversal is still available to the Device Handler.

The 0420 internal message (Financial transaction reversal advice) is used with reversals.

BASE24-teller Message Types

Message types are used inside and outside of BASE24-teller. The following table identifies the message types used inside BASE24-teller in more detail, and includes the message types as they would be sent to a host. Message types are provided for the messages as they are used internally within BASE24-teller and their ANSI and ISO equivalents used by Host Interface processes.

INT	ISO	ANSI	Description
0200	0200	8200	Financial Transaction Request — A request to approve a transaction, which if approved can be immediately applied to the customer's account for billing or statement purposes. A financial transaction response (0210) is expected in return for the 0200 message, either approving or denying the request.
0210	0210	8210	Financial Transaction Response — A response returned for a financial transaction request (0200) to approve or deny the request.
0220	0220	8220 or 8200	Financial Transaction Advice — An advice of a previously completed financial transaction. A Device Handler can send financial transaction advices to the Authorization process for transactions authorized by the controller. An Authorization process, when authorizing transactions, can send financial transaction advices to the host based on the setting (approved, declined, both, or none) of the COMPLETIONS TO HOST field in the Teller Transaction File (TTF), which can be accessed via TTF screen 1. Note: The ANSI message type is determined by the value of the Advice Response Required Flag. Refer to the Advice Response Required Flag discussion later in this section for additional information.
Not used	0221	Not used	Financial Transaction Advice Repeat — A message that is identical to a financial transaction advice, except that it denotes to the receiver that it is a possible duplicate message. This message is used when an acknowledgment was expected in response to a financial transaction advice but never received.

INT	ISO	ANSI	Description
0230	0230	8230	Financial Transaction Advice Response — An acknowledgment that a financial transaction advice or financial transaction advice repeat was received. Refer to the Advice Response Required Flag discussion later in this section for additional information.
0300	0300	8500	NBF Update Request — A request to print or reprint a passbook.
0300	0300	8510	SPF Update Request — A request to add a record to or delete a record from the SPF.
0300	0300	8520	PBF Update Request — A request to modify status information in a PBF record.
0300	0300	8530	SPF Inquiry Request — A request for information from records in the SPF.
0300	0300	8540	PBF Inquiry Request — A request for information from a record in the PBF.
0300	0300	8550	CAF Inquiry Request — A request for information from a record in the CAF.
0300	0300	8560	NBF Inquiry Request — A request for information from records in the NBF.
0300	0300	8570	CAF Update Request — A request to modify information in a CAF record.
0300	0300	8580	WHFF Update Request — A request to add a record to or delete a record from the WHFF.
0300	0300	8590	WHFF Inquiry Request — A request for information from records in the WHFF.
0310	0310	8505	NBF Update Response — The response to an NBF update request (8500).
0310	0310	8515	SPF Update Response — The response to an SPF update request (8510).

INT	ISO	ANSI	Description
0310	0310	8525	PBF Update Response — The response to a PBF update request (8520).
0310	0310	8535	SPF Inquiry Response — The response to an SPF inquiry request (8530).
0310	0310	8545	PBF Inquiry Response — The response to a PBF inquiry request (8540).
0310	0310	8555	CAF Inquiry Response — The response to a CAF inquiry request (8550).
0310	0310	8565	NBF Inquiry Response — The response to an NBF inquiry request (8560).
0310	0310	8575	CAF Update Response — The response to a CAF update request (8570).
0310	0310	8585	WHFF Update Response — The response to a WHFF update request (8580).
0310	0310	8595	WHFF Inquiry Response — The response to a WHFF inquiry request (8590).
0320	0320	8600	NBF Update Advice — Advises of a previously completed passbook print or passbook reprint transaction.
0320	0320	8610	SPF Update Advice — Advises of a previously completed request to add a record to or delete a record from the SPF.
0320	0320	8620	PBF Update Advice — Advises of a previously completed request to modify status information in a PBF record.
0320	0320	8630	SPF Inquiry Advice — Advises of a previously completed request for information from the SPF.
0320	0320	8640	PBF Inquiry Advice — Advises of a previously completed request for information from the PBF.
0320	0320	8650	CAF Inquiry Advice — Advises of a previously completed request for information from the CAF.

INT	ISO	ANSI	Description
0320	0320	8660	NBF Inquiry Advice — Advises of a previously completed request for information from the NBF.
0320	0320	8670	CAF Update Advice — Advises of a previously completed request to modify information in a CAF record.
0320	0320	8680	WHFF Update Advice — Advises of a previously completed request to add a record to or delete a record from the WHFF.
0320	0320	8690	WHFF Inquiry Advice — Advises of a previously completed request for information from the WHFF.
Not used	0321	Not used	File Inquiry/Update Advice Repeat — A message that is identical to a File Inquiry/Update advice, except that it denotes to the receiver that it is a possible duplicate message. This message is used when an acknowledgment was expected in response to a File Inquiry/Update advice but never received.
0330	0330	8605	NBF Update Advice Response — A response to an NBF update advice (8600).
0330	0330	8615	SPF Update Advice Response — A response to an SPF update advice (8610).
0330	0330	8625	PBF Update Advice Response — A response to a PBF update advice (8620).
0330	0330	8635	SPF Inquiry Advice Response — A response to an SPF inquiry advice (8630).
0330	0330	8645	PBF Inquiry Advice Response — A response to a PBF inquiry advice (8640).
0330	0330	8655	CAF Inquiry Advice Response — A response to a CAF inquiry advice (8650).
0330	0330	8665	NBF Inquiry Advice Response — A response to an NBF inquiry advice (8660).
0330	0330	8675	CAF Update Advice Response — A response to a CAF update advice (8670).

INT	ISO	ANSI	Description
0330	0330	8685	WHFF Update Advice Response — A response to a WHFF update advice (8680).
0330	0330	8695	WHFF Inquiry Advice Response — A response to a WHFF inquiry advice (8690).
0420	0420	8400 or 8420	Financial Transaction Reversal Advice — Reverses an earlier transaction or authorization. BASE24-teller can send this message in three circumstances: • A 0200 message is approved by a host, but the transaction does not complete as approved. • A 0220 message is sent to a host to notify it of a completed transaction, but the transaction does not actually complete as the host has been advised. • The teller initiates a reversal of a completed transaction. Note: The ANSI message type is determined by the value of the Advice Response Required Flag. Refer to the Advice Response Required Flag discussion later in this section for additional information.
Not used	0421	Not used	Financial Transaction Reversal Advice Repeat — A message that is identical to a financial transaction reversal advice, except that it denotes to the receiver that it is a possible duplicate message. This message is used when an acknowledgment was expected in response to a financial transaction reversal advice but never received.
0430	0430	8410 or 8430	Financial Transaction Reversal Response — An acknowledgment that a financial transaction reversal advice (0420) or financial transaction reversal advice repeat (0421) was received. (The 0421 message is an ISO message only.) Note: The ANSI message type is determined by the value of the Advice Response Required Flag. Refer to the Advice Response Required Flag discussion later in this section for additional information.

INT	ISO	ANSI	Description
0600	0600	Not used	Administrative Request — A message used to perform logons, logoffs, signons, and signoffs for each teller at each terminal. An administrative response (0610) is expected in return for the 0600 message.
0610	0610	Not used	Administrative Response — A message used to respond to an Administrative request (0600) to denote the disposition of the teller logon, logoff, signon, or signoff.
0620	0620	Not used	Administrative Advice — A message to advise of a teller logon, logoff, signon, or signoff.
Not used	0621	Not used	Administrative Advice Repeat — A message that is identical to an administrative advice, except that it denotes to the receiver that it is a possible duplicate message. This message is used when an acknowledgment was expected in response to an administrative advice but never received.
0630	0630	Not used	Administrative Advice Response — An acknowledgment that an administrative advice or administrative advice repeat was received. Refer to the Advice Response Required Flag discussion later in this section for additional information.
Not used	0800	0800	Network Management Message Request — A message used to perform echo-tests, logons, and logoffs. BASE24 sends 0800 messages only if the value in the NMM-ENABLED field in the HCF, which can be accessed via HCF screen 1, is set to 1 (enabled).
Not used	0810	0810	Network Management Message Response — A message that is returned in response to an 0800 message, regardless of the value in the NMM-ENABLED field in the HCF.

INT	ISO	ANSI	Description
Not used	Not used	8920	Store-and-Forward Request — A transaction that is being sent to the host from the BASE24 Store-and-Forward File (SAF). When the value of the Advice Response Required flag is 0 (authorizer does not acknowledge advice messages) and an advice or reversal cannot be sent to a host because communications with the host are down, the transaction is stored in the SAF for sending when host communications are restored. Store-and-forward processing is not performed when the value of the Advice Response Required flag is 1 (authorizer acknowledges advice messages).
Not used	Not used	8921	Store-and-Forward Repeat — A reissue of a store-and-forward request. A store-and-forward repeat is identical to the original request it duplicates, except that it denotes to the receiver that it is a possible duplicate message. A store-and-forward repeat is used when an acknowledgment was not received for a store-and-forward request.
Not used	Not used	8930	Store-and-Forward Response — An acknowledgment of a receipt generated by a host in response to a store-and-forward request.

Tokens

Tokens provide BASE24-teller with flexibility in the information carried in its internal messages, logged to its log files, extracted by the Super Extract process, and sent to a host via the ISO Host Interface process.

A token is a collection of related data which may or may not be needed to process a particular transaction. A token can be made up of one field or a series of related fields. If the information is needed to process a transaction, it can be added to the internal message, logged, and extracted. If the information is not needed, the token need not be involved in the transaction at all—carried in the message, logged, or extracted.

BASE24-teller makes a much greater use of tokens than BASE24-atm and BASE24-pos. BASE24-atm and BASE24-pos carry the majority of the information needed for their transactions in their standard internal messages and use tokens primarily to handle special information. However, BASE24-teller uses a standard internal message, called the Teller Standard Internal Message Header (TSTMH), that contains a limited amount of information for all transactions, then uses tokens to carry the information needed for each transaction.

When the first token is added to the message, the Header token is also added. The Header token contains a unique character and separates the remaining tokens from the standard internal message. Once the Header token is in place, subsequent tokens can be added to the message individually as required. The general structure of an internal message with message tokens is illustrated below.

BASE24-teller Internal Message with Tokens

TSTMH	Header Token	Token	Token	Token	...
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BASE24-teller Tokens

The standard BASE24-teller product can use 17 tokens. These tokens are listed and briefly described below in alphabetical order. For information on the structure of these tokens and detailed descriptions of the contents, refer to the ***BASE24 Tokens Manual***.

Account Token (Token ID TB) — Contains PBF information that is returned by the Authorization process any time the PBF is accessed successfully. One token can carry information for the *from* account, *to* account, and credit line/backup account. Information for each account includes the account status, ledger and available balances, and the total of all preauthorized holds in the Preauthorized Holds segment of the PBF.

Administrative Token (Token ID TA) — Contains user-defined administrative data that is not used by BASE24-teller for processing, but can be carried in the internal or ISO-based external message, logged to the TTLF, or extracted from the TTLF.

CAF Inquiry Token (Token ID T1) — Contains CAF information used in financial, file inquiry, and file update transactions (message types 0210, 0300, 0310, and 0320). Information includes paging indicator, card type, card status, card expiration date, and the number of accounts included in this token. One token can carry information for up to five accounts from the CAF record. Information for each account includes the account type, the account number, the account status, and an account description.

CAF Update Token (Token ID T2) — Contains CAF information used in file update transactions (message types 0300, 0310, 0320, and 0330). Information includes the card status, account status, and account description.

Credit Line Token (Token ID 13) — Contains the account type and account number of the account the BASE24-teller Authorization process used as a backup/credit line account, along with the amount transferred from the backup/credit line account to the primary account in order to authorize the transaction. This token is returned anytime the Authorization process attempts to access a backup/credit line account to authorize a transaction. It is also returned for PBF inquiry or PBF short inquiry transactions when the primary account has a backup/credit line account.

Customer Name Token (Token ID 08) — Contains information from the Customer Short Name segment of the PBF any time a PBF record with a Customer Short Name segment is accessed successfully. One token can carry information for the *from* account, *to* account, and credit line/backup account.

Financial Token (Token ID T0) — Contains information involved in financial transactions (message types 0200, 0210, 0220, 0230, 0420, and 0430). Information includes the reversal code, transaction amounts 1 through 4, store-and-forward posting date, forwarding institution ID, card-issuing institution ID, currency transaction report counter, check number, and automatic passbook printing flag.

Native Message Token (Token ID TE) — Contains the native message sent from the teller device or other user-defined data. This information is not used by BASE24-teller for processing, but can be carried in the internal or ISO-based external message, logged to the TTLf, or extracted from the TTLf.

NBF Token (Token ID T3) — Contains NBF information used in financial, file inquiry, and file update transactions (message types 0210, 0300, 0310, 0320, 0420, and 0430). Information includes the account indicator, paging indicator, starting and ending passbook balances, and the number of passbook entries included in this token. One token can carry information for up to five passbook entries from the NBF record. Information for each entry includes the transaction date and time, posting date, transaction code, transaction type, transaction amount, passbook balance, print indicator, and codes identifying the device, teller, region, branch, and city where the transaction was performed.

Override Token (Token ID TC) — Contains response codes being returned with the response to a financial, file inquiry, or file update transaction (message type 0210 or 0310). One token can carry information for the *from* account, *to* account, and credit line/backup account. Information for each account includes the account indicator and three bit maps identifying response codes that must be overridden for the transaction to be approved. The bit maps arrange the response codes according to the override level needed.

PBF Inquiry Token (Token ID T4) — Contains PBF information used in financial, file inquiry, and file update transactions (message types 0210, 0300, 0310, and 0320). One token can carry information for the *from* account, *to* account, and credit line/backup account. Information for each account includes the account indicator, confidential flag, overdraft limit, amount on hold, accrued interest, date and amount of last deposit and withdrawal, cash in and out accumulator balances, amount of credit given today, number of deposits today,

ledger and passbook balances, NBF record count and passbook indicator, customer class, cash in and out limits, currency transaction report counter, and stop pay/warning flag.

PBF Update Token (Token ID T5) — Contains PBF information used in file update transactions (message types 0300, 0310, 0320, and 0330). Information includes the account status or stop pay/warning status.

PIN Token (Token ID TD) — Contains PIN processing information used in financial, file inquiry, and file update transactions (message types 0200, 0210, 0300, 0310, 0420, and 0430). Information includes the PIN, PIN format, pad character, PIN size, number of PIN tries, the ANSI offset, and the PIN key.

SPF Inquiry Token (Token ID T6) — Contains SPF information used in financial and file inquiry transactions (message types 0210, 0300, 0310, and 0320). Information includes the account indicator, check numbers used in searching the SPF, whether to include expired records in the SPF search, the value of the stop pay/warning status flag in the PBF, and a paging indicator. One token can carry information from up to five SPF records for a single account. Information for each SPF record includes high and low check numbers, check amount, and stop payment description, date, time, and expiration date.

SPF Update Token (Token ID T7) — Contains SPF information used in file update transactions (message types 0300, 0310, 0320, and 0330). Each token can carry information for one stop payment. Information includes the high and low check numbers, check amount, description, date, time, expiration date, and the value of the stop pay/warning status flag in the PBF.

WHFF Inquiry Token (Token ID T8) — Contains WHFF information used in financial and file inquiry transactions (message types 0210, 0300, 0310, and 0320). Information includes the WHFF record type, the transaction amount, date, and time used in searching the WHFF, and whether to include expired records in the WHFF search. One token can carry information from WHFF records for the *from* account, *to* account, and credit line/backup account. Information for each account includes the value of the stop pay/warning status flag in the PBF, a paging indicator, and information from up to five WHFF records. Information for each WHFF record includes record type, amount, date, time, and expiration date.

WHFF Update Token (Token ID T9) — Contains WHFF information used in update transactions (message types 0300, 0310, 0320, and 0330). Each token can carry information for one WHFF record. Information includes record type, amount, date, time, description, and the value of the stop pay/warning status flag in the PBF.

Configuring BASE24-teller Tokens

Each BASE24-teller token is classified as standard or nonstandard. This classification is important when determining which tokens are sent to a host. All BASE24-teller tokens except the Credit Line token (token ID 13) and the Customer Name token (token ID 08) have a standard classification. Screen 1 of the External Message File (EMF) is used to specify which of the standard tokens contained in the internal message are sent to a host. If nonstandard tokens are to be sent in the external message, data element S-124 (BASE24-teller Additional Data) must be included in the external message by setting its flag in the EMF to conditional or mandatory. Screen 4 of the Token File (TKN) is used to specify which of the nonstandard tokens contained in the internal message are sent to a host via the ISO Host Interface process.

Information in the TKN does not affect the tokens carried in the internal message. The tokens BASE24-teller adds to the internal message depend on the message type, the information entered with the initial request, and the information to be returned with the response. BASE24-teller also adds tokens to inform the teller why a transaction has been declined or to provide additional information so the teller can override the initial response and continue processing the transaction.

Logging Tokens

BASE24-teller processes use the Token File (TKN) to determine which tokens are logged, extracted, or sent to a host. TKN screen 2 is used to specify which tokens contained in the internal message will be logged to the Teller Transaction Log File (TTLF). While the values entered in fields on TKN screen 2 determine which tokens are to be logged to the TTLF, these values do not control the order in which the tokens are arranged in the record that is written to the TTLF. These tokens are arranged in the order they are added to the internal message by BASE24-teller processes. The option of logging or not logging exists for all BASE24-teller tokens.

Extracting Tokens

TKN screen 3 is used to specify which tokens in the TTLF records will be extracted by the Super Extract process. This screen is also used to specify the order in which these tokens will be arranged within each TTLF extract record. The option of extracting or not extracting exists for all BASE24-teller tokens; however, a token must be logged to the TTLF before it can be extracted.

Sending Tokens to a Host

TKN screen 4 is used to specify which of the nonstandard tokens contained in the internal message will be sent to a host via the ISO Host Interface process. This screen is also used to specify the order in which the nonstandard tokens will be carried in element S-124 of the BASE24 external message. Refer to the ***BASE24 External Message Manual*** for additional information.

BASE24-teller Tokens Used in Financial Transaction Messages

There are five possible token combinations used in financial transaction messages. Within each combination, some tokens are required while the remaining tokens are optional and are part of the message only when needed. Tokens are listed here alphabetically, but appear in internal messages in the order they are added to the message by BASE24-teller processes.

Note: There are two token combinations used for financial response (0210) messages. One token combination is used when the response codes being returned in the message can be overridden. The other token combination is used when the response codes being returned in the message indicate approval or failure, neither of which can be overridden.

Financial Request (0200) Messages

Financial request (0200) messages can contain the following tokens:

Financial	Required for all transactions except transaction code 75 (PIN verify).
Header	Required whenever other tokens are included.
Native Message	Presence in the message is controlled by the Device Handler design.
PIN	Included when a PIN is entered with a card-initiated transaction.

Approved or Fatal Financial Response (0210) Messages Financial Reversal Advice (0420) Messages Financial Reversal Response (0430) Messages

Approved or fatal financial response (0210) messages, financial reversal advice (0420) messages, and financial reversal response (0430) messages can contain the following tokens:

Account	Included when account balance information is returned in a response.
Credit Line	Included when a backup/credit line account was accessed to authorize the transaction or the transaction was denied because of a credit line problem.
Customer Name	Included when the customer short name is maintained in the PBF.
Financial	Required for all transactions except transaction code 75 (PIN verify).
Header	Required whenever other tokens are included.
Native Message	Presence in the reversal advice message is controlled by the Device Handler design. In response messages, this token is present if it was included in the original request or advice.

NBF	Included when automatic passbook printing is performed.
PIN	Included when the transaction involves PIN verification.

Financial Response (0210) Messages for Possible Override

Financial response (0210) messages for possible override can contain the following tokens:

Account	Included when account balance information is returned in a response.
CAF Inquiry	Included with card-initiated transactions when multiple accounts exist or when the CAF contains a card or account status code that can be overridden.
Credit Line	Included when a backup/credit line account is accessed.
Customer Name	Included when the customer short name is maintained in the PBF.
Financial	Required for all transactions except transaction code 75 (PIN verify).
Header	Required whenever other tokens are included.
Native Message	Included if it was contained in the original request.
NBF	Included when automatic passbook printing is performed.
Override	Included when the transaction requires an override in order to be authorized.
PBF Inquiry	Included when PBF account information is returned.
PIN	Included when the transaction involves PIN verification.
SPF Inquiry	Included when stop payment information is returned.
WHFF Inquiry	Included when the transaction requires a warning-related override in order to be authorized.

Financial Advice (0220) Messages

Financial advice (0220) messages can contain the following tokens:

Financial	Required for all transactions except transaction code 75 (PIN verify).
Header	Required whenever other tokens are included.
Native Message	Presence in the message is controlled by the Device Handler design.

Financial Advice Response (0230) Messages

Financial advice response (0230) messages can contain the following tokens:

Account	Included when account balance information is returned in a response.
Credit Line	Included when a backup/credit line account was accessed to authorize the transaction.
Customer Name	Included when the customer short name is maintained in the PBF.
Financial	Required for all transactions.
Header	Required whenever other tokens are included.
Native Message	Included if it was contained in the original advice.

BASE24-teller Tokens Used in File Inquiry/Update Messages

There are four possible token combinations used in file inquiry/update messages. Within each combination, some tokens are required while the remaining tokens are optional and are part of the message only when needed. Tokens are listed here alphabetically, but appear in internal messages in the order they are added to the message by BASE24-teller processes.

File Inquiry/Update Request (0300) Messages

File inquiry/update request (0300) messages can contain the following tokens:

CAF Inquiry	Optional for transaction code 33 (CAF inquiry).
CAF Update	Required for transaction codes 73 (CAF update card status) and 74 (CAF/PBF update account status).
Header	Required whenever other tokens are included.
Native Message	Presence in the message is controlled by the Device Handler design.
NBF	Optional for transaction codes 34 (NBF inquiry), 35 (passbook print), and 36 (passbook reprint).
PBF Inquiry	Optional for transaction code 30 (PBF inquiry).
PBF Update	Required for transaction codes 82 (change PBF card status) and 83 (change PBF stop pay/warning status).
PIN	Included when a PIN is entered with a card-initiated transaction.
SPF Inquiry	Optional for transaction code 32 (SPF inquiry).
SPF Update	Required for transaction codes 80 (add stop payment) and 81 (delete stop payment).
WHFF Inquiry	Optional for transaction code 37 (WHFF inquiry).
WHFF Update	Required for transaction codes 84/89 (add/delete warning), 85/86 (add/delete hold), and 87/88 (add/delete float).

File Inquiry/Update Response (0310) Messages

File inquiry/update response (0310) messages can contain the following tokens:

Account	Included when the PBF was accessed successfully.
CAF Inquiry	Included with transaction code 33 (CAF inquiry). It is also included with other transactions involving the CAF when the Override token is returned.
CAF Update	Required for transaction codes 73 (CAF update card status) and 74 (CAF/PBF update account status).
Credit Line	Included with transaction codes 30 (PBF inquiry) and 31 (PBF short inquiry).
Customer Name	Included when the PBF was accessed successfully and the Short Name segment exists.
Header	Required whenever other tokens are included.
Native Message	Included if it was contained in the original request.
NBF	Included with transaction codes 34 (NBF inquiry), 35 (passbook print), and 36 (passbook reprint) when NBF records are available.
Override	Included when the inquiry/update requires an override in order to be processed.
PIN	Included when a PIN was entered with a card-initiated transaction.
PBF Inquiry	Included with transaction code 30 (PBF inquiry) and with other inquiry/update transactions involving the PBF when the Override token is returned.
PBF Update	Required for transaction codes 82 (change PBF card status) and 83 (change PBF stop pay/warning status).
SPF Inquiry	Required for transaction code 32 (SPF inquiry).
SPF Update	Required for transaction codes 80 (add stop payment) and 81 (delete stop payment).

WHFF Inquiry	Required for transaction code 37 (WHFF inquiry).
WHFF Update	Required for transaction codes 84/89 (add/delete warning), 85/86 (add/delete hold), and 87/88 (add/delete float).

File Update Advice (0320) Messages

File update advice (0320) messages can contain the following tokens:

CAF Inquiry	Optional for transaction code 33 (CAF inquiry).
CAF Update	Required for transaction codes 73 (CAF update card status) and 74 (CAF/PBF update account status).
Header	Required whenever other tokens are included.
Native Message	Presence in the message is controlled by the Device Handler design.
NBF	Optional for transaction codes 34 (NBF inquiry), 35 (passbook print), and 36 (passbook reprint).
PBF Inquiry	Optional for transaction code 30 (PBF inquiry).
PBF Update	Required for transaction codes 82 (change PBF card status) and 83 (change PBF stop pay/warning status).
SPF Inquiry	Optional for transaction code 32 (SPF inquiry).
SPF Update	Required for transaction codes 80 (add stop payment) and 81 (delete stop payment).
WHFF Inquiry	Optional for transaction code 37 (WHFF inquiry).
WHFF Update	Required for transaction codes 84/89 (add/delete warning), 85/86 (add/delete hold), and 87/88 (add/delete float).

File Update Advice Response (0330) Messages

File update advice response (0330) messages can contain the following tokens:

Account	Included when the PBF was accessed successfully.
CAF Inquiry	Included with transaction code 33 (CAF inquiry).
CAF Update	Required for transaction codes 73 (CAF update card status) and 74 (CAF/PBF update account status).
Customer Name	Included when the PBF was accessed successfully and the Short Name segment exists.
Header	Required whenever other tokens are included.
Native Message	Included if it was contained in the original advice.
NBF	Included with transaction codes 34 (NBF inquiry), 35 (passbook print), and 36 (passbook reprint) when NBF records are available.
PBF Inquiry	Included with transaction code 30 (PBF inquiry).
PBF Update	Required for transaction codes 82 (change PBF card status) and 83 (change PBF stop pay/warning status).
SPF Inquiry	Required for transaction code 32 (SPF inquiry).
SPF Update	Required for transaction codes 80 (add stop payment) and 81 (delete stop payment).
WHFF Inquiry	Required for transaction code 37 (WHFF inquiry).
WHFF Update	Required for transaction codes 84/89 (add/delete warning), 85/86 (add/delete hold), and 87/88 (add/delete float).

BASE24-teller Tokens Used in Administrative Messages

One token combination is used in all administrative messages. Within this combination, all tokens are optional and are part of the message only when needed. Tokens are listed here alphabetically, but appear in internal messages in the order they are added to the message by BASE24-teller processes.

Administrative Request (0600) Messages

Administrative Response (0610) Messages

Administrative Advice (0620) Messages

Administrative Advice Response (0630) Messages

Administrative request (0600), administrative response (0610), administrative advice (0620), and administrative advice response (0630) messages can contain the following tokens:

Administrative	User-defined, variable-length structure controlled by the Device Handler design.
Header	Required whenever other tokens are included.
Native Message	Presence in the message is controlled by the Device Handler design.

Advice Response Required Flag

The Advice Response Required Flag (ARR Flag) is set by the Device Handler and controls how certain transactions are processed. The value of this flag controls whether the transaction authorizer acknowledges advice (0220, 0320, 0420, and 0620) messages. It also controls which messages get logged to the Teller Transaction Log File (TTLF). For each advice message BASE24-teller processes, either the advice (0220, 0320, 0420, and 0620) message or its corresponding response (0230, 0330, 0430, and 0630) message gets logged to the TTLF.

The value of this flag, carried in the RQST.ADVC-RESP-REQ field in the TSTMH, is one of the requirements that must be determined when a Device Handler is being designed. This value is part of the Device Handler code and cannot be configured by the user. The valid values for this flag are as follows:

- 0 = Transaction authorizer does not acknowledge advice messages; advice messages are logged to the TTLF.
- 1 = Transaction authorizer acknowledges advice messages; advice response messages are logged to the TTLF.

The ARR Flag value affects processing performed by the Device Handler, Authorization, and ISO and ANSI Host Interface processes, as described below. All of the processing scenarios described below are illustrated in section 3 of this manual.

Device Handler

The Device Handler sets the value of the ARR Flag, and the Device Handler should be designed such that the flag's value coincides with other processing the Device Handler performs.

If the value of the ARR Flag is 0, the Device Handler generates and returns a native message response to the device or controller for each advice or reversal message the device or controller sends. The Device Handler then passes the advice or reversal message to the Authorization process.

If the value of the ARR Flag is 1, the Device Handler passes the advice or reversal message to the Authorization process without first generating a native message response for the device or controller. Instead, the Device Handler returns a native message response to the device or controller when it receives an advice response from the transaction authorizer, which could be the Authorization process or the host.

Authorization

The Authorization process handles an advice or reversal transaction based on three factors:

- The value of the ARR Flag.
- Where the transaction is to be authorized (Authorization process or the host).
- Whether BASE24 sends completions to the host.

The values in the AUTH LEVEL and COMPLETIONS TO HOST fields in the TTF, which are set via TTF screen 1, control where the transaction is authorized and whether completions are sent to the host.

When the value of the ARR Flag is 0 and the Authorization process is the authorizer, the Device Handler has already acknowledged each advice or reversal message with a native message response by the time the Authorization process receives the message. The Authorization process logs the advice or reversal to the TTLF. After logging the advice or reversal, the Authorization process can send a completion to the host via the Host Interface process, depending on the value in the COMPLETIONS TO HOST field in the TTF.

When the value of the ARR Flag is 0 and the host is the authorizer, the Device Handler has already acknowledged each advice or reversal message with a native message response by the time the Authorization process receives the message. The Authorization process logs the advice or reversal to the TTLF, then forwards the message to the host via the Host Interface process.

When the value of the ARR Flag is 1 and the Authorization process is the authorizer, the Authorization process generates an acknowledgment, logs it to the TTLF, and returns it to the Device Handler. The Device Handler returns a native message response to the device or controller. If the Device Handler does not receive the acknowledgment, it denies the transaction. After returning the acknowledgment to the Device Handler, the Authorization process can send a completion to the host via the Host Interface process, depending on the value in the COMPLETIONS TO HOST field.

When the value of the ARR Flag is 1 and the host is the authorizer, the Authorization process passes the advice or reversal to the host via the Host Interface process, waits for an acknowledgment from the host, and returns this acknowledgment to the Device Handler. The Device Handler returns a native message response to the device or controller. If the Device Handler does not receive the acknowledgment, it denies the transaction. The Authorization process also logs the acknowledgment it receives from the host to the TTLF.

ISO Host Interface

The ISO Host Interface process handles an advice or reversal message (0220, 0320, 0420, or 0620) based on three factors:

- The value of the ARR Flag.
- Whether the ISO Host Interface process expects logical acknowledgments from the Nucleus or text-level acknowledgments from the host.
- Whether the host is available.

The value in the ACK FROM DPC field in the HCF, which is set via HCF screen 1, controls whether the ISO Host Interface process expects logical acknowledgments from the Nucleus or text-level acknowledgments from the host. A logical acknowledgment indicates only that the Nucleus sent the advice or reversal to the host, while a text-level acknowledgment indicates that the host received the message.

Note: An ARR Flag value of 1 overrides the value in the ACK FROM DPC field in the HCF. With an ARR Flag value of 1, the ISO Host Interface process expects a text-level acknowledgment (0230, 0330, 0430, or 0630) from the host in response to an advice or reversal, regardless of the setting in the ACK FROM DPC field.

When the value of the ARR Flag is 0, the ISO Host Interface process passes the advice or reversal to the host and waits for the acknowledgment. The ISO Host Interface process does not return the acknowledgment to the Authorization process.

If the ISO Host Interface process knows that the host is unavailable, it places the advice or reversal in the Store-and-Forward File (SAF) without changing the message type, then sends the message to the host during normal store-and-forward processing.

If the ISO Host Interface process sends the advice or reversal to the host and the message times out, the ISO Host Interface process changes the message type to a repeat message (0221, 0321, 0421, or 0621), places the repeat message in the SAF, then sends the repeat message to the host during normal store-and-forward processing.

When the value of the ARR Flag is 1, the ISO Host Interface process passes the advice or reversal to the host, waits for an acknowledgment from the host, then returns this acknowledgment to the Authorization process. If the host is unavailable or does not respond, the ISO Host Interface process notifies the Authorization process that the transaction has been declined.

For additional information about how the ISO Host Interface process handles an advice or reversal message, refer to the ***BASE24 ISO Host Interface Manual***.

ANSI Host Interface

The ANSI Host Interface process handles an advice or reversal message based on three factors:

- The value of the ARR Flag.
- Whether the ANSI Host Interface process expects logical acknowledgments from the Nucleus or text-level acknowledgments from the host.
- Whether the host is available.

The value in the ACK FROM DPC field in the HCF, which is set via HCF screen 1, controls whether the ANSI Host Interface process expects logical acknowledgments from the Nucleus or text-level acknowledgments from the host. A logical acknowledgment indicates only that the Nucleus sent the advice or reversal to the host, while a text-level acknowledgment indicates that the host received the message.

Note: An ARR Flag value of 1 overrides the value in the ACK FROM DPC field in the HCF. With an ARR Flag value of 1, the ANSI Host Interface process expects a text-level acknowledgment from the host in response to an advice or reversal, regardless of the setting in the ACK FROM DPC field.

When the value of the ARR Flag is 0, the ANSI Host Interface process passes the advice or reversal to the host and waits for the acknowledgment. The ANSI Host Interface process does not return acknowledgments to the Authorization process. ANSI external message types passed to the host are as follows:

8220 = Financial transaction advice (internal message type 0220)
86x0 = File update advice (internal message type 0320)
8420 = Financial transaction reversal (internal message type 0420)

If the ANSI Host Interface process knows that the host is unavailable, it places the advice or reversal in the Store-and-Forward File (SAF), then sends the message to the host during normal store-and-forward processing. The ANSI Host Interface process places the original message type (8220, 86x0, or 8420) in the TYPE-ORIG field of the external message and uses the value 8920 in the TYP field of the external message that it places in the SAF.

If the ANSI Host Interface process sends the advice or reversal to the host and the message times out, the ANSI Host Interface process places the advice or reversal in the SAF, then sends the message to the host during normal store-and-forward processing. The ANSI Host Interface process places the original message type (8220, 86x0, or 8420) in the TYPE-ORIG field of the external message and uses the value 8921 in the TYP field of the external message that it places in the SAF.

When the value of the ARR Flag is 1, the ANSI Host Interface process changes the message type from an advice or reversal to a request, passes the message to the host, waits for an acknowledgment from the host, then returns this acknowledgment to the Authorization process. The ANSI Host Interface process sets the value 9 in the RQST.INTL-OVRRD-LVL field in the external message to distinguish between a normal request and an advice that requires a response. If the host is unavailable or does not respond, the ANSI Host Interface process notifies the Authorization process that the transaction has been declined. ANSI external message types passed to the host are as follows:

- 8200 = Financial transaction advice (internal message type 0220)
- 85x0 = File update advice (internal message type 0320)
- 8400 = Financial transaction reversal (internal message type 0420)

For additional information about how the ANSI Host Interface process handles an advice or reversal message, refer to the ***BASE24 ANSI Host Interface Manual***.

Transaction Logging

BASE24-teller Device Handlers and Authorization processes share the responsibility for logging transactions to the Teller Transaction Log File (TTLF). Three factors affect transaction logging:

- Which process is responsible for logging
- Which messages are logged
- What information is logged

Responsibility for Logging

Each financial institution uses its records in the Teller Transaction File (TTF) to specify the responsibility for logging each of its transactions to the TTLF. The Device Handler logs all transactions identified with an L (log only) in the TERM TRAN ALLOWED field in the TTF, which is set via TTF screen 1. The Authorization process is responsible for logging all other transactions.

When the account-owning financial institution and terminal-owning financial institution are not the same, only the terminal-owning financial institution can configure the Device Handler to log the transaction to the TTLF.

The Authorization process has no knowledge of transactions that have been logged by the Device Handler. Only the Authorization process can send completion messages to the host, so the TTF should not be configured for the Device Handler to log transactions requiring completions to the host.

Message Types Logged

Device Handlers and Authorization processes convert messages from the internal message into the TTLF format before logging them to the TTLF.

Note: The setting of the Advice Response Required Flag (ARR Flag) in the TSTMH identifies whether the advice message or the advice response message is logged for a transaction. Both messages are not logged for the same transaction. Refer to the Advice Response Required Flag discussion earlier in this section for more information.

The following BASE24-teller message types are logged to the TTLF:

- 0210 = Financial transaction response
- 0220 = Financial transaction advice when response is not required
- 0230 = Financial transaction advice response when response is required
- 0310 = File inquiry/update response
- 0320 = File update advice when response is not required
- 0330 = File update advice response when response is required
- 0420 = Financial transaction reversal advice when response is not required
- 0430 = Financial transaction reversal response when response is required
- 0610 = Administrative response
- 0620 = Administrative advice when response is not required
- 0630 = Administrative advice response when response is required

The following BASE24-teller message types are not logged to the TTLF:

- 0200 = Financial transaction request
- 0300 = File inquiry/update request
- 0600 = Administrative transaction request

Note: A completion (0220, 0320, 0420, or 0620) generated by the Authorization process based on the value of the COMPLETIONS TO HOST field in the TTF is not logged to the TTLF. For transactions involving multiple account selection, only the final response (0210 or 0310) is logged to the TTLF.

Information Logged

The Token File (TKN) controls which tokens in the internal message are logged to the TTLF. All tokens that can be logged to the TTLF are displayed on TKN screen 2. As each message is logged to the TTLF, tokens identified to be logged are arranged in the TTLF record in the order in which they were added to the internal message. TKN screen 3 is then used to specify which of the tokens logged to the TTLF will be extracted and the order in which they will appear in each extract record.

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Section 3

BASE24-teller Message Flows

This section contains diagrams illustrating the paths taken by BASE24-teller transactions under various types of processing situations.

These diagrams are intended to illustrate the transaction path from the teller terminal to the transaction authorizer. They include the teller terminal and controller, the host, the BASE24 Nucleus, and the BASE24-teller Device Handler, Authorization, and Host Interface processes. These diagrams are not intended to illustrate whether a transaction is approved or declined. The values from certain fields in the internal message and BASE24 database are given because they control the transaction path.

Diagram Information

The following provides some general information about the transaction message flow diagrams in this section.

Message Types Used

The following message types are used in the transaction message flow diagrams in this section.

0200 = Financial transaction request
0210 = Financial transaction response
0220 = Financial transaction advice
0230 = Financial transaction advice response
0300 = File inquiry/update request
0310 = File inquiry/update request response
0320 = File update advice
0330 = File update advice response
0420 = Financial transaction reversal advice
0430 = Financial transaction reversal advice response
0600 = Administrative transaction request
0610 = Administrative transaction response
0620 = Administrative transaction advice
0630 = Administrative transaction advice response
0x00 = Request (0200, 0300, or 0600)
0x10 = Response (0210, 0210, or 0610)
0x20 = Advice (0220, 0320, 0420, or 0620)
0x30 = Advice response (0230, 0330, 0430, or 0630)

In circumstances where BASE24-teller processing is the same for more than one message type, the diagrams use an x instead of a 2, 3, 4, or 6. For example, an 0x20 could be a 0220, 0320, 0420, or 0620.

Note: All external message types listed in this section apply to an ISO Host Interface process. Refer to “BASE24-teller Message Types” in section 2 of this manual for the external message types applicable to an ANSI Host Interface process. Administrative transactions (0600, 0610, 0620, and 0630 messages) can be sent to the host only with an ISO Host Interface process because they are not supported with an ANSI Host Interface process.

Special Diagram Notations

In addition to message types, the following special notations are used in the transaction message flow diagrams in this section.

- RQST = Request initiated by a teller terminal or controller connected to BASE24-teller. The Device Handler translates the native-mode message of the teller terminal or controller into an internal request (0x00) message.
- RESP = Response sent to a teller terminal or controller connected to BASE24-teller. The Device Handler translates the internal response (0x10 or 0x30) message into the native-mode message of the teller terminal or controller.
- ADV = Advice initiated by a teller terminal or controller connected to BASE24-teller. The Device Handler translates the native-mode message of the teller terminal or controller into an internal advice (0x20) message.
- RVSL = Reversal initiated by a teller terminal or controller connected to BASE24-teller. The Device Handler translates the native-mode message of the teller terminal or controller into an internal reversal (0420) message.
- FAI = Message failure, the result of a no-response situation with the controller and terminal or an authorizing host.

Variables Affecting Transaction Flow

Several variables have a direct effect on the transaction flows illustrated in the following diagrams.

Device Handler Process Logs Transaction

The value in the TERM TRAN ALLOWED field in the Teller Transaction File (TTF), which can be accessed via TTF screen 1, controls whether the Device Handler sends the transaction to the Authorization process or logs the transaction and responds to the controller.

Advice Response Required Flag (ARR Flag)

The value in the RQST.ADVC-RESP-REQ field in the Teller Standard Internal Message Header (TSTMH) identifies whether the Device Handler can acknowledge an advice or must wait for the acknowledgment from the transaction authorizer before responding to the controller. The value of the ARR Flag determines whether the 0x20 or 0x30 message gets logged to the Teller Transaction Log File (TTLF). For each advice message BASE24-teller processes, either the advice message or its corresponding response message gets logged to the TTLF. The value of the ARR Flag is not configurable. The Device Handler must be designed and developed to set the value of this flag.

Valid values for this flag are as follows:

- 0 = Transaction authorizer does not acknowledge advice messages; advice messages are logged to the TTLF.
- 1 = Transaction authorizer acknowledges advice messages; advice response messages are logged to the TTLF.

Completions to Host

The value in the COMPLETIONS TO HOST field in the TTF, which can be accessed via TTF screen 1, controls whether a completion is sent to the host for transactions authorized by the BASE24-teller Authorization process. Completions can be sent for approved transactions only, declined transactions only, or both. For the purpose of these diagrams, completions, when sent, are for approved and declined transactions. These completions are not logged to the TTLF.

Acknowledgments from Host

The value in the ACK FROM DPC field in the Host Configuration File (HCF), which can be accessed via HCF screen 1, controls whether the BASE24-teller Host Interface process expects an acknowledgment from the host for each transaction sent to the host. A value in the ARR Flag indicating the transaction authorizer is required to acknowledge advice messages overrides the value in the ACK FROM DPC field.

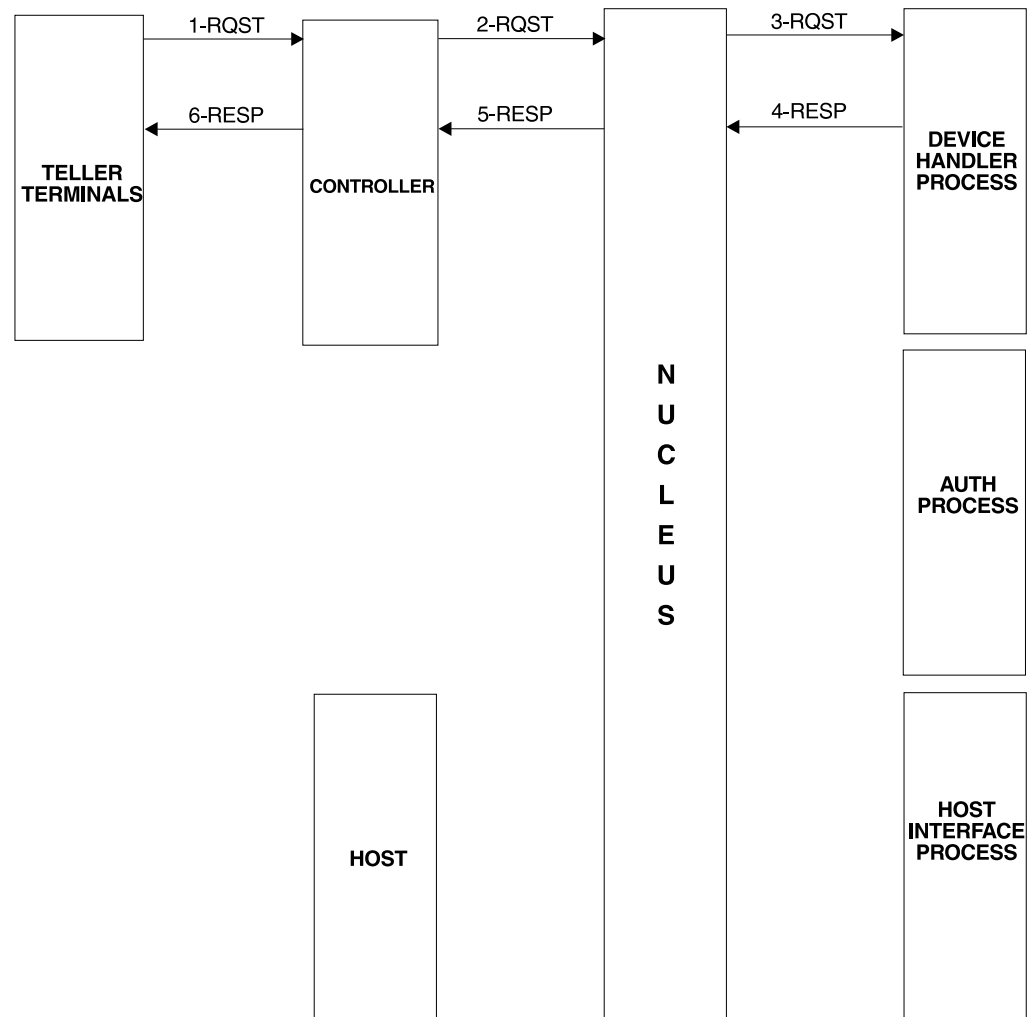
Transaction Diagrams

This section contains 14 transaction diagrams, organized as follows:

- Request logged by the Device Handler (diagram 1)
- Requests authorized by the controller (diagrams 2 and 3)
- Requests authorized by the BASE24-teller Authorization process (diagram 4)
- Requests authorized by the host (diagrams 5–7)
- Advices authorized by the host (diagrams 8–11)
- Reversal initiated by BASE24 and authorized by the BASE24-teller Authorization process (diagram 12)
- Override authorized by the BASE24-teller Authorization process (diagram 13)
- Multiple account selection authorized by the BASE24-teller Authorization process (diagram 14)

Diagram 1 — Request Logged by the Device Handler Process

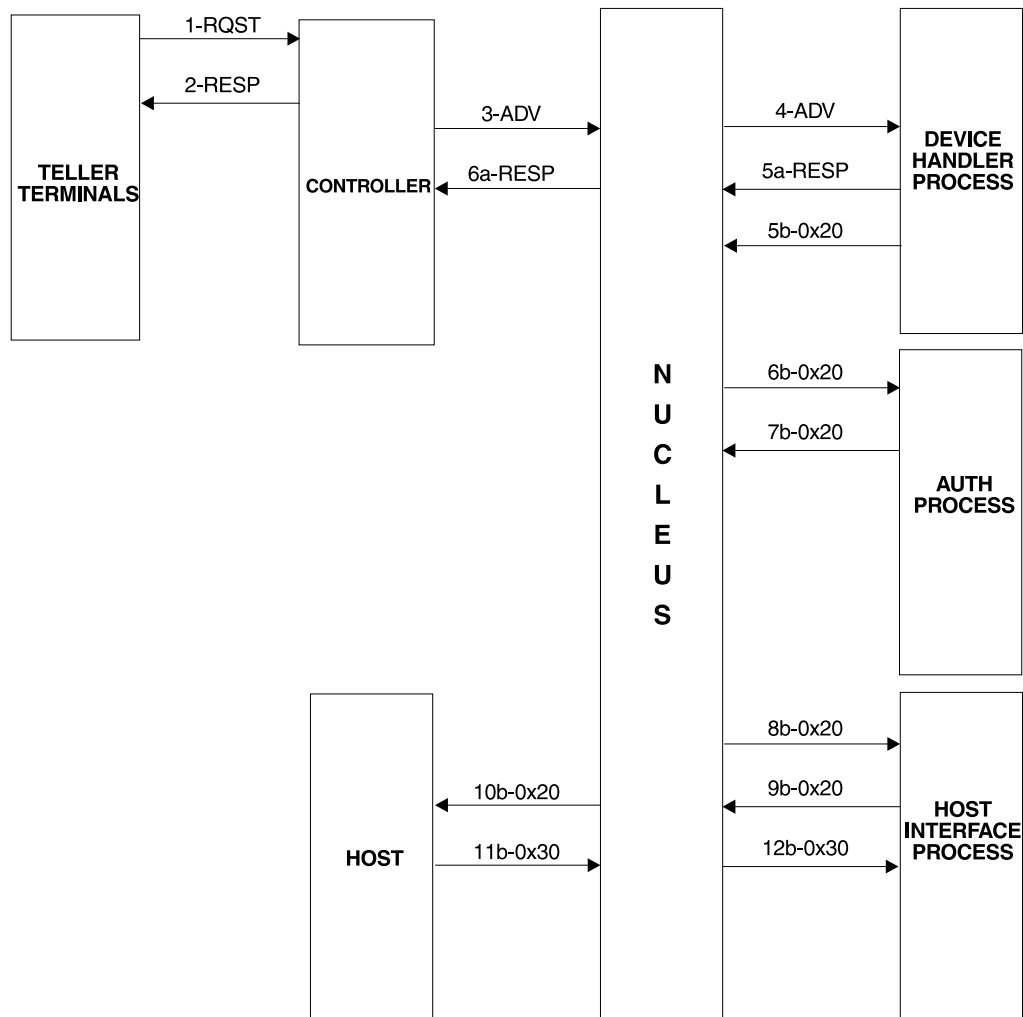
Diagram 1 illustrates the flow of a request that is logged by the Device Handler.



Steps	Processing
1, 2, 3	The device sends a native-mode request message to the Device Handler. The Device Handler logs the message to the TTLF as a 0x10 (0210, 0310, or 0610).
4, 5, 6	The Device Handler sends the native-mode response message to the device.

Diagram 2 — Request Authorized by the Controller

Diagram 2 illustrates the flow of a request that is authorized by the controller. The controller then initiates an advice for BASE24. The Device Handler is designed to return an acknowledgment to the controller and set the value of the ARR Flag to 0; the BASE24-teller Authorization process is configured to authorize the transaction; completions are sent to the host; and acknowledgments are expected from the host. In this diagram, x can be 2, 3, or 6.



Steps

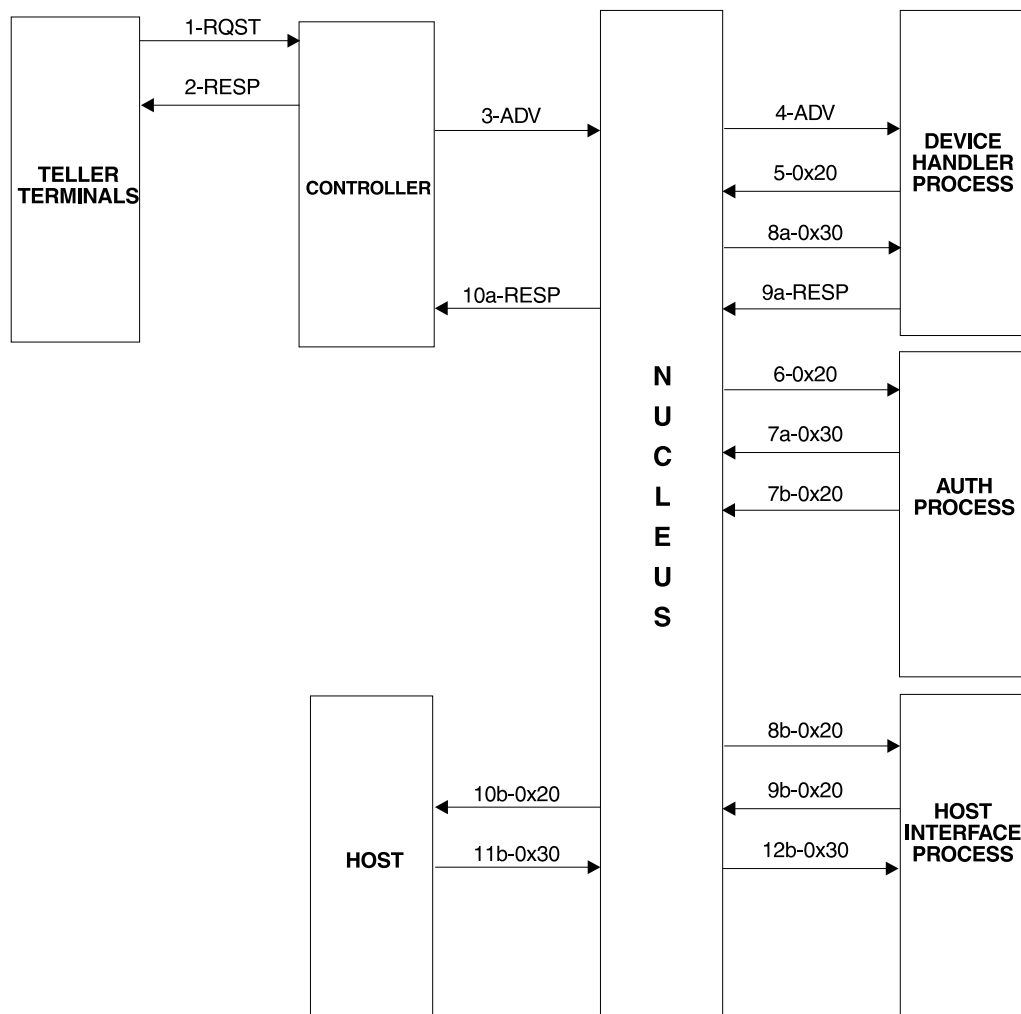
Processing

- 1 The device sends the native-mode request message to the controller for authorization.

Steps	Processing
2	The controller authorizes the request and sends the native-mode response message to the device. The controller stores the authorized request for later transmission to BASE24.
3, 4	The controller sends the native-mode advice message to the Device Handler.
5a, 6a	The Device Handler sends the native-mode response message to the controller.
5b, 6b	The Device Handler translates the native-mode message into a 0x20 that it sends to the Authorization process. The Authorization process logs the 0x20 to the TTLF.
7b, 8b	The Authorization process sends the 0x20 to the Host Interface process.
9b, 10b	The Host Interface process sends the 0x20 to the host.
11b, 12b	The host changes the 0x20 into a 0x30 that it sends to the Host Interface process.

Diagram 3 — Request Authorized by the Controller

Diagram 3 illustrates the flow of a request that is authorized by the controller. The controller then initiates an advice for BASE24. The Device Handler is designed to set the value of the ARR Flag to 1 and wait for an acknowledgment before responding to the controller; the BASE24-teller Authorization process is configured to authorize the transaction; completions are sent to the host; and acknowledgments are expected from the host. In this diagram, x can be 2, 3, or 6.



Steps

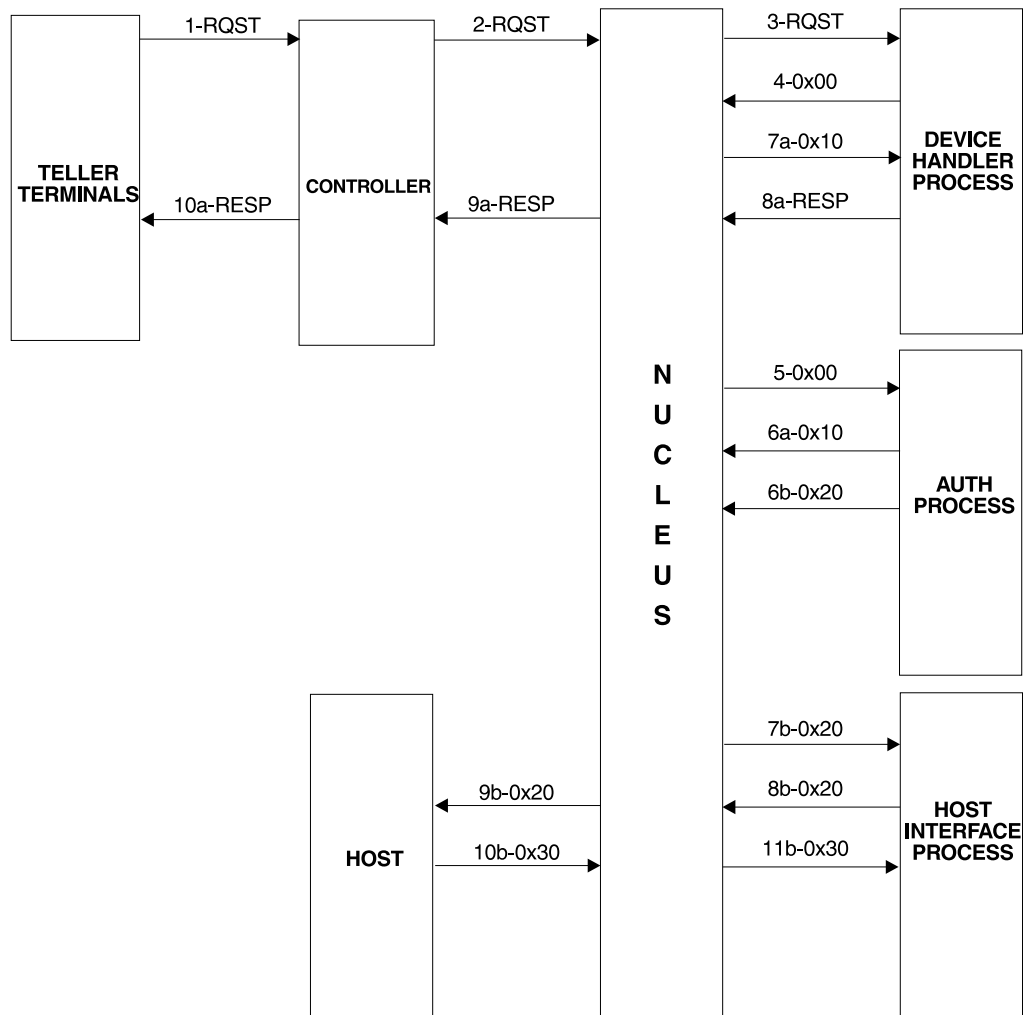
Processing

- 1 The device sends the native-mode request message to the controller for authorization.

Steps	Processing
2	The controller sends the native-mode response message to the device.
3, 4	The controller sends the native-mode advice message to the Device Handler.
5, 6	The Device Handler translates the native-mode message into a 0x20 that it sends to the Authorization process.
7a, 8a	The Authorization process changes the 0x20 into a 0x30 that it sends to the Device Handler to acknowledge the 0x20. The Authorization process logs the 0x30 to the TTLF.
9a, 10a	The Device Handler converts the 0x30 into a native-mode response message that it sends to the controller.
7b, 8b	The Authorization process generates a 0x20 that it sends to the Host Interface process. The Authorization process always sets the ARR Flag in a completion message to a value of 0, regardless of the value of the ARR Flag in the advice message from the Device Handler.
9b, 10b	The Host Interface process sends the 0x20 to the host.
11b, 12b	The host changes the 0x20 into a 0x30 that it sends to the Host Interface process.

Diagram 4 — Request Authorized by the Authorization Process

Diagram 4 illustrates the flow of a request that is authorized by the Authorization process. Completions are sent to the host and acknowledgments are expected from the host. In this diagram, x can be 2, 3, or 6.

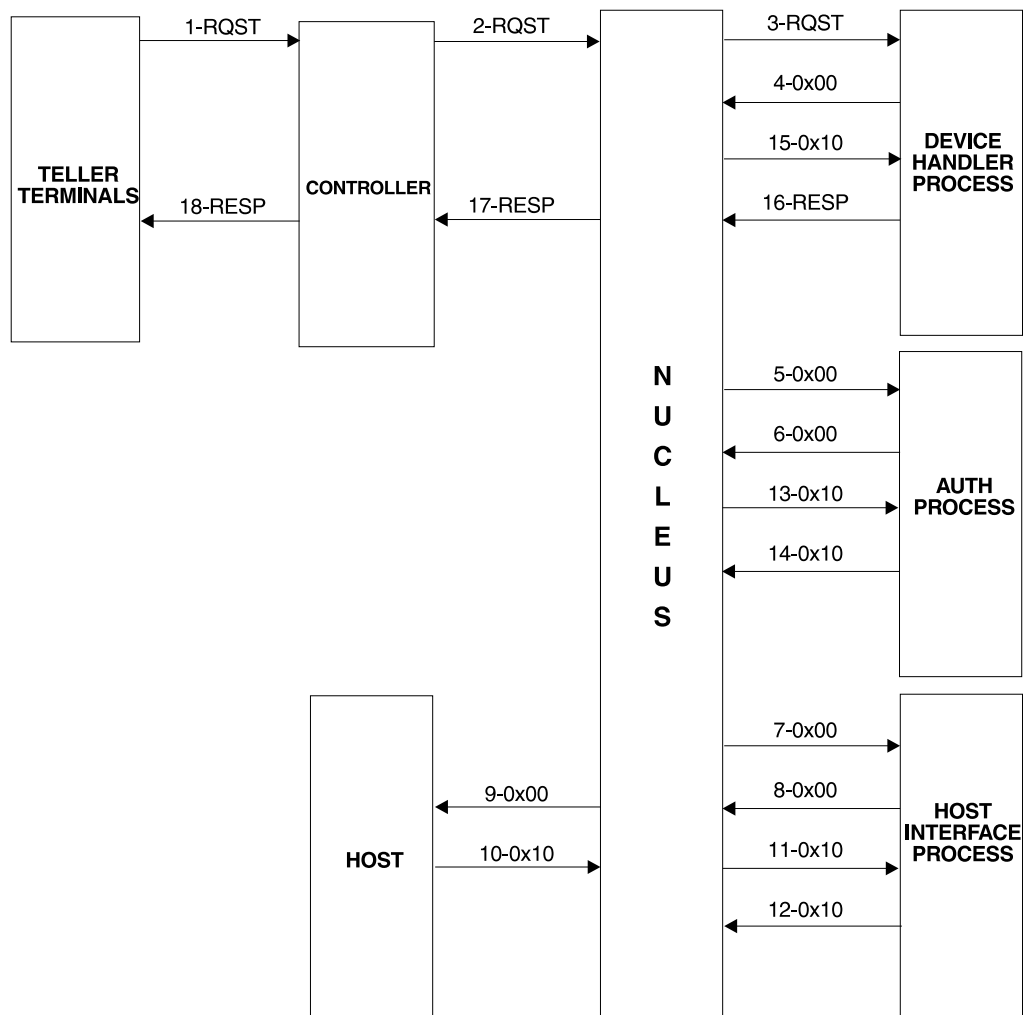


Steps	Processing
1, 2, 3	The device sends a native-mode request message to the Device Handler.
4, 5	The Device Handler translates the native-mode message into a 0x00 that it sends to the Authorization process.

Steps	Processing
6a, 7a	The Authorization process changes the 0x00 into a 0x10 that it sends to the Device Handler. The Authorization process logs the 0x10 to the TTLF.
8a, 9a, 10a	The Device Handler translates the 0x10 into a native-mode response message that it sends to the device.
6b, 7b	The Authorization process generates a 0x20 that it sends to the Host Interface process.
8b, 9b	The Host Interface process sends the 0x20 to the host.
10b, 11b	The host changes the 0x20 into a 0x30 that it sends to the Host Interface process.

Diagram 5 — Request Authorized by the Host

Diagram 5 illustrates the flow of a request that is authorized by the host. In this diagram, x can be 2, 3, or 6.

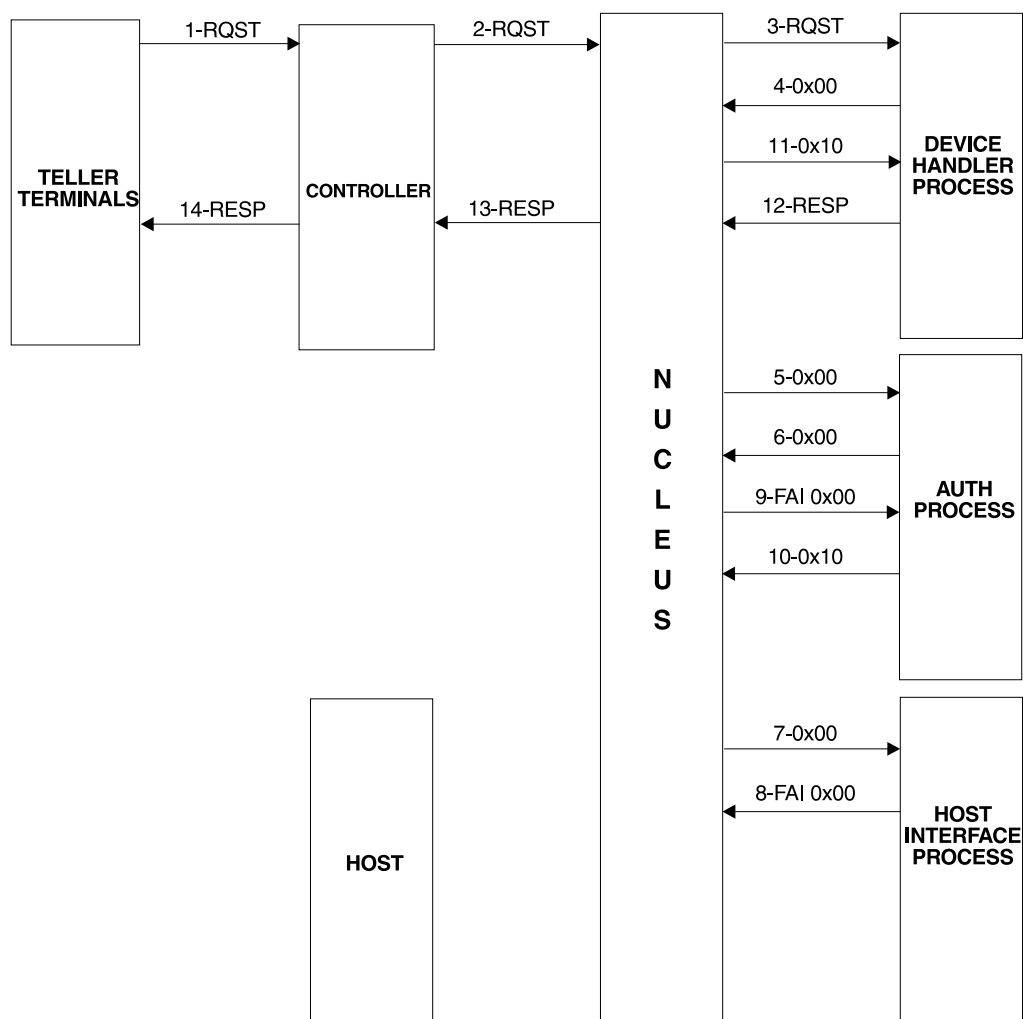


Steps	Processing
1, 2, 3	The device sends a native-mode request message to the Device Handler.
4, 5	The Device Handler translates the native-mode message into a 0x00 that it sends to the Authorization process.

Steps	Processing
6, 7	The Authorization process sends the 0x00 to the Host Interface process.
8, 9	The Host Interface process sends the 0x00 to the host.
10, 11	The host changes the 0x00 into a 0x10 that it sends to the Host Interface process.
12, 13	The Host Interface process sends the 0x10 to the Authorization process.
14, 15	The Authorization process logs the 0x10 to the TTLF and sends it to the Device Handler.
16, 17, 18	The Device Handler translates the 0x10 into a native-mode response message that it sends to the device.

Diagram 6 — Request Authorized by the Host

Diagram 6 illustrates the flow of a request that could not be authorized because the host was unavailable. The Host Interface process is already aware that the host is unavailable. The Authorization process is responsible for changing the failed 0x00 to a 0x10. In this diagram, x can be 2, 3, or 6.

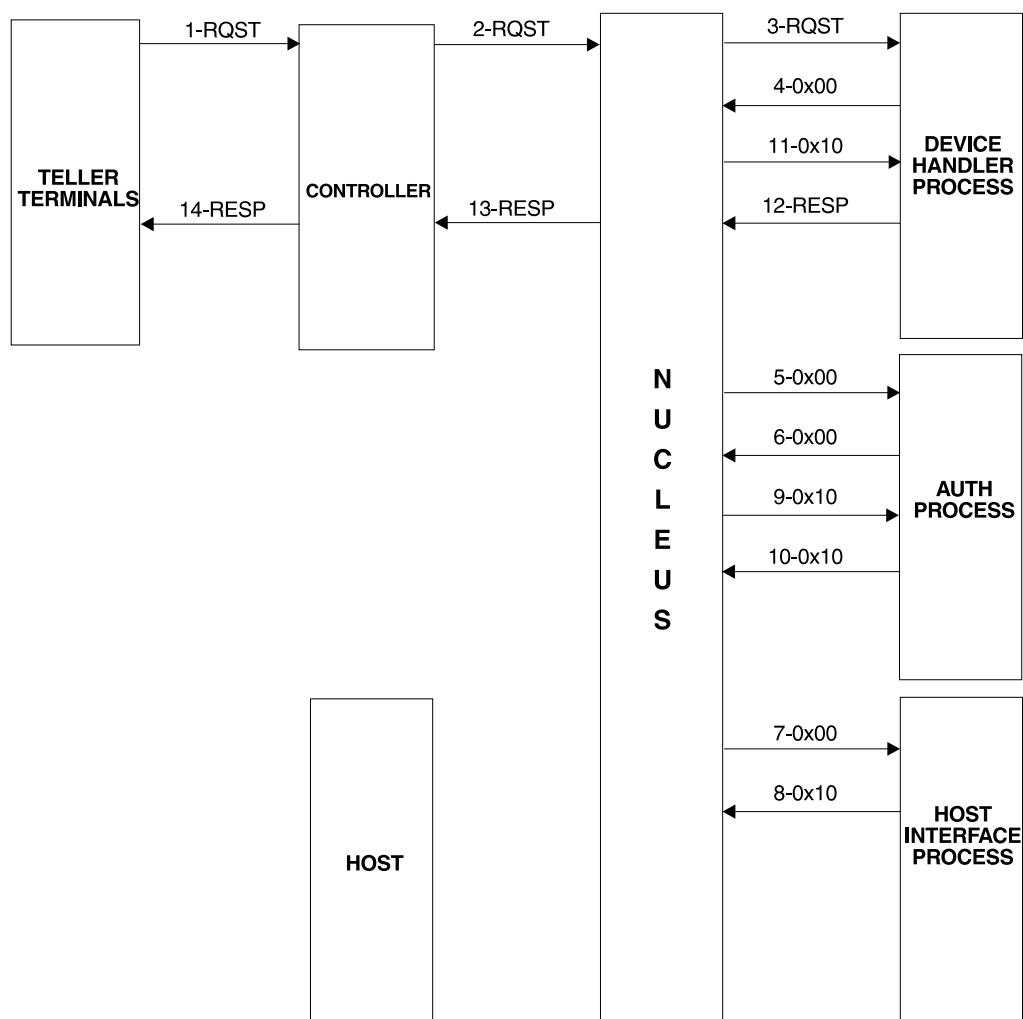


Steps	Processing
1, 2, 3	The device sends a native-mode request message to the Device Handler.
4, 5	The Device Handler translates the native-mode message into a 0x00 that it sends to the Authorization process.

Steps	Processing
6, 7	The Authorization process sends the 0x00 to the Host Interface process.
8, 9	Because the host was unavailable, the Host Interface process changes the 0x00 to a failed 0x00 and sends it to the Authorization process.
10, 11	The Authorization process changes the failed 0x00 to a 0x10, logs the 0x10 to the TTLf, and sends the 0x10 to the Device Handler.
12, 13, 14	The Device Handler translates the 0x10 into a native-mode response message that it sends to the device.

Diagram 7 — Request Authorized by the Host

Diagram 7 illustrates the flow of a request that could not be authorized because of a formatting error. The Host Interface process is responsible for changing the 0x00 into a 0x10. Host availability has no effect when a formatting error is involved. In this diagram, x can be 2, 3, or 6.

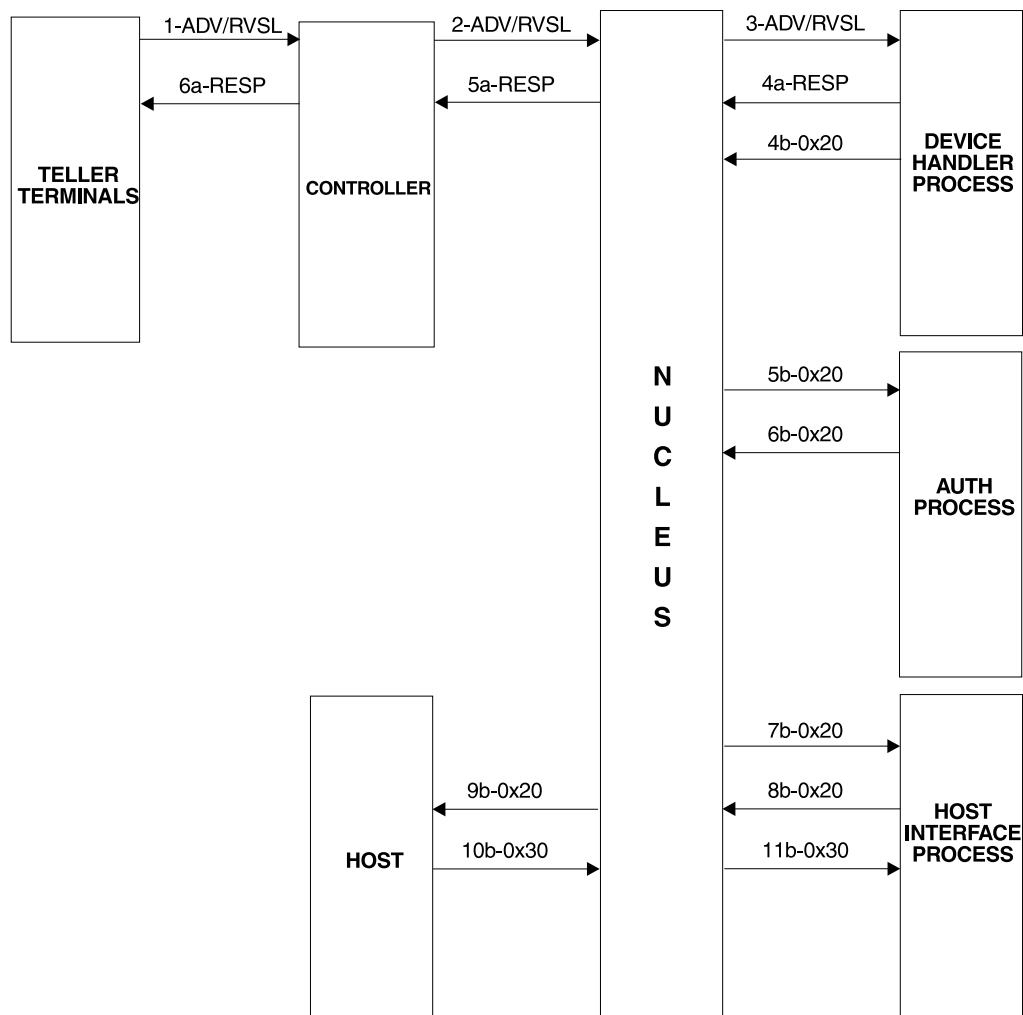


Steps	Processing
1, 2, 3	The device sends a native-mode request message to the Device Handler.
4, 5	The Device Handler translates the native-mode message into a 0x00 that it sends to the Authorization process.

Steps	Processing
6, 7	The Authorization process sends the 0x00 to the Host Interface process.
8, 9	Because the 0x00 was not formatted properly, the Host Interface process changes the 0x00 to a 0x10 and sends it to the Authorization process.
10, 11	The Authorization process logs the 0x10 to the TTLF and sends it to the Device Handler.
12, 13, 14	The Device Handler translates the 0x10 into a native-mode response message that it sends to the device.

Diagram 8 — Advice Authorized by the Host

Diagram 8 illustrates the flow of an advice authorized by the host. The host is available, the Device Handler set the ARR Flag value to 0, and acknowledgments are expected from the host. In this diagram, x can be 3, 4, or 6.

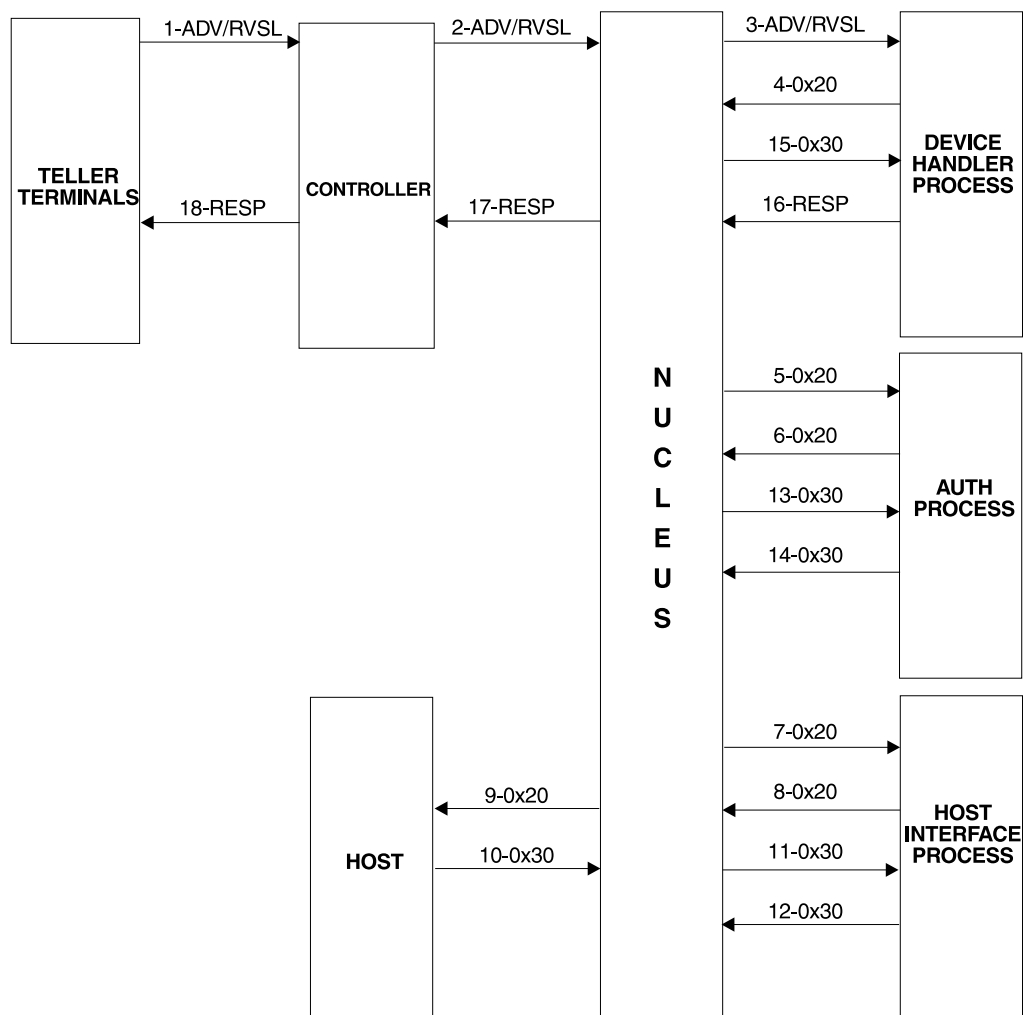


Steps	Processing
1, 2, 3	The device sends the native-mode advice or reversal message to the Device Handler.
4a, 5a, 6a	The Device Handler sends a native-mode response message to the device.

Steps	Processing
4b, 5b	The Device Handler translates the native-mode message into a 0x20 that it sends to the Authorization process.
6b, 7b	The Authorization process logs the 0x20 to the TTLF and then sends it to the Host Interface process.
8b, 9b	The Host Interface process sends the 0x20 to the host.
10b, 11b	The host changes the 0x20 into a 0x30 that it sends to the Host Interface process.

Diagram 9 — Advice Authorized by the Host

Diagram 9 illustrates the flow of an advice authorized by the host. The host is available and the Device Handler set the ARR Flag value to 1. In this diagram, x can be 3, 4, or 6.



Steps

Processing

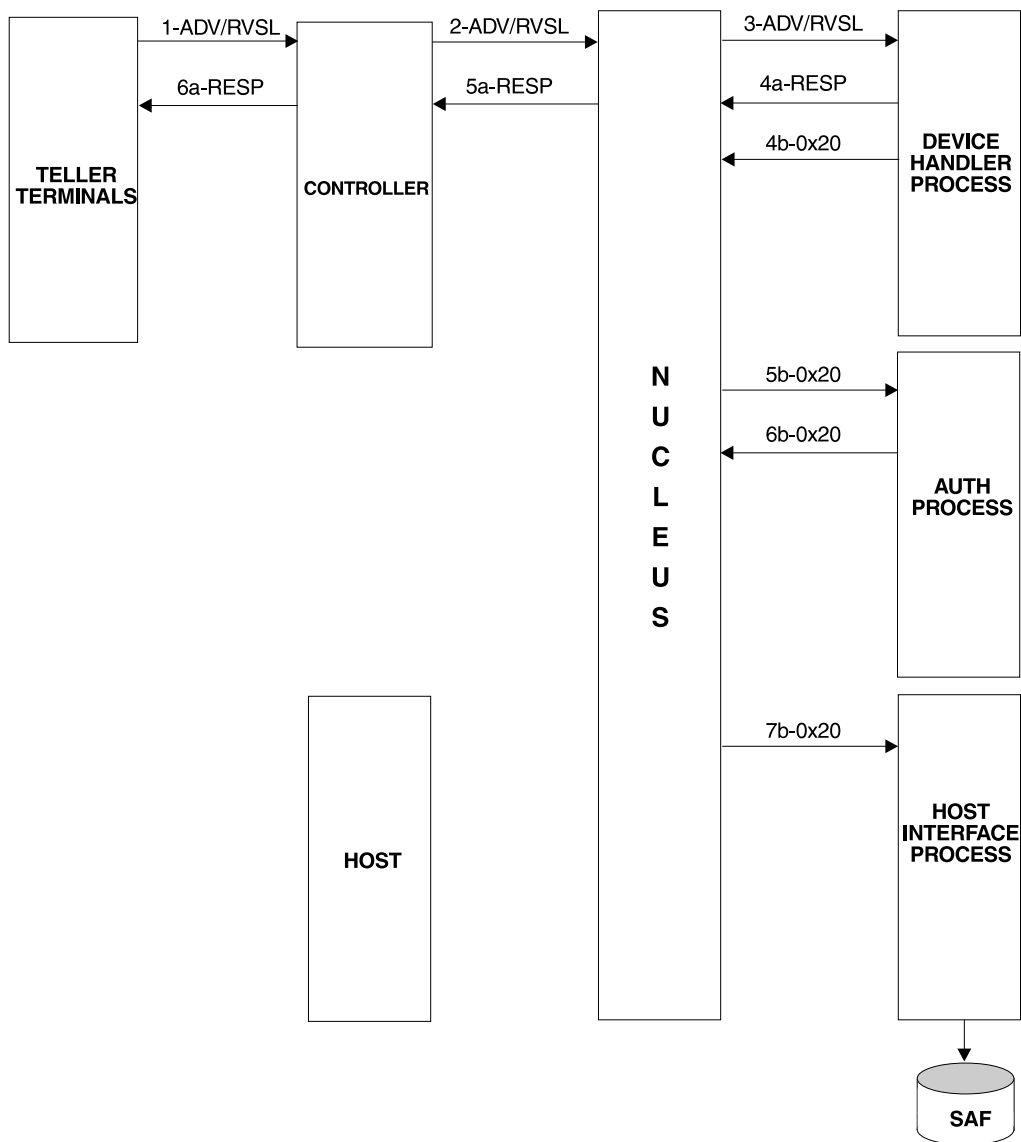
1, 2, 3

The device sends a native-mode advice or reversal message to the Device Handler.

Steps	Processing
4, 5	The Device Handler translates the native-mode advice or reversal message into a 0x20 that it sends to the Authorization process. The Device Handler must wait for an acknowledgment from the Authorization process before responding to the controller and device.
6, 7	The Authorization process sends the 0x20 to the Host Interface process.
8, 9	The Host Interface process sends the 0x20 to the host.
10, 11	The host changes the 0x20 into a 0x30 that it sends to the Host Interface process.
12, 13	The Host Interface process sends the 0x30 to the Authorization process.
14, 15	The Authorization process logs the 0x30 to the TTLF and then sends it to the Device Handler.
16, 17, 18	The Device Handler translates the 0x30 into a native-mode response message that it sends to the device.

Diagram 10 — Advice Authorized by the Host

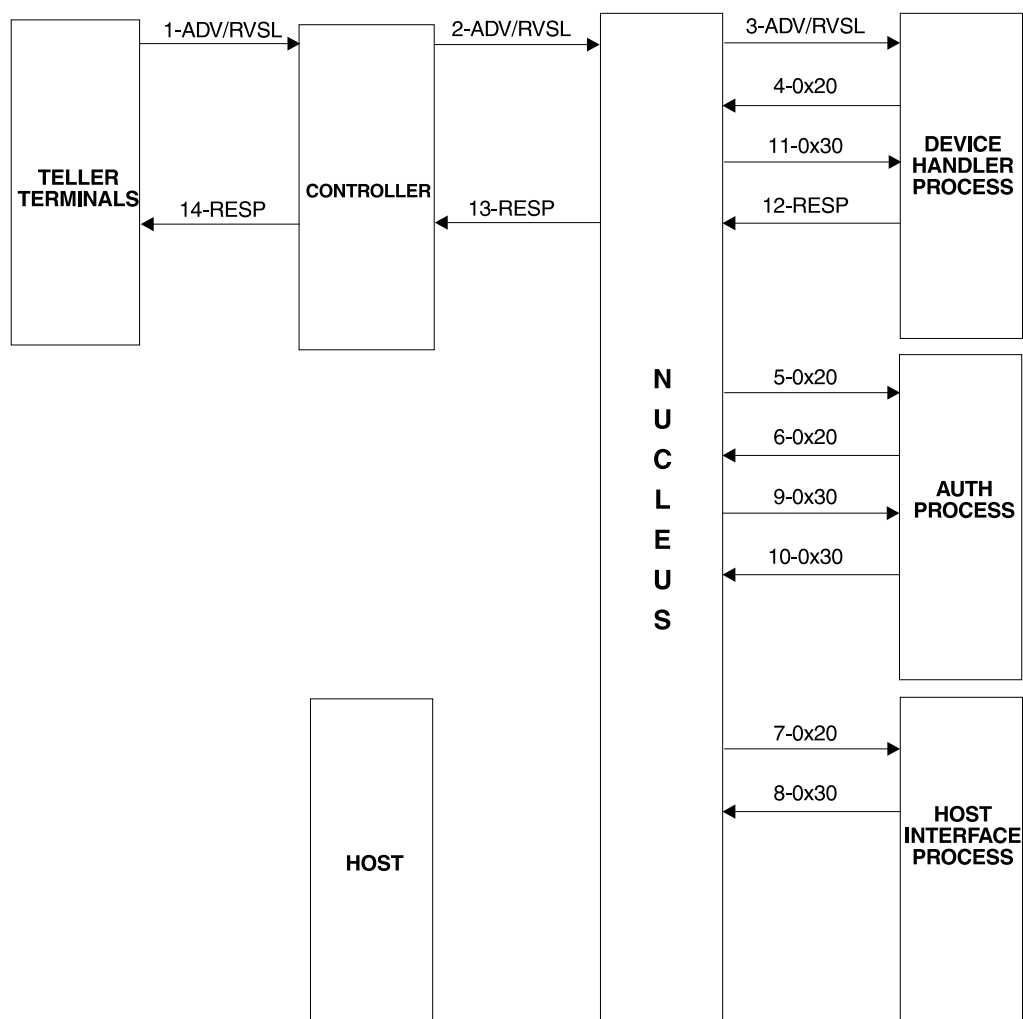
Diagram 10 illustrates the flow of an advice to be authorized by the host. The host is unavailable, but transactions can be logged to a Store-and-Forward File (SAF) for later processing because the Device Handler set the ARR Flag to a value of 0. In this diagram, x can be 3, 4, or 6.



Steps	Processing
1, 2, 3	The device sends a native-mode advice or reversal message to the Device Handler.
4a, 5a, 6a	The Device Handler sends the native-mode response message to the device.
4b, 5b	The Device Handler translates the native-mode message into a 0x20, sets the ARR Flag to a value of 0, and sends the 0x20 to the Authorization process.
6b, 7b	The Authorization process logs the 0x20 to the TTLF and then sends it to the Host Interface process. The host is unavailable, so the Host Interface process logs the 0x20 to the SAF for updating the host records at a later time.

Diagram 11 — Advice Authorized by the Host

Diagram 11 illustrates the flow of an advice to be authorized by the host. The host is unavailable and the Device Handler set the ARR Flag to a value of 1, so the transaction is declined by the Host Interface process. In this diagram, x can be 3, 4, or 6.

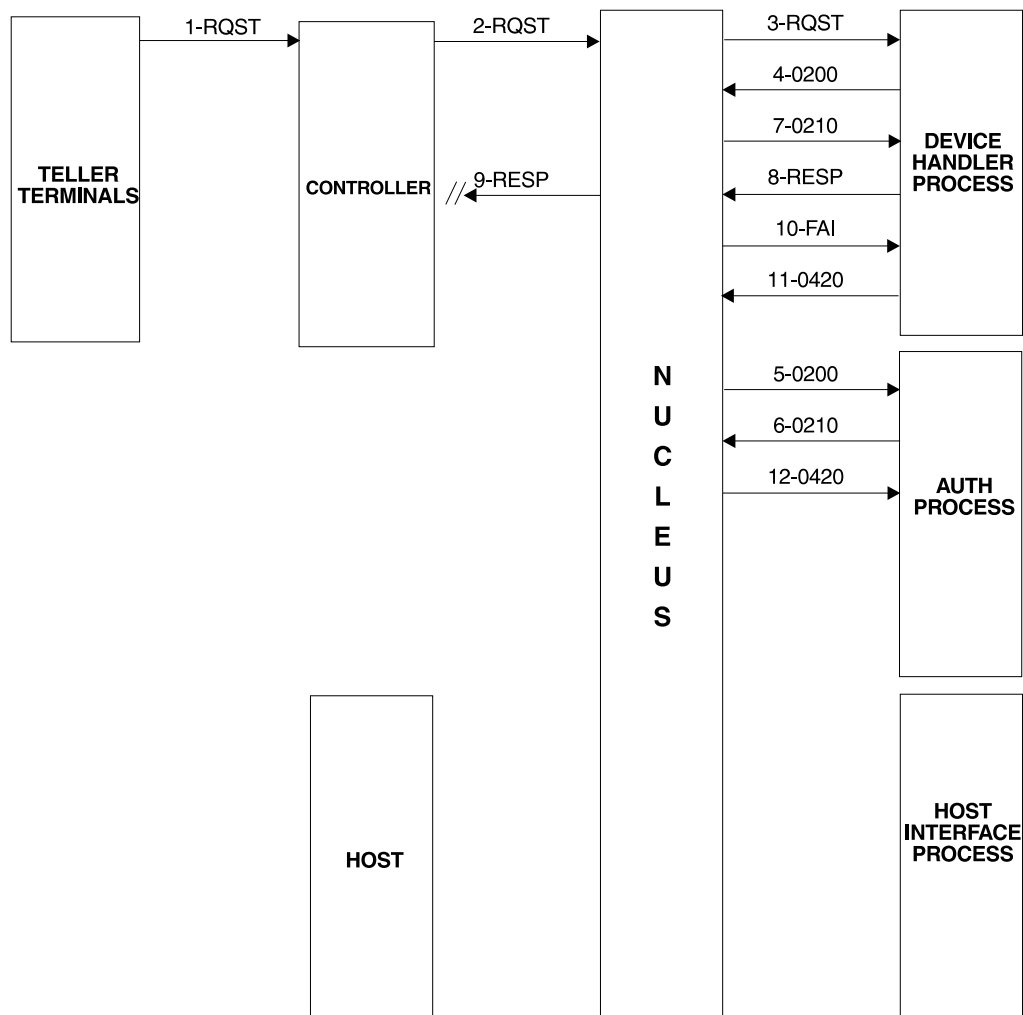


Steps	Processing
1, 2, 3	The device sends a native-mode advice or reversal message to the Device Handler.

Steps	Processing
4, 5	The Device Handler translates the native-mode advice or reversal message into a 0x20, sets the ARR Flag to a value of 1, and sends the 0x20 to the Authorization process. The Device Handler must wait for an acknowledgment from the Authorization process before responding to the controller and device.
6, 7	The Authorization process sends the 0x20 to the Host Interface process.
8, 9	The Host Interface process declines the transaction, changing the 0x20 into a 0x30 that it sends to the Authorization process.
10, 11	The Authorization process logs the 0x30 to the TTLF and then sends it to the Device Handler.
12, 13, 14	The Device Handler translates the 0x30 into a native-mode response message that it sends to the device.

Diagram 12 — Reversal Initiated by BASE24

Diagram 12 illustrates the flow of a reversal initiated by BASE24 because it could not return the response to the controller.

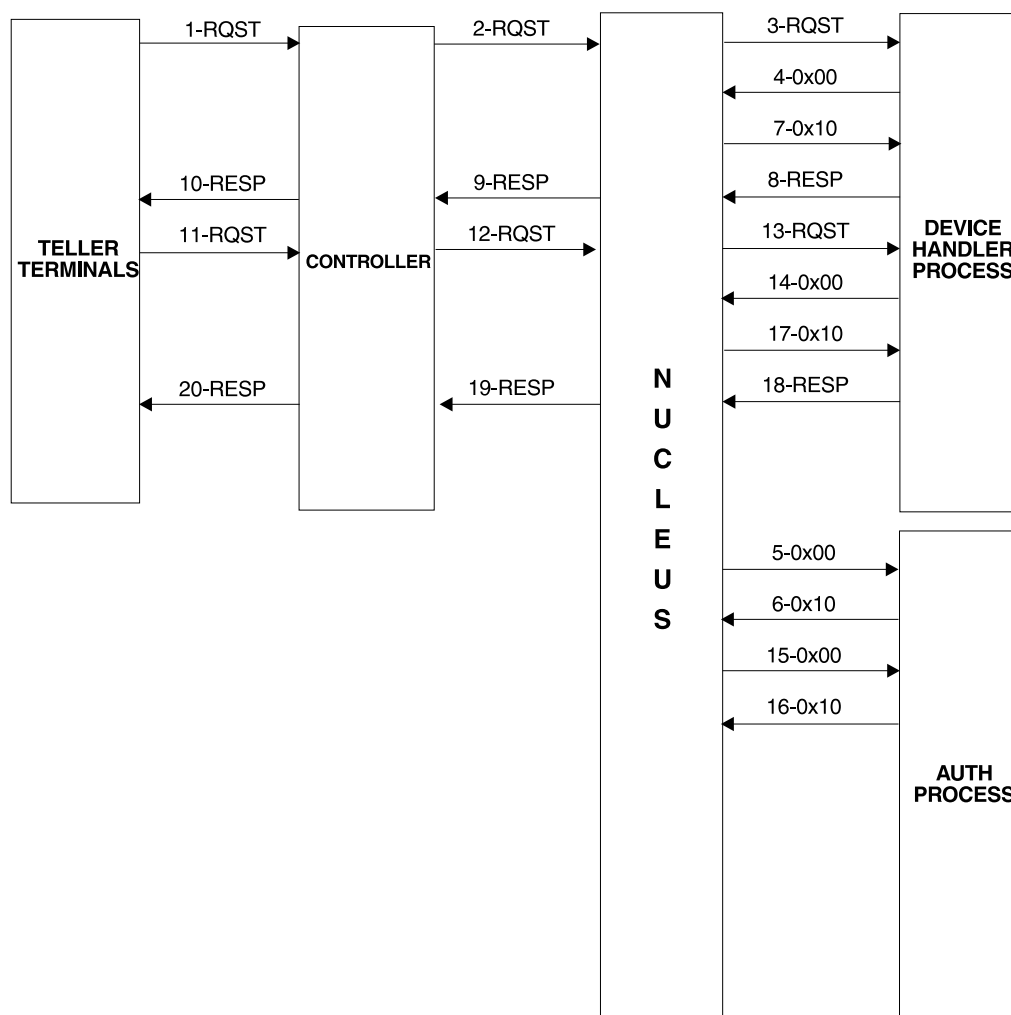


Steps	Processing
1, 2, 3	The device sends a native-mode request message to the Device Handler.
4, 5	The Device Handler translates the native-mode message into a 0200 that it sends to the Authorization process.

Steps	Processing
6, 7	The Authorization process changes the 0200 into a 0210 that it sends to the Device Handler. The Authorization process logs the 0210 to the TTLF.
8, 9	The Device Handler translates the 0210 into a native-mode response message that it sends to the controller and device.
10	Unable to communicate with the controller, the Nucleus sends a failure message to the Device Handler.
11, 12	The Device Handler translates the failure message into a 0420 that it sends to the Authorization process. Authorization process reverses the effect of the original transaction and logs the 0420 to the TTLF.

Diagram 13 — Override Authorized by the Authorization Process

Diagram 13 illustrates the flow of a request authorized by the Authorization process with one or more response codes that can be overridden. In this diagram, x can be 2 or 3.

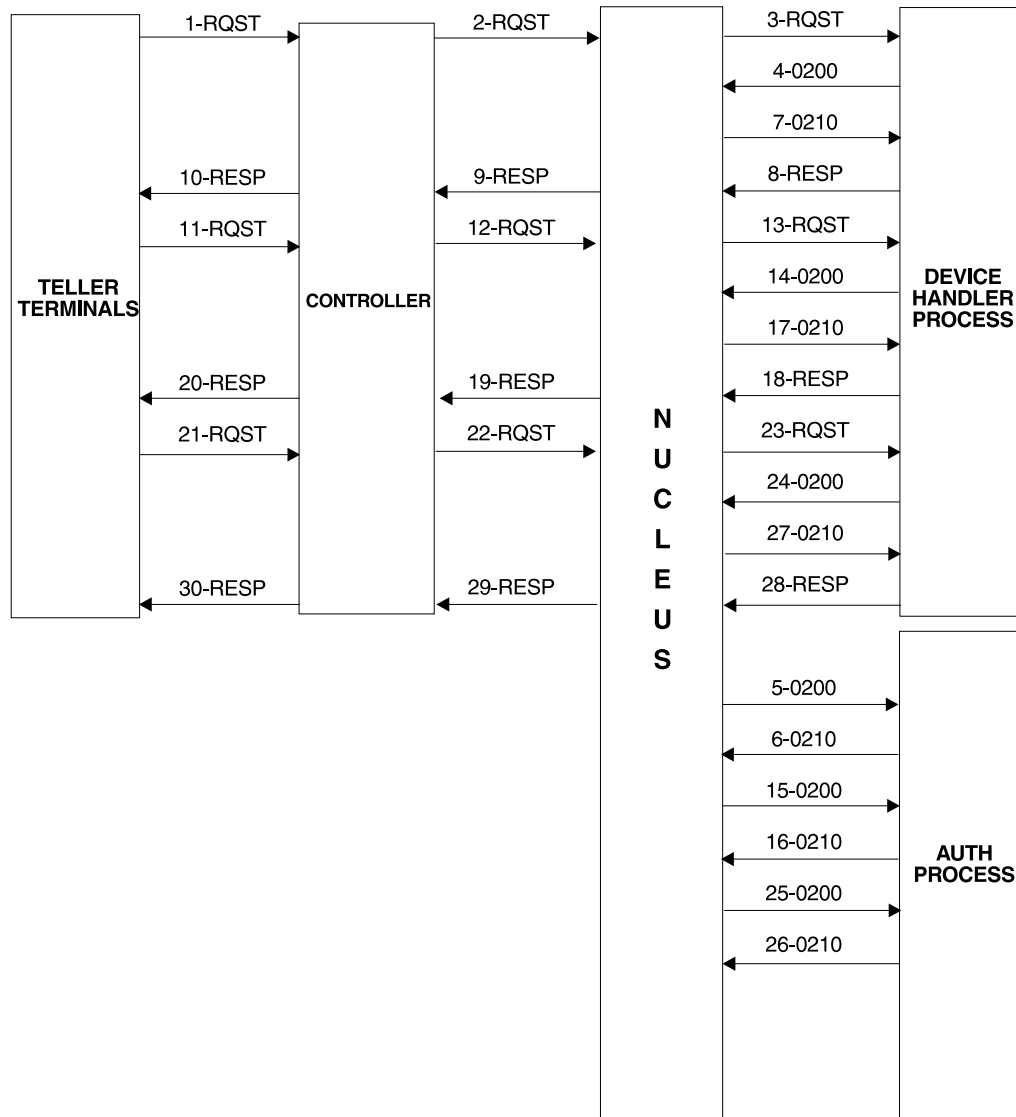


Steps	Processing
1, 2, 3	The device sends a native-mode request message to the Device Handler.
4, 5	The Device Handler translates the native-mode message into a 0x00 that it sends to the Authorization process.

Steps	Processing
6, 7	The Authorization process changes the 0x00 into a 0x10 that includes response code F0Z (override required) and an Override token containing the response codes that can be overridden. The Authorization process then sends the 0x10 to the Device Handler.
8, 9, 10	The Device Handler translates the 0x10 into a native-mode response message and returns it to the device.
11, 12, 13	The teller provides an initial override level that is equal to or greater than the override required for each of the response codes in the first response. The native-mode request message containing the new initial override level sends from the device to the Device Handler. BASE24-teller treats this native-mode request message as a new transaction.
14, 15	The Device Handler translates the native-mode message into a 0x00 that it sends to the Authorization process.
16, 17	The Authorization process changes the 0x00 into a 0x10, logs it to the TTLF, and then sends it to the Device Handler.
18, 19, 20	The Device Handler translates the 0x10 into a native-mode response message that it sends to the device.

Diagram 14 — Multiple Account Selection

Diagram 14 illustrates the flow of a financial request involving multiple account selection. While shown here for a financial request, multiple account selection can also occur with file inquiry/update requests. Multiple account selection is supported only for card-initiated transactions, and is necessary when the customer's Cardholder Authorization File (CAF) record contains more than one account of the type requested for the transaction.



The card-issuing institution can support multiple account selection in three ways:

- Require an account number with the transaction (i.e., The institution does not support multiple account selection).
- Return the available account numbers to the teller terminal so the customer can make a selection.
- Use the first applicable account in the CAF record.

The teller terminal used to perform the transaction must support multiple account selection if account numbers are returned for the customer to make a selection. Refer to “Account Selection” in section 4 of this manual for additional information.

Steps	Processing
1, 2, 3	The device sends a native-mode request message to the Device Handler.
4, 5	The Device Handler translates the native-mode message into a 0200 and sends it to the Authorization process.
6, 7	The Authorization process changes the 0200 into a 0210 and sends it to the Device Handler. Because the CAF contains more than one account of the requested type, this 0210 includes a CAF Inquiry token and response code F3Z (multiple account selection). The CAF Inquiry token can carry five account numbers. The Authorization process does not log this 0210 to the TTLF.
8, 9, 10	The Device Handler translates the 0210 into a native-mode response message and sends it to the device. This message contains the account numbers that were returned to the Device Handler in the CAF Inquiry token. The teller reviews the account numbers in this message. If this message contains the desired account number, processing continues with step 21. If this message does not contain the desired account number and additional account numbers are available, processing continues with step 11.
11, 12, 13	The teller requests additional account numbers. The device sends a native-mode request message to the Device Handler.

Steps	Processing
14, 15	The Device Handler translates the native-mode message into a 0200 and sends it to the Authorization process. The Device Handler sets a flag (RQST.MULT-ACCT-RQST-FLG) in the message header to identify this 0200 as part of another transaction.
16, 17	The Authorization process changes the 0200 into a 0210 and sends it to the Device Handler. Like the 0210 earlier, this 0210 includes a CAF Inquiry token and response code F3Z (multiple account selection). The Authorization process does not log this 0210 to the TTLF because of the multiple account flag in the message header. The CAF Inquiry token in this message can carry five additional account numbers.
18, 19, 20	The Device Handler translates the 0210 into a native-mode response message and sends it to the device. This message contains the additional account numbers that were returned to the Device Handler in the CAF Inquiry token.
21, 22, 23	The teller selects one of the account numbers returned in the response. The device sends a native-mode request message to the Device Handler.
24, 25	The Device Handler translates the native-mode message into a 0200 and sends it to the Authorization process.
26, 27	The Authorization process changes the 0200 into a 0210 and sends it to the Device Handler. The Authorization process logs this 0210 to the TTLF.
28, 29, 30	The Device Handler translates the 0210 into a native-mode response message and sends it to the device.

Section 4

BASE24-teller Transaction Processing Controls

This section describes the basic BASE24-teller transaction processing controls that are available. This section should be used with section 5, which describes how BASE24-teller performs the various types of processing.

Processing Control Options

The options presented in this section deal specifically with BASE24-teller transaction processing controls. Additional configuration issues such as hardware characteristics, terminal locations, and BASE24 file and process names also need to be addressed. Refer to the **BASE24 Base Files Maintenance Manual**, **BASE24-teller Files Maintenance Manual**, and **BASE24 Logical Network Configuration File Manual** for information regarding these additional configuration issues. Also refer to these manuals for the screen layouts and valid values for setting the transaction processing options presented in this section.

BASE24-teller offers processing control options in the following areas:

- Banking relationships, including on-us and not-on-us processing, interbank routing, and account number modification.
- Transaction destinations, including message logging, transaction authorizers, and completion messages.
- Store-and-forward processing.
- Card handling, including account selection and transaction screening.
- Transaction security, including PIN and card verification, PIN encryption, and message authentication.
- Account processing, including customer classes, balance updates, cash limits, preauthorized holds, passbooks, stop payments, warnings, holds, deposit float, and customer short name.
- Overdraft control and protection, including overdraft limit and credit line/backup accounts.
- Operator level controls, which include the minimum operator level needed to perform each transaction code or override each response code at a teller terminal and the maximum level of override that can be performed at each teller terminal.

Banking Relationships

The relationship between the financial institution that owns a customer's account and the financial institution handling the customer's transaction impacts the way BASE24-teller processes the transaction.

On-us and Not-on-us Transactions

A transaction performed at the financial institution that owns the account is an *on-us* transaction, while a transaction performed at a financial institution that does not own the account is a *not-on-us* transaction. BASE24-teller uses the distinction between on-us and not-on-us transactions to make two determinations: whether the transaction is supported and the operator level required to override any error messages received while the transaction is being processed.

Supported Transactions

BASE24-teller uses the distinction between on-us and not-on-us transactions to determine whether a transaction is supported for one financial institution's accountholder when it is performed at another financial institution's terminal.

The value in the TERM TRAN ALLOWED field of the Teller Transaction File (TTF), which can be accessed via TTF screen 1, controls processing for our accountholder at our terminal. The value in the OUR CUST/NOT-ON-US TERM field of the TTF, which can be accessed via TTF screen 1, controls processing for our accountholder at another institution's terminal. The following table illustrates the three possible combinations for transaction support.

Option	Our Accountholder at Our Terminal (On-us)	Our Accountholder at Another Institution's Terminal (Not-on-us)
1	Yes	Yes
2	Yes	No
3	No	No

A transaction can be supported for the account-owning financial institution's accountholder at the institution's own terminal with or without being supported when the accountholder attempts to perform the transaction at another financial institution's terminal.

In Release 5.0, BASE24-teller cannot acquire transactions from terminals attached to the host. As a result, a transaction cannot be supported for an accountholder at another financial institution's terminal if the transaction is not supported for the accountholder at his or her own financial institution's terminal.

Note: When TTF and ORF records exist for the accountholder's financial institution and the terminal owner's financial institution, the BASE24-teller Authorization process uses the TTF and ORF records belonging to the accountholder's financial institution to determine whether the transaction is supported. The BASE24-teller Device Handler uses only the terminal owner's TTF to determine whether Device Handler log-only transactions are supported.

Required Operator Level

BASE24-teller uses the distinction between on-us and not-on-us transactions to determine the minimum operator level required for tellers to override certain response codes when handling transactions.

The value in the OUR CUST/OUR TERM OVERRIDE field of the Override Response File (ORF), which can be accessed via ORF screen 1, sets this level for on-us transactions. The value in the OUR CUST/NOT-ON-US TERM OVERRIDE field of the ORF, which can be accessed via ORF screen 1, sets this level for not-on-us transactions. This distinction allows a financial institution to require different levels of authority to override a response code depending on where its customers conduct their business. Refer to "Operator Level Controls" at the end of this section for additional information.

Note: When TTF and ORF records exist for the accountholder's financial institution and the terminal owner's financial institution, BASE24-teller uses the TTF and ORF records belonging to the accountholder's financial institution to determine the minimum operator level that is required to override the response codes.

Interbank Routing

Support for not-on-us transactions is also known as interbank routing because more than one financial institution is involved. Interbank routing allows an accountholder to conduct transactions at a financial institution other than the one that owns his or her account. The transaction information is passed from the financial institution that is serving the accountholder to the financial institution that owns the account.

Interbank routing permits transactions involving one or more accounts owned by one financial institution to be performed at a terminal belonging to another financial institution. Regardless of whether interbank routing is involved, however, a single financial institution must own all of the accounts involved in a transaction. For example, Institution A must own the *from* and *to* accounts in a transfer transaction (transaction code 40).

Note: BASE24-teller supports interbank routing for all message types except administrative messages (0600s).

Account Routing File (ARF)

BASE24-teller uses the Account Routing File (ARF) in two ways when interbank routing is supported:

- To identify the FIID of the accountholder's financial institution.
- To modify the accountholder's account number in situations when the account number on the customer's document does not match the account number on the financial institution's records.

BASE24-teller uses the ARF only when interbank routing is supported. The BASE24-atm add-on Self-Service Banking Check Application also uses the ARF. Refer to the device-specific BASE24-atm self-service banking (SSB) manual for information about self-service banking.

Banking Relationships

Banking relationship is the term used by BASE24-teller to describe an arrangement between two financial institutions permitting the accountholder of one financial institution to conduct transactions on his or her account at a terminal owned by the other financial institution. A banking relationship must exist between the financial institutions involved in interbank routing. In BASE24-teller, each financial institution can participate in up to 24 banking relationships, with no limit to the number of financial institutions in each of the banking relationships.

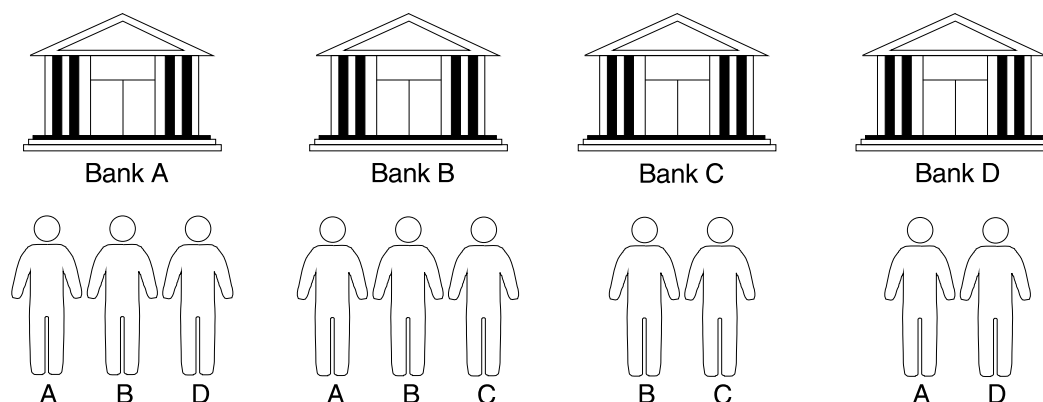
The value in the INTERBANK ROUTING field of the Institution Definition File (IDF), which can be accessed via IDF screen 25, controls whether a financial institution supports interbank routing. An ARF must be configured if any institution in the logical network has the value in this field set to Y (yes, interbank routing is supported). During initialization, the BASE24-teller Authorization process sets a fatal error flag in its internal IDF table for each institution with a Y in the INTERBANK ROUTING field if the ARF has not been configured.

One financial institution's accountholder can perform a transaction at another financial institution's terminal when the accountholder's financial institution and the terminal owner's financial institution participate in the same banking relationship. Codes in the BANKING RELNSHP field in the IDF record of the accountholder's financial institution identify each of the relationships in which that institution participates. This field can be accessed via IDF screen 25. Codes in the BANKING RELNSHP field in the Teller Terminal Data File (TTDF) of the terminal owner identify each of the relationships in which that institution participates. This field can be accessed via TTDF screen 1. The BANKING RELNSHP fields in the IDF and TTDF use the values A through Z and 1 through 9 to identify the banking relationships. Each field can contain a maximum of 24 one-character codes.

Banks can have one-way or two-way relationships. For example, banks A, B, C, and D could have the following relationships defined in their IDF and TTDF records.

	Bank A	Bank B	Bank C	Bank D
IDF Relationships	1, G	1, J	J	G
TTDF Relationships	1, G	1, J	J	G

These are two-way relationships because each bank has the same relationships defined in its IDF and TTDF records (e.g., Bank A has relationships 1 and G in its IDF record and relationships 1 and G in each of its TTDF records). In these relationships, the customers of both banks can conduct their business at the other bank's terminals, as illustrated below.



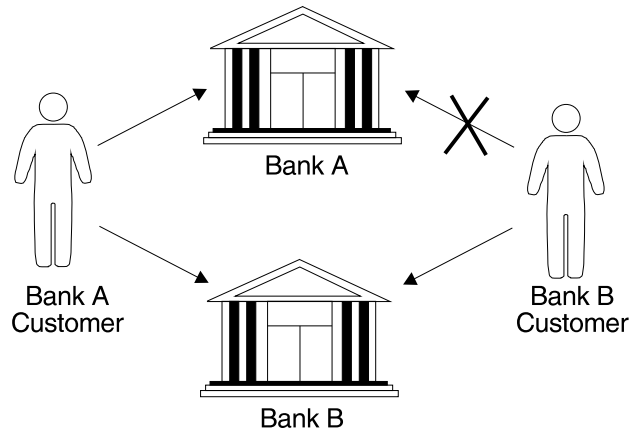
Customers from Bank A can use terminals at Bank B because both banks have relationship 1 or use terminals at Bank D because both banks have relationship G. Customers from Bank B can use terminals at Bank A because both banks have relationship 1 or use terminals at Bank C because both banks have relationship J. Customers at Bank C can use terminals at Bank B, and customers at Bank D can use terminals at Bank A. Banking relationships do not affect a customer's ability to use the terminals at his or her own financial institution.

Financial institutions can also have one-way banking relationships. For example, banks A and B could have the following relationships defined in their IDF and TTDF records.

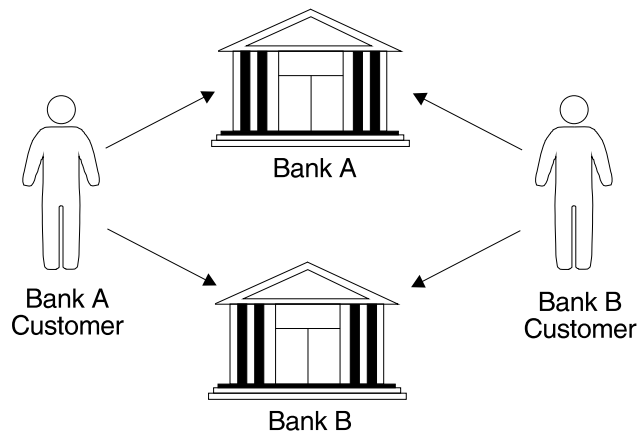
	Bank A	Bank B
IDF Relationships	1, D, H	H
TTDF Relationships	1, D	H

One-way relationships mean each bank does not necessarily have the same relationships defined in its IDF and TTDF records (e.g., Bank A has relationships 1, D, and H in its IDF record but has only relationships 1 and D in each of its TTDF records). In these relationships, one bank's customers may or may not be

able to conduct business at a second bank's terminals just because the second bank's customers can conduct business at the first bank's terminals, as illustrated below.



Customers of Bank A can conduct their business at Bank B's terminals. However, the customers of Bank B cannot conduct their business at Bank A's terminals. If Bank A adds relationship H to its TTDF records, Bank B's customers then would be able to use Bank A's terminals, as illustrated below.



Note: Since banking relationships are defined in the TTDF, each terminal can have its own set of relationships. These relationships can be based on physical location or teller responsibility. For example, a financial institution has branches in several communities but the teller terminals in only one of the communities should be in a particular banking relationship. In the case of teller responsibility, participation in a banking relationship can be limited to the terminal in a special customer service window.

Identifying the Accountholder's Financial Institution

When interbank routing is supported and a plastic card is not used to initiate the transaction, the BASE24-teller Authorization process can use information from the Account Routing File (ARF) in two ways to identify the FIID of the accountholder's financial institution.

In one method, the account type, account number length, one or more digits within the account number, and the corresponding FIID are entered on the account number routing version of the ARF record. The account type is based on the transaction code and the account number length is computed by the Authorization process. The digits within the account number must be a specific value located in a specific position and uniquely identify each FIID.

In the other method, the account type, account number length, a routing code, and the corresponding FIID are entered on the bank routing code version of the ARF record. The account type is based on the transaction code and the account number length is computed by the Authorization process. When digits within the account number cannot be used to uniquely identify each FIID, a routing code assigned to each FIID is entered by the teller each time he or she is processing a not-on-us item.

Card-Initiated Transactions

The source from which BASE24-teller obtains the FIID of an accountholder's financial institution depends on whether interbank routing is supported by the terminal-owning FIID and whether the transaction is initiated with a plastic card. When interbank routing is supported and a transaction is initiated without a plastic card, BASE24-teller uses the ARF to identify the FIID of the accountholder's financial institution. The teller provides the information necessary to select the correct ARF record (i.e., an account number with unique digits or an account number and routing code). BASE24-teller uses the FIID from the ARF record.

When interbank routing is supported and a transaction is initiated with a plastic card, BASE24-teller obtains the FIID of the accountholder's financial institution from the Card Prefix File (CPF) instead of the ARF. The primary account number (PAN) on the card includes a prefix and customer account number. BASE24-teller uses the prefix and the length of the PAN to select the correct CPF record. The CPF record contains the FIID of the accountholder's financial institution.

When interbank routing is not supported by the terminal-owning FIID, BASE24-teller uses the TTDF record for the terminal where the transaction is initiated to identify the FIID of the accountholder's financial institution.

Account Number Modification

BASE24-teller maintains records for customer accounts in Positive Balance Files (PBFs). Each record in a PBF must have a unique key, or value that distinguishes one record from another. The financial institution's FIID and the account type are added to the account number to form the key to the PBF. Normally these values are sufficient to uniquely identify each PBF record; however, mergers and acquisitions can result in accounts formerly identified by two or more FIIDs being identified by the same FIID.

For example, if the institution known to BASE24 as FIID A acquires the institution known as FIID B, the FIID value for all of the second institution's PBF records would be changed from B to A. If the PBF records from the two FIIDs can be combined without any records having the same key, there are no problems. However, more than likely, the two institutions issued at least some of the same account numbers. Since duplicate PBF record keys are not permitted, FIID A has two options:

- Reissue one account from each pair of duplicates. A new PBF record is created and all the customer records such as checks or passbooks must be replaced immediately.
- Create a new PBF record for one account from each pair of duplicates and have BASE24 modify the duplicate customer account number so that it matches the key to the new PBF record. Checks and passbooks can be changed when it is practical to do so.

As it processes transactions, the BASE24-teller Authorization process uses information from the ARF to modify a customer account number so that it matches the key to a PBF record. The bank routing code version of the ARF record contains instructions for modifying the customer account number. (This is the same ARF record that BASE24-teller uses to determine the FIID based on the account type, account number length, and routing code entered by the teller.) One or more digits can be inserted at a specific location within the customer account number to make it unique and match a PBF record key. Digits can be added to, but cannot replace, digits in the existing customer account number.

Card-Initiated Transactions

BASE24-teller can modify account numbers for transactions initiated with or without a plastic card. The ARF contains all account number modification information. No distinction is made between transactions initiated with a plastic card and transactions initiated without a plastic card. If the customer's account number is included with the transaction, the BASE24-teller Authorization process can obtain instructions for modifying the customer account number from the bank routing code version of the ARF record.

Note: If an institution is using the ARF to modify account numbers in card-initiated transactions, the ACCOUNT NUMBER fields on CAF screens 3 and 4 must contain the modified account numbers.

Transaction Destinations

BASE24-teller can send the transactions it receives to one of three possible destinations: Device Handler, Authorization process, or host. The Authorization process can perform two functions with each transaction it receives: authorize and log the transaction or just log the transaction. Each institution can specify one of the following options for each BASE24-teller transaction code:

- Log to the Teller Transaction Log File (TTLF) by the Device Handler.
- Log to the TTLF by the BASE24-teller Authorization process. The Authorization process does not attempt to approve or decline the transaction.
- Authorize by the BASE24-teller Authorization process using the Positive Balance Authorization method. The Authorization process also logs the transaction to the TTLF.
- Authorize by the host. The BASE24-teller Authorization process logs the transaction to the TTLF when it receives the response from the Host Interface process.

Values in the Teller Transaction File (TTF) control how BASE24-teller processes each transaction it receives. Another value in the TTF controls when the Authorization process sends a completion message to the host. Each financial institution has records in the TTF for every transaction code it can process using BASE24-teller. These fields can be accessed via TTF screen 1.

Logging by Device Handler

The value in the TERM TRAN ALLOWED field of the terminal owner's TTF controls whether a transaction is allowed at a terminal and if allowed, whether the transaction is logged to the TTLF by the Device Handler or passed to the Authorization process. Any BASE24-teller transaction can be logged by the Device Handler; however, a transaction that is logged by the Device Handler is not passed to the Authorization process or the host. Only the Authorization process can update or have access to the following files:

- Cardholder Authorization File (CAF)
- No Book File (NBF)
- Positive Balance File (PBF)
- Stop Payment File (SPF)
- Warning/Hold/Float File (WHFF)

No distinction is made between on-us and not-on-us accountholders for transactions logged by the Device Handler.

When the value in the TERM TRAN ALLOWED field in the TTF indicates that a transaction is not allowed, the transaction is dropped without being logged to the TTLF. Transactions that are supposed to be logged to the TTLF by the Device Handler are also dropped without being logged if an error occurs while the Device Handler is translating the native message into the internal message.

Transaction codes identified in appendix C as log-only transactions (e.g., 130000 is a cash CD log-only transaction and 160000 is a miscellaneous debit log-only transaction) are given that name because they do not update the CAF, NBF, or PBF, regardless of settings in the TTF. Log-only transactions cannot update the CAF, NBF, PBF, SPF, or WHFF because the transaction codes do not specify the *from* account type or *to* account type. These transactions can be logged to the TTLF by the Device Handler or the Authorization process and can also be passed to the host.

Logging by Authorization Process

If a transaction has not been logged to the TTLF by the Device Handler, the Authorization process checks the value in the AUTH LEVEL field in the account-owning institution's TTF records to determine whether the Authorization process should only log the transaction or should perform additional processing. Any BASE24-teller transaction code can be logged to the TTLF by the Authorization process.

Authorization by BASE24-teller Authorization Process

If a transaction is to be authorized, the value in the AUTH LEVEL field in the account-owning institution's TTF controls whether the BASE24-teller Authorization process authorizes the transaction or passes the transaction to a host for authorization. BASE24-teller uses the Positive Balance Authorization method for all transactions it authorizes.

The Authorization process logs all transactions it authorizes itself or passes to a host for authorization. When the transaction is logged depends on the authorizer, the message type, and other configuration options. The message flow diagrams in section 3 illustrate this timing.

Any BASE24-teller transaction code can be configured to be authorized by the Authorization process; however, processing of transaction codes identified in appendix C as log-only (e.g., 130000 and 160000) depends on the setting in the AUTH LEVEL field of the TTF record and whether a card is used to initiate the transaction.

When a log-only transaction is initiated with a card and the AUTH LEVEL field in the TTF is set for authorization by BASE24, the BASE24-teller Authorization process can check the card expiration date, verify the PIN, and verify the card before logging the transaction to the TTLF. These checks are known as transaction screening. Refer to “Transaction Screening” later in this section for additional information.

When a log-only transaction is initiated with a card and the AUTH LEVEL field in the TTF is set for log only, the BASE24-teller Authorization process logs the transaction to the TTLF without performing any transaction screening. The AUTH LEVEL field in the TTF must be set for authorization by BASE24 to perform transaction screening.

When a log-only transaction is initiated without a card and the AUTH LEVEL field in the TTF is set for authorization by BASE24 or set for log only, the BASE24-teller Authorization process logs the transaction to the TTLF without performing any screening checks. Cards are required to perform screening checks.

Authorization by Host

If the value in the AUTH LEVEL field in the account-owning institution’s TTF indicates the transaction is authorized by the host, the BASE24-teller Authorization process passes the transaction to the host. Any BASE24-teller transaction code can be configured to be authorized by the host. The BASE24-teller Authorization process cannot stand in for the host to authorize a transaction when the host is unavailable, so transactions configured to be authorized by the host are declined when the host is unavailable.

Values in the TELLER ROUTING INFORMATION fields in the IDF, which can be accessed via IDF screen 28, identify the data processing center (DPC) that is to receive the transaction. Each financial institution can specify up to five different DPCs to receive its transactions, based on the type of account.

Completion Messages

For transactions authorized by the Authorization process, the value in the COMPLETIONS TO HOST field in the account-owning institution's TTF indicates when the Authorization process should generate a completion message and pass it to the host. There are four options for sending completion messages:

- Do not send completion messages.
- Send completion messages only for approved transactions.
- Send completion messages only for declined transactions.
- Send completion messages for approved and declined transactions.

The Authorization process checks the first position of the response code in the RESP-HDR.RESP-CDE field of the Teller Standard Internal Message Header (TSTMH) to determine whether a transaction has been approved (a value of A) or declined (a value of F for fatal or a value of O for override possible). If the authorization result matches the option specified in the COMPLETIONS TO HOST field in the TTF, the Authorization process generates a completion message. A completion message is not sent when the Authorization process is unable to locate a TTF record for the transaction code or when the transaction has been declined because of a message edit error (response code F01).

Store-and-Forward Processing

When communications to a host system are down for maintenance or other processing needs, BASE24-teller can continue to process the transactions designated for authorization on BASE24, storing completion messages for these transactions in a data file. These completion messages can be transmitted to the host when communications are reestablished. This function is known as store-and-forward processing.

BASE24-teller does not perform stand-in authorization. BASE24-teller declines transactions designated for host authorization when the host is unavailable.

When communications are reestablished, BASE24-teller can forward the stored transactions to the host either interspersed with live transactions or consecutively. Institutions can specify the method used to send the store-and-forward messages to the host by setting the STORE AND FORWARD processing flag on screen 1 of the Host Configuration File (HCF). Institutions can also specify the maximum number of store-and-forward messages that can be outstanding to a host at one time and the number of times the Host Interface process should attempt to send the same store-and-forward message before logging it to the error log and deleting it from the Store-and-Forward File (SAF). These limits are set via the MAXIMUM OUTSTANDING SAFS and MAX SAF RETRY fields on HCF screen 1.

For detailed information on store-and-forward processing to the host system, refer to the ***BASE24 ISO Host Interface Manual*** or the ***BASE24 ANSI Host Interface Manual***.

Card Handling Options

A teller can initiate all BASE24-teller transactions except those listed below without a plastic card by manually entering the account number at the teller terminal.

- 33 = Cardholder Authorization File (CAF) inquiry
- 73 = Change CAF card status
- 74 = Change CAF and Positive Balance File (PBF) account status
- 75 = Verify PIN

Depending on the teller terminal capabilities and the BASE24-teller configuration, all BASE24-teller transactions (including transaction codes 33, 73, 74, and 75) can be initiated by passing a plastic card through a magnetic stripe reader attached to the teller terminal. The value of the TLR-PROCESS-CRD-PIN param in the Logical Network Configuration File (LCONF) controls whether BASE24-teller accepts plastic cards.

The characteristics of any card used with BASE24-teller must be defined in the Card Prefix File (CPF), which can be accessed via CPF screen 1. The CARD TRACK INFORMATION, PAN ACCESS TYPE, and EXP CHECK TYPE fields must be completed for each card prefix. CARD TRACK INFORMATION fields include the type of card and the position of information contained in Track 2 of the card. A position, or offset, can be specified for the cardholder's member number, PIN verification information, and the card expiration date. The value in the PAN ACCESS TYPE field identifies whether the cardholder's member number is located on the card. The value in the EXP CHECK TYPE field identifies whether BASE24-teller should check the card expiration date and where the expiration date is located.

A CAF record must exist for each card used with BASE24-teller. The CAF record contains the card type and card status, which can be accessed via CAF screen 1, and the account number and status of each account associated with the card, which can be accessed via CAF screens 3 and 4. While the BASE24-teller Authorization process does not use any of the activity limits or accumulators in the CAF, it will clear the accumulators when it posts the first transaction to an account on a new calendar day.

Note: BASE24-teller supports account types 02 through 09, 14 through 19, and 33 through 39 only when cards are used to access PBF accounts. Care must be taken when using cards because the BASE24-teller Authorization process treats all account types in a range to be the same when determining whether a PBF account number is unique. When making this determination, account types 01 through 09 are in the same range, account types 11 and 14 through 19 are in the same range,

and account types 31 and 33 through 39 are in the same range. As a result, the accounts shown below are considered unique when they are entered in the PBF and the CAF. However, the BASE24-teller Authorization process considers the accounts to be duplicates because types 31 and 33 are in the same range.

Description	Account A	Account B
FIID	BNK0	BNK0
Account Number	123456	123456
Account Type	31	33

Account Selection

The CAF record for one card can contain up to 16 accounts, so the teller must be able to select the appropriate account number when a card with more than one account is used to initiate the transaction. The value in the MULTIPLE ACCT SELECT field in the Teller Terminal Data File (TTDF), which can be accessed via TTDF screen 1, determines whether the teller terminal can handle multiple account selection. The value in the ACCOUNT SELECT INDICATOR field in the IDF, which can be accessed via IDF screen 25, determines how the financial institution wants the BASE24-teller Authorization process to handle a transaction when the CAF record contains more than one account of the type specified by the transaction code. The Authorization process can perform one of the following actions:

- Deny the transaction if the account number is not specified.
- Return the CAF accounts to the teller so the customer can select the appropriate account. If the MULTIPLE ACCT SELECT field in the TTDF indicates the teller terminal cannot handle multiple account selection, the transaction will be declined.
- Use the first applicable CAF account. This is an open primary account identified by a value of 3, M, N, O, P, Q, or R in the STATUS fields of the CAF. These fields can be accessed via CAF screen 3 or 4. If no open primary account exists, the Authorization process uses the first account in the CAF record of the account type specified by the transaction code that permits the requested transaction.

Transaction Screening

Transaction screening can be performed in situations where BASE24 shares authorization responsibilities with the host. In such situations, the BASE24-teller Authorization process can make preliminary standard checks on the transaction and decline those that do not meet the preliminary requirements. This prevents unnecessary processing by the host.

BASE24-teller can perform transaction screening only when configured to do so by the institution and when the institution is authorizing transactions on the host. BASE24-teller automatically performs these checks as part of its authorization routine when transactions are authorized offline by BASE24-teller.

The BASE24-teller Authorization process can perform the following transaction screening checks:

- Check the card expiration date
- Verify the PIN
- Verify the card for a card-initiated transaction

BASE24-teller does not perform the other transaction screening checks available with BASE24-atm and BASE24-pos (i.e., transaction limits and card status). BASE24-teller does not perform any transaction screening checks for transaction code 750000 (verify PIN) when it is being passed to the host for authorization.

The BASE24-teller Authorization process can perform transaction screening checks in two situations. When a card-initiated transaction is to be authorized by the host, the Authorization process can perform these checks before passing the transaction to the host. By declining a transaction that fails these checks instead of passing the transaction to the host, BASE24-teller can help reduce unnecessary host processing and communication costs.

When a log-only transaction code (e.g., 110000, 120000, 130000) is initiated with a card, the Authorization process can perform these checks before logging the transaction to the TTLF. In order for the Authorization process to perform these checks, the AUTH LEVEL field in the TTF records for the log-only transaction codes must be set to authorization by BASE24. These checks are not performed when the AUTH LEVEL field in the TTF records for log-only transaction codes is set to log only.

Expiration Date

The expiration date check compares the card expiration date to the current date to determine whether the card has expired. The card expiration check is controlled by the value in the CHECK IF HOST ONLINE EXP DATE field in the IDF, which can be accessed via IDF screen 2. BASE24-teller performs the card expiration date check for 0200, 0220, 0300, and 0320 messages.

PIN Verification

The PIN verification check determines whether cardholders enter the correct PINs when initiating transactions. BASE24-teller can perform the PIN verification check for 0200 and 0300 messages. The PIN verification check is controlled by the value in the PIN field in the IDF, which is accessed via IDF screen 2. It is also controlled by the CHECK IF HOST ONLINE PIN field in the CPF, which is accessed via CPF screen 2.

The PIN field in the IDF establishes PIN prescreening at the institution level, meaning that all transactions initiated with cards issued by that institution should or should not be prescreened for valid PINs. However, an institution may want to prescreen all PINs except for those belonging to cards with specific prefixes or prescreen only those cards with certain prefixes. In either of these cases, the value in the CHECK IF HOST ONLINE PIN field in the CPF overrides the value in the PIN field in the IDF.

Card Verification

Card verification ensures that the card presented by the customer for a transaction is valid. Track 2 on a card can contain a value BASE24 calls the Card Verification Digits (CVD). The card issuer generates the CVD using card verification keys, the primary account number, the card expiration date, and the card's service code. To verify a card, BASE24 uses the same card verification keys and Track 2 information to compute another CVD. If the two CVDs match, BASE24 considers the card to be valid.

BASE24-teller can perform the card verification check for 0200 and 0300 messages. The card verification check is controlled by the value in the CHECK IF HOST ONLINE CV field in the CPF, which can be accessed via CPF screen 2.

Transaction Security

BASE24-teller offers four forms of transaction security: PIN verification, card verification, PIN encryption, and message authentication. The information presented here summarizes the transaction security features. Refer to the ***BASE24 Transaction Security Manual*** for additional information on each of these features.

Transaction security is established at the institution level, the card prefix level, and the teller terminal level. Institution-level parameters are contained in the IDF and are accessed via IDF screen 2. Card prefix-level parameters are contained in the CPF and are accessed via CPF screen 2. Teller terminal-level parameters are contained in the TTDF and are accessed via TTDF screen 2. Institution-level and card prefix-level parameters are also contained in the Key File (KEYF), Key Authorization File (KEYA), and the External Message File (EMF). Refer to the ***BASE24 Transaction Security Manual***, ***BASE24 Base Files Maintenance Manual***, and ***BASE24 External Message Manual*** for specific configuration instructions.

Transaction security can be performed by the teller controller, Device Handler, Authorization process, or the host. However, some transaction security features (e.g., message authentication) can be performed only by the host.

The BASE24-teller ISO Host Interface process can carry the PIN and other transaction security-related information in its external message. However, the BASE24-teller ANSI Host Interface process does not carry this information in its external message. As a result, the BASE24-teller ISO Host Interface process, but not the BASE24-teller ANSI Host Interface process, supports the transaction security features presented here.

PIN Verification

The BASE24-teller Authorization process can perform PIN verification for card-initiated transactions authorized by BASE24-teller or as a transaction screening step for card-initiated transactions authorized by a host. The value of the TLR-PROCESS-CRD-PIN param in the LCONF controls whether BASE24-teller should expect PINs with its card-initiated transactions.

PINs can be verified for financial transaction requests (0200 messages) and file inquiry/update requests (0300 messages); however, the Device Handler typically does not capture PINs for transactions it logs without sending them to the Authorization process, so PINs cannot be verified for those transactions.

If PINs are used, BASE24 can use software or security devices to perform PIN verification. BASE24 supports four methods of PIN verification:

- DES (IBM 3624) PIN verification
- Diebold PIN verification
- Identikey PIN verification
- VISA PVV PIN verification

The teller controller can also verify PINs. When PIN verification is performed by the teller controller, BASE24-teller must not receive any transactions containing invalid PINs from the controller.

BASE24-teller supports PINs as a method for identifying customers. Transaction code 750000 (verify PIN) is the only transaction code that actually requires a PIN when it is performed. It verifies the PIN without initiating any additional processing. The teller must initiate another transaction code to perform any additional processing.

The PIN is optional with all other transaction codes. BASE24-teller permits the teller to continue processing the transaction by overriding the response codes that indicate a PIN is missing or invalid.

The PIN offset is the value BASE24-teller uses when verifying the PIN. When the BASE24-teller Authorization process is verifying the PIN and authorizing the transaction, the PIN offset can be kept on Track 2 of the card or in the cardholder's CAF record. The PIN offset must be kept on Track 2 of the card when PIN verification is part of the transaction screening the BASE24-teller Authorization process performs before passing the transaction to a host for authorization.

BASE24-teller uses the BAD PIN TRIES field in the cardholder's CAF record to track bad PIN tries. Updating the BAD PIN TRIES field in the CAF is a step that the BASE24-teller Authorization process performs while authorizing a card-activated transaction. BASE24-teller does not track bad PIN tries when PIN verification is performed by the teller controller or the host.

Card Verification

Card verification can be performed on card-initiated transactions, and can be performed with or without PIN verification. Cards can be verified for financial transaction requests (0200 messages) and file inquiry/update requests (0300 messages); however, the Authorization process will not verify cards for log-only transactions. Card verification cannot be performed from manually entered card information; the Track 2 data on the card must be captured via a magnetic stripe reader. BASE24 uses security devices to perform card verification.

PIN Encryption

PINs for card-initiated transactions can be sent encrypted or in the clear. Dynamic key management support for PIN encryption is provided between BASE24-teller and the host with the ISO Host Interface process, but is not provided between BASE24-teller and the teller device or controller.

Message Authentication

Message authentication is a method used to monitor the integrity of data flowing between EFT devices and the transaction authorizer. A message can change as the result of tampering or because of communications difficulties. Message authentication identifies whether or not the message has changed. BASE24-teller supports message authentication for messages exchanged between BASE24 and the host via the ISO Host Interface process, but not between BASE24-teller and the teller device or controller.

Cardholder Usage Accumulation

BASE24 keeps card usage and amount totals in the Cardholder Authorization File (CAF) if the Positive Balance Authorization method is used; however, the only card usage that BASE24-teller keeps in the CAF is bad PIN tries. BASE24-atm and BASE24-pos keep a complete set of card usage and amount totals in the CAF. BASE24-atm and BASE24-pos also keep card usage and amount totals in the Usage Accumulation File (UAF) if the Negative Authorization with Usage Accumulation method is used.

BASE24-atm, BASE24-pos, and BASE24-teller clear all card usage and amount totals in an account's CAF record when processing the account's first transaction of a new accumulation period. All three products use the dates in the LAST RESET DATE field on CAF screen 3 and the BEGINNING DATE field on Institution Definition File (IDF) screen 4 to determine whether a new accumulation period has begun.

The BASE24-teller Authorization process updates an account's CAF record and clears usages and totals for the first transaction of a new accumulation period when processing the following transactions:

- An approved financial transaction.
- A financial transaction that is declined because of a bad PIN.
- An approved file inquiry transaction, regardless of the file, if the message contains a PIN token.
- An approved file update transaction, regardless of the file, if the message contains a PIN token.
- An approved CAF update transaction.

BASE24-atm and BASE24-pos maintain the BEGINNING DATE or LAST RESET DATE fields; however, BASE24-teller does not maintain either of these fields at the start of a new usage accumulation period. Therefore, BASE24-teller cannot recognize that a new usage accumulation period has begun or clear card usage and amount totals other than bad PIN tries unless the institution is using BASE24-atm or BASE24-pos in addition to BASE24-teller.

Clearing Bad PIN Tries without BASE24-atm or BASE24-pos

When an institution uses BASE24-teller without BASE24-atm or BASE24-pos, the only method available for clearing bad PIN tries is the entry of a valid PIN. The PIN TRIES RESET OPTION field on Card Prefix File (CPF) screen 2 or IDF screen 2 must be set to one of the following options:

- Reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- Reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.

Clearing Bad PIN Tries with BASE24-atm or BASE24-pos

When an institution uses BASE24-teller with BASE24-atm or BASE24-pos, bad PIN tries can be cleared by the entry of a valid PIN or by starting a new usage accumulation period. The PIN TRIES RESET OPTION field on Card Prefix File (CPF) screen 2 or IDF screen 2 can be set to any of the available options.

Account Processing Options

BASE24-teller offers several options regarding customer account processing, including customer classes, balance updates, cash limits, preauthorized holds, passbooks, stop payments, warnings, holds, deposit float, and customer short name.

Customer Class

The TELLER CUSTOMER CLASS TABLE in the IDF, which can be accessed via IDF screen 24, allows each financial institution to establish up to ten classes defining the maximum amount of cash that can be withdrawn in one day and the portion of each deposit that will be available for immediate withdrawal. This table enables a financial institution to limit its exposure from withdrawals of uncollected funds.

The CUSTOMER CLASS field in the PBF, which can be accessed via PBF screen 10, is used to assign each customer account to one of the ten classes. All BASE24-teller accounts must be assigned to one of the classes. Deposit and split deposit transactions (transaction codes 20 and 21) use all of the information for the assigned class, while all transactions in which cash is disbursed use the cash out limit information for the assigned class.

BASE24-teller uses values in the CASH IN INDICATOR and CASH OUT INDICATOR fields in the IDF, which can be accessed via IDF screen 25, with values in the TELLER CUSTOMER CLASS TABLE to compute the available funds and amount on hold balances for deposit transactions. Section 5 provides a table that illustrates these computations.

Available Funds and Amount on Hold Balances

The CASH IN INDICATOR and CASH OUT INDICATOR fields in the IDF control how the Authorization process handles the cash portion of deposit transactions when updating available funds and amount on hold balances in the PBF. The total amount of the deposit is always added to the ledger balance and the amount of cash disbursed is always added to the cash out accumulator.

BASE24-teller uses values in the CASH IN INDICATOR and CASH OUT INDICATOR fields and values in the TELLER CUSTOMER CLASS TABLE to compute the available funds and amount on hold balances for deposits. Refer to “Deposit Processing” in section 5 of this manual for examples of these computations.

Cash In Indicator

The value in the CASH IN INDICATOR field determines how much of the cash received with a transaction is added to the available funds balance and how much is added to the amount on hold balance. Options include the following:

- Add the entire cash in amount to the available balance.
- Add part of the cash in amount to the available and add the remainder of the cash in amount to the amount on hold balance.

When the cash in amount is split between the available funds and amount on hold balances, the portion added to each balance is based on the values in the TELLER CUSTOMER CLASS TABLE fields.

Cash Out Indicator

The value in the CASH OUT INDICATOR field determines how much of the cash disbursed with a transaction is subtracted from the available funds balance and added to the amount on hold balance. Options include the following:

- Do not adjust the available funds and amount on hold balances by the cash out amount.
- Subtract the entire cash out amount from the available funds balance and add it to the amount on hold balance.
- Reduce the cash out amount by the cash in amount, then subtract the net amount from the available funds balance and add it to the amount on hold balance. In this situation, the cash in and cash out amounts are ignored if the cash in amount is greater than the cash out amount.

Cash Limits

Two PBF fields and one IDF field permit a financial institution to limit the amount of cash received or disbursed for each of its accounts during a calendar day. The CASH IN LIMIT field on PBF screen 10 can be used to limit cash received, the CASH OUT LIMIT field on PBF screen 10 can be used to limit cash disbursed. The MAXIMUM CASH OUT field in the TELLER CUSTOMER CLASS TABLE on IDF screen 24 also can be used to limit the amount of cash disbursed from each of the accounts in a class. If the IDF and PBF cash out limits are set, both of them are applied to a transaction, with the IDF limit being applied before the PBF limit. These cash receipt and disbursement checks are in addition to the available funds and amount on hold calculations discussed earlier.

Preauthorized Holds

When using BASE24-pos with BASE24-teller, a financial institution has the option of considering preauthorized holds when authorizing transactions through BASE24-teller. A preauthorized hold is created when a merchant using BASE24-pos posts a hold transaction against a customer's checking, savings, or credit account before the purchase occurs in order to verify that funds will be available when the merchant posts the actual transaction. For example, a hotel clerk could place a preauthorized hold for the estimated amount of the customer's bill on the credit card when he or she checks in to ensure that funds will be available in the account to pay the bill when the customer is ready to check out.

BASE24-pos preauthorized purchase transactions and reversals of preauthorized purchase completion transactions add preauthorized holds to the appropriate account. BASE24-pos preauthorized purchase completion transactions and reversals of preauthorized purchase transactions can remove preauthorized holds. Preauthorized holds can also have expiration dates, after which they are no longer considered. Preauthorized holds can be maintained in the CAF, NEG, PBF, or CAF and PBF; however, BASE24-teller checks for preauthorized holds only in the PBF, so this must be considered when configuring BASE24-pos.

The option of considering PBF preauthorized holds for a BASE24-teller transaction is controlled at the transaction level using the INCLUDE PRE-AUTH HOLDS field in the TTF, which can be accessed via TTF screen 1. The customer's available balance does not reflect any preauthorized holds, but when the TTF record for the transaction being authorized indicates preauthorized holds should be considered, the Authorization process begins with the account's available balance and then subtracts the total of all active preauthorized holds to obtain an adjusted available balance. This adjusted available balance is used to authorize the transaction.

The BASE24-teller transaction codes that can include preauthorized holds when authorizing the transaction are identified in appendix C. Preauthorized holds cannot be placed on NOW, credit line/backup, installment loan, mortgage loan, or commercial loan accounts because these account types are not recognized by BASE24-pos. Preauthorized holds are checked only on transactions that reduce the available balance on noncredit accounts or available credit on credit accounts. They are not checked on log-only transactions or transactions that do not reduce these balances.

Passbooks

BASE24-teller supports the use of passbooks on the following account types:

- Savings accounts
- Individual retirement accounts (IRAs)
- Certificates of deposit (CDs)
- NOW accounts
- Installment loans
- Mortgage loans
- Commercial loans

Each of these accounts can be handled as a passbook or nonpassbook account. With a passbook account, a financial institution provides the customer with a passbook as evidence of the account and records each transaction in the passbook. With a nonpassbook account, a financial institution typically uses cards or coupons as evidence of the account and usually provides the customer with a receipt as a record of each transaction. For these accounts, the value in the PASSBOOK INDICATOR field in the PBF, which can be accessed via PBF screen 10, controls whether a passbook is being used with the account. Transaction codes supporting the use of passbooks are identified in appendix C.

When passbooks are issued on accounts, situations arise when transactions may be posted to the account, but the passbook is not available for updating. BASE24-teller uses the No Book File (NBF) as a temporary storage place for these transactions. If a passbook is being used with an account, the account also has an NBF record for each transaction posted to the account.

BASE24-teller treats all passbook account transactions as if the passbook is unavailable, adding an NBF record for each transaction posted to the account. NBF records are maintained from the time transactions occur until they are printed

in the passbook. NBF records can originate from transactions authorized by BASE24-teller and transactions authorized or generated by the host. Transactions authorized by BASE24 are added to the NBF when they are logged to the TTLF. The TTLF is extracted to update records on the host and transactions from the host are added to the NBF through partial or full-file refresh. During the period of time starting when the TTLF extract begins and ending when the NBF refresh is complete, BASE24-teller does not consider the NBF to be current. The PASSBOOK PRINT FLAG field in the IDF, which can be accessed via IDF screen 25, controls whether passbook print, passbook reprint, and automatic passbook print transactions can be performed while the NBF is not current.

Up to three NBF records can be written for a transaction. If a transaction involves a *from* account, a *to* account, and a credit line/backup account and all three accounts have passbooks, an NBF record is written for each account.

The TLR-MAX-NBF-REC-CNT param in the LCONF controls the number of NBF records that can exist on an account before a response code will be returned to inform the teller that too many NBF records exist. This response code, which can be overridden, gives the teller the option of canceling a passbook print or automatic passbook print transaction when continuing the transaction would tie up the teller terminal for an excessive amount of time.

The AUTOMATIC PASSBOOK PRINTING field in the TTDF, which can be accessed via TTDF screen 1, identifies whether a teller device supports automatic passbook printing. BASE24-teller supports automatic passbook printing for transactions that involve one account and are typically performed with the customer present so the passbook is available. The customer's current transaction is still added to the NBF so that any prior NBF records can be added to his or her passbook before the current transaction. This is done to maintain the passbook entries in chronological order. Transaction codes that can include automatic passbook printing are identified in appendix C.

Stop Payments

As part of the processing performed for some of the check cashing transactions, BASE24-teller searches the Stop Payment File (SPF) for stop payment information before authorizing the transaction. A stop payment order can be placed on a check written by one of a financial institution's own customers to prevent the check from being cashed by anyone.

Stop payment records can be added to the SPF based on a check number, a check number and amount, or a range of check numbers. Transaction codes 110100, 112100, 120100, 122100, 170100, and 172100 are the only transactions for which

BASE24 checks the SPF. Some or all of these transaction codes must be included in the TTF if the SPF is going to be used. Section 5 describes how BASE24-teller uses the SPF when processing check cashing transactions.

The STOP PAY/WARNING STATUS field in the PBF, which can be accessed via PBF screen 10, identifies whether a stop pay exists on the account.

Warnings

A warning is one of the record types in the Warning/Hold/Float File (WHFF). This record type can be used with any account type to notify the teller of a condition that should be investigated before finishing the transaction.

The SEARCH THE WHFF field in the TTF, which can be accessed via TTF screen 1, controls whether the Authorization process checks the WHFF before authorizing a financial transaction. The STOP PAY/WARNING STATUS field in the PBF, which can be accessed via PBF screen 10, identifies whether a warning exists on the account.

Holds and Deposit Float

Hold and deposit float are the remaining record types in the WHFF. Holds limit the funds available to a customer and deposit float entries increase the funds available to a customer.

The value of the TLR-WRITE-HLDS-DEP-FLOATS param in the LCONF controls whether WHFF records are written for add hold or add deposit float transactions and removed for delete hold or delete deposit float transactions. The value of this param does not affect the adjustments these transactions make to available funds and amount on hold balances in the PBF.

Customer Short Name

The CUSTOMER SHORT NAME field in the PBF, which is accessed via PBF screen 5, can contain a shortened version of the customer's name or name and address. The BASE24 database does not contain a customer's complete name and address record; however, maintaining an abbreviated version of this information allows BASE24 to include it in responses to a teller terminal. Having the customer short name available enables the teller to personalize his or her conversation with the customer or more quickly select the appropriate account when one card

accesses multiple accounts. Each institution can determine whether its PBF records will include a segment containing the customer's short name. The NAME (PBF SHORT) FIID AUTH FILE SEGMENT INDICATOR on IDF screen 5 controls the presence of short name information in the PBF record.

Short names are carried in the Customer Name token. When customer information is maintained on the host instead of BASE24, BASE24 uses the Customer Name token to carry the customer short name in the external message between the host and BASE24. The ISO Host Interface process must be used in order to carry this information. The ANSI Host Interface process does not include the customer short name in its external message.

Overdraft Control and Protection

BASE24-teller offers overdraft control and protection. Overdraft control limits how far an account can be overdrawn while overdraft protection keeps an account from becoming overdrawn by transferring funds to it from another account.

Overdraft Control

The OVERDRAFT LIMIT field in the PBF, which can be accessed via PBF screen 1, contains the amount by which a debit transaction (e.g., withdrawal, transfer, payment, cashed check) can exceed the available balance and still be approved. Transaction codes that check the overdraft limit are identified in appendix C. An overdraft limit can be specified for any account except installment, mortgage, and commercial loan accounts. BASE24-teller ignores the overdraft limit when an overdraft limit and an overdraft protection account are specified for the same account.

Overdraft Protection

Automatic overdraft protection goes by two names—credit line account or backup account—depending on the type of account being used to provide the protection. Credit line indicates a credit account (credit card or line of credit) is providing the protection while backup account indicates a noncredit account (checking, savings, or NOW) is providing the protection. Overdraft protection can be set up for checking, savings, and NOW accounts. Transaction codes that can check for automatic overdraft protection are identified in appendix C.

When credit line/backup account information is maintained on the host, BASE24 uses a token to carry credit line/backup account information between it and the host. An ISO Host Interface process must be used in order to carry credit line/backup account information because the ANSI Host Interface process does not include this information in its external message.

Each institution can determine whether its PBF records will include a segment containing information to identify the account a customer is using for overdraft protection. The CREDIT LINE FIID AUTH FILE SEGMENT INDICATOR field in the IDF, which can be accessed via IDF screen 5, controls the presence of overdraft protection account information in the PBF record.

Note: Segments supported by a logical network must be identified in the Product Indicator Table (PITABLE) before they will appear on IDF screen 5.

The TRANSFER METHOD and INCREMENT AMOUNT fields in the IDF, which can be accessed via IDF screen 6, control how the Authorization process calculates the amount of a transfer from the overdraft protection account to the account in need of the funds.

The ACCOUNT TYPE and ACCOUNT NUMBER fields in PBF, which can be accessed via PBF screen 6, identify the account from which funds are transferred when the customer's first account does not contain sufficient funds. If the account with insufficient funds is accessed via card, that account and the credit line/backup account do not have to be accessible (i.e., included on CAF screens 3 and 4) from the same card.

Operator Level Controls

BASE24-teller uses levels to determine whether an operator can post a transaction or override a response code at a teller terminal, as well as the maximum level of override that can be performed from a given teller terminal. The available levels include the following:

- None
- Teller
- Supervisor
- Manager
- Restricted

There is nothing unique about the teller, supervisor, and manager levels. The importance is the relationship the levels have to each other. The supervisor can perform more transactions or override more response codes than the teller, and the manager can perform more transactions or override more response codes than the supervisor. One of these levels is assigned to each teller terminal operator to control what he or she can do.

Restricted means no one can perform the transaction or override the response code. None means there are no restrictions on who can perform the transaction or that all teller terminal operators can override the response code.

The procedure for determining the operator level varies depending on the teller device being used. One device might use a sign-on transaction to identify a teller and a key switch to identify a supervisor's approval of a transaction. Another device might use cards to identify tellers, supervisors, and managers. The highest level of operator present, whether identified by a key, a card, or other procedure, is passed from the device to BASE24-teller in the native message.

BASE24-teller uses response codes to identify the result of processing each transaction it receives. These response codes are divided into three categories: approved, fatal, and override possible. Approved response codes indicate a transaction has been completed successfully, and fatal response codes indicate the reason a transaction could not be completed. A teller terminal operator must accept the information presented by approved and fatal response codes. The override-possible response codes also indicate a transaction could not be completed, but provide additional information to a teller terminal operator and give the operator an opportunity to complete the transaction. This opportunity to complete the transaction is known as an override.

The ANSI Host Interface can provide only one response code and a limited amount of supporting information (e.g., account balances), while the ISO Host Interface can provide all applicable response codes and supporting information. The ANSI Host Interface also uses the Release 3.4 response codes instead of the current release response codes. Refer to the ***BASE24 ANSI Host Interface Manual*** for the Release 3.4 response codes and their descriptions.

Minimum Teller Level

The MINIMUM TELLER LEVEL field in the TTF, which can be accessed via TTF screen 1, is used to control the operators that can perform each transaction code. Any of the five operator levels can be assigned to each transaction code. The operator attempting to perform the transaction must have at least the level specified in this field in order to perform the transaction. An operator level of none allows any operator to perform the transaction, while an operator level of restricted eliminates the transaction's use by any operator.

If this minimum is greater than the operator level, the transaction is declined with a fatal response code.

Response Code Override Level

BASE24-teller supports nearly 100 three-character response codes. Each code identifies a reason the transaction was processed in a particular manner. Response codes can be placed in three categories. The first character of each response code identifies the category in which it belongs: A for approved, F for fatal, or O for override possible. Refer to appendix B for a complete description of all BASE24-teller response codes.

Each of the response codes has its own record in the Override Response File (ORF). Records in the ORF are arranged in groups called profiles, with each profile having a unique name and containing one complete set of response codes. Profile names allow multiple financial institutions that have assigned matching override levels to their response codes to share the same records in the ORF. Otherwise, each institution would have its own records in the ORF for every response code it uses.

The PROFILE field in the ORF for each response code contains the profile name given to each group of ORF records. This field can be accessed via ORF screen 1. The ORF PROFILE field in the IDF, which can be accessed via IDF screen 25, contains the group of ORF records used by a financial institution. The profile

name in the IDF field must match the profile name in the ORF records for a complete set of BASE24-teller response codes. Multiple IDF records can contain the same profile name.

For each response code in the ORF, the value in the OUR CUST/OUR TERM OVERRIDE field in the ORF, which can be accessed via ORF screen 1, establishes the operator level required to override the response code for an on-us transaction.

For each response code in the ORF, the value in the OUR CUST/NOT-ON-US TERM OVERRIDE field, which can be accessed via ORF screen 1, establishes the operator level required to override the response code when our customer is performing the transaction at another financial institution's terminal.

While the override levels in the ORF can be changed for all response codes, changing the override level of approved or fatal response codes does *not* affect the processing performed by BASE24-teller. BASE24-teller uses the override levels in the ORF only for the override-possible response codes.

Any of the five operator levels can be assigned to the two override fields in each ORF record. The individual attempting to override the transaction must have at least the level specified in this field in order to complete the transaction. An override level of none allows any operator to override the response code while an override level of restricted allows no operator to override the response code.

If the override level is less than the operator level, the transaction is completed and returned to the operator with an approved response code. The override-possible response code is not returned to the operator.

If the override level is greater than the operator level but is less than or equal to the maximum override level for the terminal, the operator is given the opportunity to complete the transaction by obtaining a higher operator level. One or more override-possible response codes can be returned to the operator.

If the override level is greater than the operator level and greater than the maximum override level for the terminal, the operator is not given the opportunity to override the transaction. The transaction is declined with a fatal response code.

Note: The processing performed after comparing this override level to the operator level also depends on the override level of any other response codes that are returned with the transaction. If more than one response code is returned with a transaction, the response code with the highest override level is used in the comparisons to the operator level and maximum override level.

Maximum Override Level

The MAX TERMINAL OVERRIDE LEVEL field in the Teller Terminal Data File (TTDF), which can be accessed via TTDF screen 1, determines the highest level of override that a teller can perform at a teller device in response to an override-possible response code. One teller terminal may only have one level of override that is accomplished by a supervisor turning on a key switch. Another teller terminal may have two levels because it uses magnetic stripe cards that can be encoded to identify more operator levels.

Before an override-possible response code is returned, it is compared to this maximum override level. Only three operator levels (teller, supervisor, and manager) can be used in this field.

If the required override level is greater than the operator level but is less than or equal to this maximum override level, the operator is given the opportunity to complete the transaction by obtaining a higher operator level.

If the required override level is greater than this maximum override level, the operator is not given the opportunity to override the transaction. The transaction is declined with a fatal response code.

Section 5

BASE24-teller Transaction Routing and Authorization

This section summarizes some of the processing BASE24-teller performs during transaction routing and authorization, and should be used with section 4, which describes the controls available for all of the transaction processing performed by BASE24-teller. This section includes the following processing topics:

- Interbank routing and account number modification
- Passbooks
- Stop pays, warnings, holds, and deposit float
- Credit line and backup accounts
- Overrides
- Deposits

Interbank Routing and Account Number Modification

The relationship between the financial institution that owns a customer's account and the financial institution handling the customer's transaction impacts the way BASE24-teller processes the transaction.

Interbank routing is necessary when one financial institution owns an account and a different financial institution owns the terminal where a transaction involving the account is processed.

Account number modification is necessary when the account number that appears on a customer's check or passbook does not match the account number that identifies the customer's account on the BASE24 database. This situation occurs when mergers or acquisitions result in financial institutions being combined and sharing a database. Multiple customers may end up having the same account number. To eliminate duplicates on the BASE24 database, one customer from each pair of customers with duplicate account numbers must be assigned a new account number. When a customer's account number is changed, the financial institution can immediately replace all customer records such as checks or passbooks or have BASE24 modify the account number that appears on customer records to match the account number used on the BASE24 database.

BASE24-teller uses the Account Routing File (ARF) to perform interbank routing and account number modification for all transactions initiated without a plastic card.

Card-initiated transactions also can use the ARF. However, BASE24-teller first obtains the accountholder's FIID from the Card Prefix File (CPF) record identified by the card used to initiate the transaction. Then, if the customer's account number is included with the transaction, the BASE24-teller Authorization process can obtain instructions for modifying the customer account number from the bank routing code version of the ARF record. The FIIDs in the CPF and ARF records must match.

The ARF contains two record formats: account number routing and bank routing code. The account number routing format is used when the accountholder's FIID can be determined based on the account type, account number length, and a value within the account number itself. The bank routing code format is used when a routing code must be provided with the account type and account number length to determine the accountholder's FIID. When a routing code is provided, the ARF record can also contain information to modify the account number in situations

where the account number appearing on the customer's check or passbook does not match the account number that identifies the customer's account on the Positive Balance File (PBF).

When a routing code is provided, BASE24 uses values in the ACCOUNT NUMBER INSERT VALUE and ACCOUNT NUMBER INSERT POSN fields of the ARF record to create the account number for searching the BASE24 database. For example, if the value in the ACCOUNT NUMBER INSERT VALUE field is 54 and the value in the ACCOUNT NUMBER INSERT POSN field is 03, an instrument with the account number 666666 would be matched to PBF account number 66546666 of the institution identified by the FIID in the ARF record.

No digits are lost from the existing account number when creating the account number used to search the BASE24 database. In the example, values in positions 3 through 6 of the existing account number are moved two places to the right (to positions 5 through 8) to make room for the insert value in positions 3 and 4.

Bank Routing Code Format

The screen on the following page illustrates an ARF summary screen displaying bank routing code records for three banks that support interbank routing. In this example, the account numbers defined within the logical network are not unique so these institutions are using bank routing codes to route their transactions. The tellers enter a bank routing code of 111 for all of Bank 1's account numbers, a bank routing code of 222 for all of Bank 2's account numbers, and a bank routing code of 333 for all of Bank 3's account numbers. Tellers may also enter bank routing codes of 4 through 9 for Bank 1's savings accounts. The account numbers used by all three banks are six digits in length.

Values in the ACCT TYPE, ACCT LGTH, BNK RTG CDE (HI), and BNK RTG CDE (LO) fields, arranged in this order, serve as the key to bank number routing records in the ARF and determine the order of the records on the ARF summary screen. The BASE24-teller Authorization process builds internal tables from all of the ARF records. The BASE24-teller Authorization process searches for ARF records with an exact match on account type and length, then uses the first ARF record that contains a bank routing code or range of bank routing codes matching the bank routing code received with the transaction.

The Authorization process repeats the process using ARF records with an account type of ** and an exact match on account length if no ARF records with an exact match on account type and length provide a match on the bank routing code.

ARF Summary Screen — Bank Routing Code Format

In the following examples, references are to entries on the ARF screen shown below. ARF screen entries are identified by numbers in parentheses in the first column of the screen.

```

BASE24-BASE  ACCOUNT ROUTING FILE      LLLL      YY/MM/DD  HH:MM  03 OF 03

      RECORD TYPE: 01 (BANK ROUTING CODE)
      ACCOUNT TYPE:                      ACCOUNT LENGTH:

      ACCOUNT ROUTING FILE SUMMARY

      ACCT  ACCT      (HI)      (LO)      ACCT NUM      ACCT NUM      ACCT
      TYPE  LGTH  BNK RTG CDE  BNK RTG CDE  INSERT POSN  INSERT VALUE  FIID
(1)  **    06    00000000111          04      555          BNK1
(2)  **    06    00000000222          01      777          BNK2
(3)  **    06    00000000333          01      888          BNK3
(4)  11    06    00000000009  00000000004  04      555          BNK1

***** BASE24 *****
      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
      F7-SELECT RECORD      F9-NEXT PAGE      F11-PREV PAGE      F12-HELP

```

Example 1

The teller performs a withdrawal from checking transaction (transaction code 100100) and enters account number 123456 and the bank routing code of 333.

- The BASE24-teller Authorization process searches the ARF for an entry with the following values:
 - Record type 01 (because a bank routing code was entered by the teller).
 - Account type 01 (because the transaction code had a *from* account type of 01).
 - Account length 06 (determined from the account number itself).
 - Bank routing code 333 (entered by the teller).
- The Authorization process does not locate an ARF entry that matches the teller's entry.

3. The Authorization process searches the ARF again for an entry with the following values:
 - Record type 01 (because a bank routing code was entered by the teller).
 - Account type ** (using ** as a wildcard entry because account type 01 was not in the table).
 - Account length 06 (determined from the account number itself).
 - Bank routing code 333 (entered by the teller).
4. The Authorization process finds a match with entry 3 on the ARF screen.
5. The Authorization process continues processing with the following values:
 - FIID of BNK3 (because BNK3 is the FIID in entry 3 on the ARF screen).
 - PBF account number of 888123456 (because the account number insert position of 01 and the account number insert value of 888 in entry 3 on the ARF screen mean insert 888 beginning at position 1 of the account number entered by the teller).

Example 2

The teller performs a deposit to savings transaction (transaction code 200011) and enters account number 123456 and the bank routing code of 5.

1. The BASE24-teller Authorization process searches the ARF for an entry with the following values:
 - Record type 01 (because a bank routing code was entered by the teller).
 - Account type 11 (because the transaction code had a *to* account type of 11).
 - Account length 06 (determined from the account number itself).
 - Bank routing code 5 (entered by the teller).
2. The Authorization process finds a match with entry 4 on the ARF screen since bank routing code 5 falls between 4 and 9.

3. The Authorization process continues processing with the following values:
 - FIID of BNK1 (because BNK1 is the FIID in entry 4 on the ARF screen).
 - PBF account number of 123555456 (because the account number insert position of 04 and the account number insert value of 555 in entry 4 on the ARF screen mean insert 555 beginning at the fourth position of the account number entered by the teller).

Account Number Routing Format

The screen on the following page illustrates the account number routing format of the ARF summary screen containing records for three banks that support interbank routing. All account numbers defined within the logical network are unique so these institutions are using account number routing to route their transactions. The tellers do not have to enter any extra information to identify the bank that owns the account. All of Bank 1's account numbers are 6 or 8 digits in length and start with the identifier 111. All of Bank 2's account numbers are 6 digits in length and contain the identifier 2 in the fourth position. All of Bank 3's account numbers are 6 or 8 digits in length and end with the identifier 3.

Values in the ACCT TYPE, ACCT LGTH, ACCT NUM MATCH POSN, and ACCT NUM MATCH VALUE fields, arranged in this order, serve as the key to account number routing records in the ARF and determine the order of the records on the ARF summary screen. The BASE24-teller Authorization process builds internal tables from all of the ARF records. The BASE24-teller Authorization process searches for ARF records with an exact match on account type and length, then uses the ARF record containing the account number match position and value that most closely matches the account number received with the transaction. For example, a match of one character in the first position of the account number is a closer match than a match of one character in the third position; a match of three characters in the third, fourth, and fifth positions is a closer match than a match of two characters in the third and fourth positions.

The Authorization process repeats the process using ARF records with an account type of ** and an exact match on account length if no ARF records with an exact match on account type and length provide a match on the account number match position and value.

ARF Summary Screen — Account Number Routing Format

For the following examples, references are to entries on the ARF screen shown below. ARF screen entries are identified by numbers in parentheses in the first column of the screen.

```

BASE24-BASE  ACCOUNT ROUTING FILE      LLLL      YY/MM/DD  HH:MM  03 OF 03

      RECORD TYPE: 02 (ACCOUNT NUMBER ROUTING)
      ACCOUNT TYPE:                      ACCOUNT LENGTH:

      ACCOUNT ROUTING FILE SUMMARY

      ACCT      ACCT      ACCT NUM      ACCT NUM      ACCT
      TYPE      LGTH      MATCH POSN      MATCH VALUE      FIID
(1)    **        06          01        111          BNK1
(2)    **        06          04          2           BNK2
(3)    **        06          06          3           BNK3
(4)    **        08          01        111          BNK1
(5)    **        08          08          3           BNK3

***** BASE24 *****
      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F7-SELECT RECORD      F9-NEXT PAGE      F11-PREV PAGE      F12-HELP
  
```

Example 1

The teller performs a withdrawal from checking transaction (transaction code 100100) and enters account number 111456.

1. The BASE24-teller Authorization process searches the ARF for an entry with the following values:
 - Record type 02 (because a bank routing code was not entered by the teller).
 - Account type 01 (because the transaction code had a *from* account type of 01).
 - Account length 06 (determined from the account number itself).
2. The Authorization process does not locate an ARF entry that matches the teller's entry.

3. The Authorization process searches the ARF again for an entry with the following values:
 - Record type 02 (because a bank routing code was not entered by the teller).
 - Account type ** (using ** as a wildcard entry because account type 01 was not in the table).
 - Account length 06 (determined from the account number itself).
4. The Authorization process finds matches with entries 1, 2, and 3 on the ARF screen based on account type ** and account length 06.
5. The Authorization process uses the account number match position and account number match value entries from ARF entries 1, 2, and 3 to find the correct FIID.
6. The Authorization process finds 111 beginning in the first position of the account number, which matches ARF entry 1.
7. The Authorization process continues processing with the following values:
 - FIID of BNK1 (because BNK1 is the FIID in entry 1 on the ARF screen).
 - PBF account number of 111456 (the account number originally entered by the teller).

Example 2

The teller performs a deposit to savings transaction (transaction code 201100) and enters account number 113256.

1. The BASE24-teller Authorization process searches the ARF for an entry with the following values:
 - Record type 02 (because a bank routing code was not entered by the teller).
 - Account type 11 (because the transaction code had a *from* account type of 11).
 - Account length 06 (determined from the account number itself).
2. The Authorization process does not locate an ARF entry that matches the teller's entry.

3. The Authorization process searches the ARF again for an entry with the following values:
 - Record type 02 (because a bank routing code was not entered by the teller).
 - Account type ** (using ** as a wildcard entry because account type 11 was not in the table).
 - Account length 06 (determined from the account number itself).
4. The Authorization process finds matches with entries 1, 2, and 3 on the ARF screen based on account type ** and account length 06.
5. The Authorization process uses the account number match position and account number match value entries from ARF entries 1, 2, and 3 to find the correct FIID.
6. The Authorization process finds 2 in the fourth position of the account number, which matches ARF entry 2.
7. The Authorization process continues processing with the following values:
 - FIID of BNK2 (because BNK2 is the FIID in entry 2 on the ARF screen).
 - PBF account number of 113256 (the account number originally entered by the teller).

Example 3

The teller performs a withdrawal from savings transaction (transaction code 101100) and enters account number 111253.

1. The BASE24-teller Authorization process searches the ARF for an entry with the following values:
 - Record type 02 (because a bank routing code was not entered by the teller).
 - Account type 11 (because the transaction code had a *from* account type of 11).
 - Account length 06 (determined from the account number itself).
2. The Authorization process does not locate an ARF entry that matches the teller's entry.

3. The Authorization process searches the ARF again for an entry with the following values:
 - Record type 02 (because a bank routing code was not entered by the teller).
 - Account type ** (using ** as a wildcard entry because account type 11 was not in the table).
 - Account length 06 (determined from the account number itself).
4. The Authorization process finds matches with entries 1, 2, and 3 on the ARF screen based on account type ** and account length 06.
5. The Authorization process uses the account number match position and account number match value entries from ARF entries 1, 2, and 3 to find the correct FIID.
6. The Authorization process finds 111 beginning in the first position of the account number, which matches ARF entry 1.

The account number also matches ARF entries 2 and 3. However, the Authorization process uses ARF entry 1 because ARF entries with the same record type, account type, and account length are arranged in the internal table according to the length of the account number match values. This way, the ARF entries are checked for the best possible match first. The search stops with the first match.

7. The Authorization process continues processing with the following values:
 - FIID of BNK1 (because BNK1 is the FIID in entry 1 on the ARF screen).
 - PBF account number of 111253 (the account number originally entered by the teller).

Passbook Processing

Passbook processing allows a financial institution to record financial transactions in a passbook for the customer. Since transactions can be processed with or without the passbook, BASE24-teller uses the No Book File (NBF) to store the transactions for later printing in the passbook. For many of the financial transactions posted with the passbook present, BASE24-teller can print any transactions stored in the NBF and bring the passbook up to date automatically without any extra actions by the teller.

The PBF contains the number of unprinted records stored in the NBF for an account and the ending balance in the customer's passbook the last time it was updated. The PBF is updated each time BASE24-teller adds records to the NBF or prints NBF records in the customer's passbook.

Supported Transactions

Any financial transaction performed on BASE24-teller involving account types that support passbooks are added to the NBF if the account is configured as a passbook account. These transactions are identified in appendix C. Automatic passbook printing can be performed for a subset of these transactions. The transactions that can include automatic passbook printing are also identified in appendix C.

If automatic passbook printing is supported for one or more transaction codes, the Device Handler must be designed to determine whether the passbook is present and set the value of the AUTO-PASSBOOK-PRNT field in the Financial token accordingly. The BASE24-teller Authorization process uses the value in the AUTO-PASSBOOK-PRNT field in the Financial token to determine whether the passbook is present. When the passbook is present, the Authorization process adds the financial transaction to the NBF and automatically returns all outstanding NBF transactions, including the current transaction, to the Device Handler to update the passbook. When the passbook is not present, the Authorization process adds the current transaction to the NBF and does not return any outstanding NBF transactions to the Device Handler.

One or more of the following methods can be used to identify whether a passbook is present, depending on Device Handler design.

- The teller can enter one transaction code when posting a transaction with the passbook and another transaction code when posting the same transaction without the passbook. The Device Handler maps both teller-entered transaction codes to the same BASE24-teller transaction code. The Device Handler sets the AUTO-PASSBOOK-PRNT field in the Financial token based on the transaction code it receives from the teller terminal.
- The teller can include the passbook balance in the original request when posting a transaction with the passbook and exclude the passbook balance from the original request when posting a transaction without the passbook. The Device Handler sets the AUTO-PASSBOOK-PRNT field in the Financial token based on whether the transaction it receives from the teller terminal includes a passbook balance.
- The teller can control the value of an automatic passbook print flag in the original request to indicate whether a transaction is being posted with or without the passbook. The Device Handler sets the AUTO-PASSBOOK-PRNT field in the Financial token based on the value of the automatic passbook print flag included with the transaction it receives from the teller terminal.

NBF records are added for all passbook accounts involved in a BASE24-teller transaction. This can include both accounts for a transfer, split deposit, or payment transaction. It can also include the backup account when it is used to complete a transaction.

Card-initiated transactions performed on BASE24-atm or BASE24-pos are not added to the NBF. If necessary, these transactions could be added to the NBF based on information contained in host files during refresh processing. Financial transactions performed on the host when BASE24-teller is not involved can also be added to the NBF during refresh processing. Crediting interest or assessing service charges are examples.

Using Passbooks with Loan Accounts

BASE24-teller supports passbooks for installment loans, mortgage loans, and commercial loans. However, the BASE24-teller Authorization process adds the transaction to the NBF without updating PBF ledger or available balances for transactions involving these account types. As a result, the passbook balance in the NBF usually will not match the ledger balance in the PBF until both files are refreshed. This is because the entire transaction amount is usually not applied to the principal balance of these account types. A portion of the transaction amount is interest that must be collected from the payment, then the account balance is reduced by the portion of the transaction amount that remains after the interest is collected.

For example, a loan has a balance of \$10,000, the customer makes a \$300 payment, and the teller immediately requests a passbook print transaction or the payment includes automatic passbook printing. Before printing the transaction in the passbook, the BASE24-teller Authorization process computes what the passbook balance should be after the \$300 payment is posted and compares it to the account's ledger balance. The passbook balance is the balance entered by the teller with the transaction, reduced by the payment amount. The ledger balance is not reduced by the payment amount because the transaction does not update the PBF. As a result, the teller receives response code O71 (PBF/NBF balance difference).

	Passbook Balance	Ledger Balance
Before payment	10,000	10,000
Payment	– 300	No Change
After payment	9,700	10,000

In this example, if \$200 of the payment should be collected as interest, the loan application on the host decreases the account balance by only \$100 (\$300 minus \$200) instead of the \$300 shown in the NBF. By refreshing the NBF and PBF with information from host files after the payment has been posted properly, the passbook balance in the NBF and the ledger balance in the PBF reflect the correct information.

	Passbook Balance	Ledger Balance
Before payment	10,000	10,000
Payment	– 100	– 100
After payment	9,900	9,900

Note: If the host application applies the full \$300 payment to the account balance instead of splitting it between the account balance and interest, only the PBF needs to be refreshed. The NBF already reflects a \$9,700 balance and the PBF would also reflect a \$9,700 balance after it is refreshed.

Transaction Processing

A customer can present his or her passbook to the teller for any financial transaction that includes passbook support, as well as the passbook print and reprint transactions, and a teller can perform an NBF inquiry transaction to obtain passbook information. For the financial transactions that do not support automatic passbook printing, BASE24-teller adds the transaction to the NBF and the teller must then initiate a passbook print transaction to record the transaction in the passbook. For the financial transactions that support automatic passbook printing, the teller must inform BASE24-teller that the passbook is present. BASE24-teller then posts the financial transaction and automatically initiates the passbook print transaction. Automatic passbook printing is supported only for financial request (0200) messages.

During a passbook print transaction or automatic passbook printing performed with a financial transaction, BASE24-teller performs similar steps. The actual steps depend on the design of the Device Handler.

1. The BASE24-teller Authorization process compares the passbook balance entered by the teller with the PBF passbook balance. If the two balances do not match, the BASE24-teller Authorization process declines the transaction with a response code that can be overridden. Depending on the Device Handler design, the amount of the difference can also be returned to the teller.
2. The BASE24-teller Authorization process returns NBF records in the NBF token in multiples of five. The BASE24-teller Authorization process changes the print status in the NBF for each record it places in the NBF token and updates the passbook balance and unprinted NBF records counter in the PBF as it returns each token.
3. The Device Handler requests additional NBF entries when a flag in the internal message indicates more NBF entries are available.

Stop Pay, Warning, Hold, and Deposit Float Processing

Stop payment records in the Stop Payment File (SPF) identify checks that the customer has requested the financial institution not to pay. Records in the Warning/Hold/Float File (WHFF) contain three different types of information: warning, hold, and deposit float. Warning records inform the teller of circumstances affecting an account to protect the teller from completing transactions that should be declined. Hold transactions are used to limit the funds available to a customer and deposit float transactions are used to increase the funds available to a customer.

Stop pay and warning transactions add records that can be checked each time a transaction is performed, while hold and deposit float transactions adjust the PBF balances that are used in performing financial transactions.

PBF Stop Pay/Warning Status

BASE24-teller uses the STOP PAY/WARNING STATUS field in the PBF, which can be accessed via PBF screen 10, to determine whether the SPF or WHFF contain information for the account that is being processed. This PBF field identifies whether or not stop payments, warnings, or both have been issued.

Each time the SPF or WHFF is updated to add or delete a stop payment or warning via a teller transaction, file maintenance, or BASE24-from host maintenance, BASE24 updates this PBF field to indicate whether stop payments or warnings exist. When stop pay and warning records are added or deleted, the teller or file maintenance operator has the opportunity to accept BASE24's information or manually enter a value for this PBF field.

The BASE24-teller Authorization process uses this field to determine whether the SPF or WHFF should be checked when processing transactions. The SPF is checked for stop pay records when the STOP PAY/WARNING STATUS field indicates that stop payments or warnings exist before authorizing the following check cashing transaction codes:

- | | |
|----------|----------|
| • 110100 | • 122100 |
| • 112100 | • 170100 |
| • 120100 | • 172100 |

The WHFF is checked for warning records before authorizing financial transactions identified by the SEARCH THE WHFF field in the Teller Transaction File (TTF), which can be accessed via TTF screen 1, when the STOP PAY/WARNING STATUS field in the PBF indicates that warnings exist. The WHFF is not checked for log-only transactions, regardless of the value in the SEARCH THE WHFF field.

Stop Payments

The SPF can contain records for checks identified by the following information:

- Check number
- Check number and amount
- Range of check numbers

Because SPF records can be added for individual checks or for groups of checks, the potential exists for multiple SPF records pertaining to the same check. For example, one SPF record could be added for check number 125 and one could be added for the check numbers between 120 and 130, resulting in check number 125 being in the SPF twice. To avoid this situation, the teller or operator should check for existing SPF records before adding records to the SPF.

Each SPF record includes an expiration date. When processing inquiries, the Device Handler can set the CHK-FOR-EXPIRED-RECS field in the SPF Inquiry token to indicate whether expired SPF records should be included with the active SPF records returned to the teller. If the Device Handler does not set the CHK-FOR-EXPIRED-RECS field or the SPF Inquiry token is not included with the request, active and expired SPF records are returned to the teller. When expired records are not to be returned, the Authorization process compares the date in the SYS.ORIG.TRAN-DAT field of the TSTMH to the date in the EXPIRATION DATE field in the SPF, which can be accessed via SPF screen 1, to determine whether a record has expired.

When processing a check cashing transaction, the Authorization process uses the following steps to search the SPF for an unexpired record containing information for the check being cashed:

1. Search for an exact match on the check number and transaction amount.
2. Search for an exact match on the check number only.
3. Search for a range that includes the check number.

The search ends when the first SPF record for the check number is found. If additional SPF records contain the same check number, they are not identified. When the Authorization process finds an SPF record or the host returns a stop payment response code, the Authorization process returns one of the following override-possible response codes to the teller:

O50 = Match on check number and amount

O51 = Match on check number

O52 = Match on check range

Information from the SPF Inquiry token, which includes most information contained in the SPF record, is returned to the teller with the response code.

An SPF record can be considered obsolete when it expires or the customer requests that it be canceled. Once a record is added to the SPF, it remains there until a specific action is taken to remove it. Four methods are available for removing obsolete records from the SPF.

- Delete all obsolete records via full-file refresh.
- Delete an individual record via BASE24-from host maintenance.
- Delete an individual record via file maintenance.
- Delete an individual record via teller terminal (transaction code 81).

Warnings

The WHFF can contain records with warning information for the following accounts involved in a transaction:

- The *from* account
- The *to* account
- The credit line/backup account

When the SEARCH THE WHFF field in the TTF, which can be accessed via TTF screen 1, indicates the WHFF should be checked for warnings, the Authorization process uses the value of the STOP PAY/WARNING STATUS field in the PBF for all accounts involved in the transaction to determine whether the WHFF contains warnings for any of the accounts being processed. The Authorization process can check the WHFF for warnings when processing any BASE24-teller financial transaction. When the Authorization process finds a warning record in

the WHFF or the host returns a warning response code, the Authorization process returns override-possible response code O60 (warning record(s) found) to the teller.

Information from the WHFF Inquiry token, including information contained in the WHFF record, is returned to the teller with the response code.

Each WHFF record includes an expiration date. When processing inquiries, the Device Handler can set the CHK-FOR-EXPIRED-RECS field in the WHFF Inquiry token to indicate whether expired WHFF warning records should be included with any active WHFF warning records returned to the teller. If the Device Handler does not set the CHK-FOR-EXPIRED-RECS field or the WHFF Inquiry token is not included with the request, active and expired WHFF warning records are returned to the teller. When expired records are not to be returned, the Authorization process compares the date in the SYS.ORIG.TRAN-DAT field of the TSTMH to the date in the EXPIRATION DATE field in the WHFF, which can be accessed via WHFF screen 1, to determine whether a record has expired.

A WHFF warning record can be considered obsolete when it expires or the customer requests that it be canceled. Once a warning record is added to the WHFF, it remains there until a specific action is taken to remove it. Four methods are available for removing obsolete warning records from the WHFF:

- Delete all obsolete records via full-file refresh.
- Delete an individual record via BASE24-from host maintenance.
- Delete an individual record via file maintenance.
- Delete an individual record via teller terminal (transaction code 89).

Holds

Holds limit the funds available to a customer. Holds can be added to the following accounts involved in a transaction:

- The *from* account
- The *to* account
- The credit line/backup account

Adding Holds

Hold entries, when added, decrease the PBF available funds balance and increase the PBF amount on hold balance, as shown below. Hold entries do not affect the PBF ledger balance. The following example illustrates adding a hold for \$1,000.

	Available Funds Balance	Amount On Hold Balance	Ledger Balance
Before transaction	5,000	500	5,500
Hold added	– 1,000	+ 1,000	No change
After transaction	4,000	1,500	5,500

The BASE24-teller Authorization process uses the PBF available funds balance when it determines whether an account contains sufficient funds to approve a transaction. The decrease in the PBF available funds balance may cause insufficient funds-related response codes (e.g., O44 or O45) to be returned when a transaction would otherwise be approved. In this example, a \$5,000 withdrawal could be made before the hold was added but only a \$4,000 withdrawal could be made with the hold in place.

The presence of a hold entry in the WHFF does not generate any particular response codes. This is because an add hold transaction can adjust the PBF account balances without adding a record to the WHFF. The setting of the TLR-WRITE-HLDS-DEP-FLOATS param in the LCONF determines whether a hold entry is written to or removed from the WHFF.

Removing Holds

Normally, the amount added by the hold transaction remains in the PBF amount on hold balance until the hold is removed. The transaction to remove the hold entry then increases the PBF available funds balance and decreases the PBF amount on hold balance, as shown in the following example.

	Available Funds Balance	Amount On Hold Balance	Ledger Balance
Before transaction	4,000	1,500	5,500
Hold deleted	+ 1,000	– 1,000	No change
After transaction	5,000	500	5,500

Transactions that add or delete hold records via the teller device or BASE24-from host maintenance can automatically update the PBF available funds and amount on hold balances; however, adding or deleting records through a WHFF refresh does not update the balances in the PBF. The change in PBF balances must be performed in a PBF refresh.

Note: PBF balances are not adjusted when WHFF hold records expire. The change in balances must be included in a PBF refresh.

Removing Holds in Excess of PBF Balance

If the same hold information is not used to refresh the WHFF and PBF, there may be situations when the amount of the holds contained in the WHFF exceeds the PBF amount on hold balance. Continuing the \$1,000 hold example, if the PBF amount on hold balance is less than \$1,000 when it is time to remove the hold, the BASE24-teller Authorization process adjusts the amount of the hold removal transaction so that it leaves a PBF amount on hold balance of zero instead of letting the balance go negative. The teller enters a transaction amount of \$1,000, but if the PBF amount on hold balance is only \$500, the Authorization process adjusts the transaction amount from \$1,000 to \$500, as shown below.

	Available Funds Balance	Amount On Hold Balance	Ledger Balance
Before transaction	4,000	500	4,500
Hold deleted	+ 500	– 500	No change
After transaction	4,500	0	4,500

Expiration Dates

Each WHFF record includes an expiration date. When processing inquiries, the Device Handler can set the CHK-FOR-EXPIRED-RECS field in the WHFF Inquiry token to indicate whether expired WHFF hold records should be included with any active WHFF hold records returned to the teller. If the Device Handler does not set the CHK-FOR-EXPIRED-RECS field or the WHFF Inquiry token is not included with the request, active and expired WHFF records are returned to the teller. When expired records are not to be returned, the Authorization process compares the date in the SYS.ORIG.TRAN-DAT field of the TSTMH to the date in the EXPIRATION DATE field in the WHFF, which can be accessed via WHFF screen 1, to determine whether a record has expired.

Deposit Float

Deposit float entries increase the funds available to a customer. The WHFF can contain deposit float records for the following accounts involved in a transaction:

- The *from* account
- The *to* account
- The credit line/backup account

Adding Deposit Float

Deposit float entries, when added, increase the PBF available funds balance, as shown below. Deposit float entries do not affect the PBF amount on hold balance or PBF ledger balance. The following example illustrates adding a deposit float entry of \$1,000.

	Available Funds Balance	Amount On Hold Balance	Ledger Balance
Before transaction	5,000	500	5,500
Float added	+ 1,000	No change	No change
After transaction	6,000	500	5,500

The BASE24-teller Authorization process uses the PBF available funds balance when it determines whether an account contains sufficient funds to approve a transaction. The increase in the PBF available funds balance can allow a transaction to be approved that would otherwise be declined with insufficient funds-related response codes (e.g., O44 or O45). In this example, a \$5,000 withdrawal could be made before the deposit float was added and a \$6,000 withdrawal could be made with the deposit float in place.

The presence of a deposit float entry in the WHFF does not generate any particular response codes. This is because an add deposit float transaction can adjust the PBF available funds balance without adding a record to the WHFF. The setting of the TLR-WRITE-HLDS-DEP-FLOATS param in the LCONF determines whether a deposit float entry is written to or removed from the WHFF.

Removing Deposit Float

When the amount added by the deposit float transaction remains in the PBF available funds balance until the deposit float entry is removed, the transaction to remove the deposit float entry decreases the PBF available funds balance, as shown below.

	Available Funds Balance	Amount On Hold Balance	Ledger Balance
Before transaction	6,000	500	5,500
Float deleted	– 1,000	No change	No change
After transaction	5,000	500	5,500

Transactions that add or delete deposit float records via the teller device or BASE24-from host maintenance can automatically update the PBF available funds balance; however, adding or deleting records through a WHFF refresh does not update the balances in the PBF. The change in PBF balances must be performed in a PBF refresh.

Note: PBF balances are not adjusted when WHFF deposit float records expire. The change in balances must be included in a PBF refresh.

Removing Deposit Float in Excess of PBF Balance

If the same deposit float information is not used to refresh the WHFF and PBF, there may be situations when the amount of the deposit float records contained in the WHFF exceeds the PBF available funds balance. Continuing the \$1,000 deposit float example, if the PBF available funds balance is less than \$1,000 when it is time to remove the deposit float entry, the BASE24-teller Authorization process creates a negative PBF available funds balance, as shown below.

	Available Funds Balance	Amount On Hold Balance	Ledger Balance
Before transaction	700	100	600
Float deleted	– 1,000	No change	No change
After transaction	– 300	100	600

Expiration Dates

Each WHFF record includes an expiration date. When processing inquiries, the Device Handler can set the CHK-FOR-EXPIRED-RECS field in the WHFF Inquiry token to indicate whether expired WHFF deposit float records should be included with any active WHFF deposit float records returned to the teller. If the Device Handler does not set the CHK-FOR-EXPIRED-RECS field or the WHFF Inquiry token is not included with the request, active and expired WHFF records are returned to the teller. When expired records are not to be returned, the Authorization process compares the date in the SYS.ORIG.TRAN-DAT field of the TSTMH to the date in the EXPIRATION DATE field in the WHFF, which can be accessed via WHFF screen 1, to determine whether a record has expired.

Credit Line and Backup Account Processing

Checking, savings, and NOW accounts can use a credit line or backup account to provide overdraft protection. Credit, line of credit, checking, savings, and NOW accounts can be used to provide overdraft protection. Transaction codes that can use a credit line or backup account to complete the transaction are identified in appendix C.

The BASE24-teller Authorization process returns information about a customer's primary account to the teller in response to a transaction request. When a credit line or backup account is being used to provide overdraft protection for that primary account, BASE24-teller adds information about the credit line or backup account to the primary account information it returns to the teller. This additional information includes warnings returned in the WHFF Inquiry token, response codes returned in the Override token, and customer information returned in the Account token, Customer Name token, and PBF Inquiry token.

Transfer Amount

The amount transferred from a credit line/backup account is controlled by the values in the IDF, which can be accessed via IDF screen 6. This screen has two TRANSFER METHOD and INCREMENT AMOUNT fields, depending on the type of account used as the credit line/backup account. Credit account types include credit and line of credit. Noncredit account types include checking, savings, and NOW. Transfers can be made in an amount that leaves a zero balance in the primary account (i.e., the account that has the overdraft protection) or in the increment amount specified in the IDF.

Using an overdraft of \$75.50 as an example, an exact amount transfer is \$75.50. With an overdraft of \$75.50 and an increment amount of \$50, the transfer is \$100 because 100 is the smallest multiple of 50 (e.g., 50, 100, 150) that is equal to or greater than the overdraft amount. In this example, a \$50 transfer creates a negative balance of \$25.50 and a \$100 transfer creates a positive balance of \$24.50.

Number of Credit Line/Backup Accounts

An account can have one credit line/backup account. No transfer is made if the credit line/backup account does not have sufficient available funds to make the necessary transfer. Even if the account used as a credit line/backup account has its own credit line/backup account, the second credit line/backup account is not considered when determining whether the transfer can be made.

For example, an accountholder has checking accounts 11111, 22222, and 33333. Checking account 11111 has checking account 22222 as its backup account, checking account 22222 has checking account 33333 as its backup account, and these accounts have the balances shown below:

Account	Balance
11111	\$200
22222	\$100
33333	\$500

If the accountholder writes a \$400 check on account 11111, the BASE24-teller Authorization process attempts to transfer \$200 from account 22222 to cover the shortage. Account 22222 only has \$100, so the transfer cannot be made, even though account 22222 would have enough if funds were first transferred to it from account 33333.

Transfers or Payments to Credit Line/Backup Accounts

A situation could arise where an accountholder does not have sufficient funds in his or her account to make a payment or transfer and the account to receive the funds is also the credit line/backup account. No transfer is made when this occurs.

Using accounts 11111 and 22222 from the earlier example, a transfer or payment of \$250 from account 11111 to account 22222 is denied with response code F08 (unable to process) because account 22222 is the account receiving the funds and is also the credit line/backup account for account 11111. The BASE24-teller Authorization process declines this transaction even though account 22222 has a \$100 balance, and account 11111 needs only \$50 transferred to it to cover the transaction. A transfer or payment of \$150 can be made from account 11111 to account 22222 because a \$150 transfer or payment can be made from account 11111 without using account 22222 as a credit line/backup account to cover the transaction.

Reversals

When a transfer is made from a credit line/backup account and it becomes necessary to reverse the transaction that caused the transfer, the timing of the reversal determines whether the transfer is also reversed.

If the reversal is performed immediately after the original transaction, the transfer and the original financial transaction are reversed. This reversal can be teller-initiated or BASE24-initiated.

If another transaction separates the original transaction and the reversal, only the original transaction is reversed. The funds transferred as part of the original transaction remain in the primary account.

Note: When the BASE24-teller Authorization process receives a teller-initiated reversal of a transaction that involved a transfer from a credit line/backup account, the reversal message must contain the Credit Line token. The Authorization process declines such a reversal with response code F01 (message edit error) if the Credit Line token is missing.

Override Processing

BASE24-teller provides the teller with all override-possible response codes that prevent a transaction from being approved before requesting an override. Response codes for the *from* and *to* accounts are provided for two-sided transactions. Response codes for the credit line/backup account are also provided when a credit line/backup account is involved.

Override processing is based on the three operator level controls discussed in section 4:

Minimum teller level — Establishes the operator level required to perform each transaction code. This level is set in the TTF.

Response code override level — Establishes the operator level required to override each override-possible response code. This level is set in the Override Response File (ORF) and can be established depending on whether the transaction involved is on-us or not-on-us.

Maximum terminal override level — Establishes the highest level of operator override that can be returned from a particular teller device in response to an override-possible response code. This level is set in the Teller Terminal Data File (TTDF).

These three controls determine what a given operator can do, what an operator must be able to do to complete a transaction, and whether the terminal can support the operator's action. The way BASE24-teller uses these controls is illustrated in the "Transaction Processing" discussion later in this section.

All response codes returned by BASE24-teller are in one of three categories:

Approved response codes — These response codes indicate the successful completion of a transaction.

Override-possible response codes — These response codes indicate the transaction cannot be completed without an override from the teller terminal operator.

Fatal response codes — These response codes indicate the transaction cannot be completed. An override-possible response code is treated as a fatal response code when the response code's override level is greater than the maximum terminal override level of the device where the transaction has been performed.

Override Bit Maps

Transaction processing results in one approval or fatal response code that is returned to the Device Handler in the RESP-HDR.RESP-CDE field in the Teller Standard Internal Message Header (TSTMH). When processing involves overrides, the fatal response code F0Z (override needed) is returned in the TSTMH field and the Override token contains bit maps identifying all applicable response codes. The Device Handler then can return all applicable response codes to the controller or teller terminal, allowing the teller to review them before deciding whether to perform an override.

There are nine bit maps containing 64 bits each in the Override token. The bit maps separate the response codes into the following groups:

- Teller level overrides for the *from* account
- Supervisor level overrides for the *from* account
- Manager level overrides for the *from* account
- Teller level overrides for the *to* account
- Supervisor level overrides for the *to* account
- Manager level overrides for the *to* account
- Teller level overrides for the credit line/backup account
- Supervisor level overrides for the credit line/backup account
- Manager level overrides for the credit line/backup account

The bits in all nine bit maps identify specific override-possible response codes and a given bit represents the same response code, regardless of the bit map in which it appears. For example, the second bit represents response code O31 in all nine bit maps.

If response code O31 is defined as a manager level override in its ORF record and applies to the *from* account, the Authorization process sets the second bit in the manager level bit map for the *from* account.

If response code O31 is defined as a teller level override in its Override Response File (ORF) record and applies to the *to* account, the Authorization process sets the second bit in the teller level bit map for the *to* account.

If response code O31 is defined as a supervisor level override in its ORF record and applies to the credit line/backup account, the Authorization process sets the second bit in the supervisor level bit map for the credit line/backup account.

The response code associated with each bit is defined in the Response Code Bit Map Table in the BASE24-teller Tables File. This file is located on the BASE24-teller source subvolume (e.g., *volume*.TR50SRC.TRTABLES). Each record in the file is an unformatted line of text that contains one override-possible response code (e.g., O31). Only override-possible response codes, up to a maximum of 64 response codes, can be included in the file. If there are 25 override-possible response codes defined for BASE24-teller, there should be 25 entries in this file.

All BASE24-teller processes use the Response Code Bit Map Table in the BASE24-teller Tables File to determine what response code each bit in the Override token represents. This file should be modified only when new override-possible response codes are defined, and can be maintained with any Tandem text editor. When this file is modified, all BASE24-teller processes that can use the Override token must be recompiled. This includes Device Handler, Authorization, and ANSI and ISO Host Interface processes, BASE24-teller Report Programs, and the Teller Transaction Log File (TTLF) server.

Transaction Processing

When processing a transaction, the BASE24-teller Authorization process starts with a response code of A00 (approved transaction), then modifies the response code based on the conditions it encounters. The Authorization process performs the following steps:

1. Determines the account-owning FIID so it can use the appropriate TTF and ORF records.
2. Checks the TTF for the transaction being processed, depending on whether the transaction is on-us or not-on-us.
3. Compares the minimum teller level in the TTF to the level of the operator initiating the transaction. The transaction is declined with response code F11 (invalid teller level) if the operator level is less than the minimum teller level.
4. Determines appropriate response codes.

If no fatal or override-possible response codes are encountered, the Authorization process returns the message to the Device Handler as a response (0x10) with the appropriate approval response code in the RESP-HDR.RESP-CDE field in the TSTMH after logging it to the TTLF. Processing is complete for this transaction.

For the first fatal response code encountered, the Authorization process returns the message to the Device Handler as a response with the fatal response code in the RESP-HDR.RESP-CDE field in the TSTMH after logging it to the TTLF. Processing is complete for this transaction.

For each override-possible response code encountered, assuming no fatal response code has been encountered yet, processing continues with the next step.

5. Checks the ORF for the response code override level, depending on whether the transaction is on-us or not-on-us.
6. Compares the response code override level to the current operator level and, if necessary, the maximum terminal override level.

If the response code level is less than or equal to the current operator level, automatic override occurs. If all response code levels are less than or equal to the current operator level, the Authorization process returns the message to the Device Handler as a response (0x10) with response code A03 (approved with override) in the RESP-HDR.RESP-CDE field in the TSTMH after logging it to the TTLF. Processing is complete for this transaction.

If the response code level is greater than the current operator level and greater than the maximum terminal override level, it is treated like a fatal response code. The Authorization process returns the message to the Device Handler as a response with the response code in the RESP-HDR.RESP-CDE field in the TSTMH after logging it to the TTLF. The response code returned in the RESP-HDR.RESP-CDE field in the TSTMH is the first response code encountered that has an override level that exceeds the maximum terminal override level. Processing is complete for this transaction.

If the response code level is greater than the current operator level, but less than or equal to the maximum terminal override level, an override can be performed. The bit for the response code is set in one of the override bit maps, depending on which account (*to*, *from*, or credit line/backup) caused the response code and the override level (teller, supervisor, or manager) required to complete the transaction. Processing continues with the next step after all override-possible response codes have been identified.

7. Adds the Override token plus one or more of the following tokens to the internal message.
 - CAF Inquiry token is added if card-related bits are turned on.
 - PBF Inquiry token is added if PBF-related bits are turned on.
 - SPF Inquiry token is added if SPF-related bits are turned on.
 - WHFF Inquiry token is added if WHFF-related bits are turned on.

8. Returns the message to the Device Handler as a response with response code F0Z (override needed) in the RESP-HDR.RESP-CDE field in the TSTMH after logging it to the TTLF. The Device Handler returns the message to the controller and teller device with the override-possible response codes and inquiry information.

If the teller chooses not to override the transaction, BASE24-teller considers processing complete.

If the teller chooses to override the transaction, he or she must resubmit the transaction after raising his or her operator level high enough to override all of the override-possible response codes returned with the initial transaction. BASE24-teller treats an override as a new transaction, not an extension of the current transaction, so the Authorization process returns to step 1 to process the new transaction.

Deposit Processing

BASE24-teller uses the CUSTOMER CLASS field in the PBF, which can be accessed via PBF screen 10, to assign each account to one of the ten classes that can be defined in the TELLER CUSTOMER CLASS TABLE in the Institution Definition File (IDF), which can be accessed via IDF screen 24. Each class in the TELLER CUSTOMER CLASS TABLE in the IDF contains the following values that BASE24-teller uses to determine the available funds and amount on hold resulting from deposits.

- PERCENT OF DEPOSIT
- MAXIMUM DEPOSIT CREDIT
- MAXIMUM NUMBER DEPOSITS

Each class in the TELLER CUSTOMER CLASS TABLE in the IDF also contains a MAXIMUM CASH OUT value that BASE24-teller uses to limit the amount of cash disbursed in one day, but this value does not affect the available funds and amount on hold computations.

Values in the CASH IN INDICATOR and CASH OUT INDICATOR fields in the IDF, which can be accessed via IDF screen 25, control how BASE24-teller treats checks, cash in, and cash out when determining the available funds and amount on hold resulting from deposits.

Available Funds Limits

BASE24-teller can limit the increase in the available funds balance resulting from deposits a customer makes during a calendar day. Limits can be based on transaction amount, number of deposits, or both.

BASE24-teller uses the values in the PERCENT OF DEPOSIT, CASH IN INDICATOR, and CASH OUT INDICATOR fields to compute the available funds and amount on hold portions of deposit and split deposit transactions, then uses the values in the MAXIMUM DEPOSIT CREDIT and MAXIMUM NUMBER DEPOSITS fields to adjust the available funds and amount on hold computations if necessary. The values in these fields do not stop a deposit from being processed.

Deposit Credit Limit

The AMOUNT DEPOSIT CREDIT field in the PBF, which can be accessed via PBF screen 10, is used to accumulate the deposits added to the available funds balance during a calendar day. Values in the CASH IN INDICATOR and PERCENT OF DEPOSIT fields of the PBF control the deposit credit computation. The formulas and examples are provided later in this section. The computation result is compared to the value in the MAXIMUM DEPOSIT CREDIT field.

If the deposit credit computation result, when added to the AMOUNT DEPOSIT CREDIT amount, does not cause the AMOUNT DEPOSIT CREDIT amount to exceed the MAXIMUM DEPOSIT CREDIT amount, the initial available funds and amount on hold computations are not adjusted.

If the deposit credit computation result, when added to the AMOUNT DEPOSIT CREDIT amount, would cause the AMOUNT DEPOSIT CREDIT amount to exceed the MAXIMUM DEPOSIT CREDIT amount, the initial available funds amount is decreased and the initial amount on hold is increased by the excess deposit credit amount.

If the AMOUNT DEPOSIT CREDIT amount already exceeds the MAXIMUM DEPOSIT CREDIT amount, the entire deposit amount is added to the amount on hold instead of the available funds balance, regardless of the initial computation.

Deposit Number Limit

The NUMBER OF DEPOSITS field in the PBF, which can be accessed via PBF screen 10, is used to track the number of deposits made during a calendar day. Each time a deposit is made, BASE24-teller adds 1 to the value in the NUMBER OF DEPOSITS field, then compares the values in the NUMBER OF DEPOSITS and MAXIMUM NUMBER DEPOSITS fields.

If the NUMBER OF DEPOSITS value is less than or equal to the MAXIMUM NUMBER DEPOSITS value, the initial available funds and amount on hold computations are not adjusted.

If the NUMBER OF DEPOSITS value already exceeds the MAXIMUM NUMBER DEPOSITS value, the value in the CASH IN INDICATOR field of the PBF controls the deposit credit computation, as follows:

- If the value in the CASH IN INDICATOR field is Y, the cash received amount, if any, is added to the available funds balance and the remainder of the deposit amount is added to the amount on hold, regardless of the initial computation.
- If the value in the CASH IN INDICATOR field is N, the entire deposit amount is added to the amount on hold instead of the available funds balance, regardless of the initial computation.

Formulas for Available Funds, Amount on Hold, and Deposit Credit Amount

The formulas for available funds, amount on hold, and deposit credit amount shown on the next page depend on the values in the PERCENT OF DEPOSIT, CASH IN INDICATOR, and CASH OUT INDICATOR fields. The following values are used in these formulas:

- A1 = **AMT-1 or AMT-4 in the Financial token** — In a regular deposit transaction, AMT-1 is the original transaction amount. In a split deposit transaction, AMT-1 is the first account deposit amount and AMT-4 is the second account deposit amount. In a split deposit transaction, the available funds and amount on hold computations are performed for each account. The original transaction amount is always added to the ledger balance.
- A2 = **AMT-2 in the Financial token** — The amount of cash received. In a split deposit transaction, the amount of cash received for the first and second accounts depends on the amount of cash received (A2) compared to the amount being deposited in the first account (A1), as follows:
- If the amount of cash received is less than or equal to the amount being deposited in the first account, the entire cash received amount is added to the first account's CASH IN balance.
 - If the amount of cash received is greater than the amount being deposited in the first account, the amount being deposited in the first account is added to the first account's CASH IN balance and the balance of the cash received is added to the second account's CASH IN balance.
- A3 = **AMT-3 in the Financial token** — The amount of cash disbursed. In a split deposit, the entire cash out amount is always added to the first account's CASH OUT balance.
- % = **PERCENT OF DEPOSIT from the TELLER CUSTOMER CLASS TABLE on IDF screen 24** — The percentage, which is expressed as a decimal in the formula, of the deposit to be given immediate credit.

Cash In	Cash Out	Added to Available Balance
Y	Y	$A2 - A3 + (\% * (A1 - A2))$
Y	N	$A2 + (\% * (A1 - A2))$
N	Y	$(-1 * A3) + (\% * A1)$
N	N	$\% * A1$
N	X	<i>If A3 is greater than A2: $A2 - A3 + (\% * A1)$</i>
		<i>If A3 is less than or equal to A2: $\% * A1$</i>

Cash In	Cash Out	Added to Amount on Hold
Y	Y	$A1 - (A2 - A3 + (\% * (A1 - A2)))$
Y	N	$A1 - (A2 + (\% * (A1 - A2)))$
N	Y	$A1 + A3 - (\% * A1)$
N	N	$A1 - (\% * A1)$
N	X	<i>If A3 is greater than A2: $A1 - (A2 - A3 + (\% * A1))$</i>
		<i>If A3 is less than or equal to A2: $A1 - (\% * A1)$</i>

Cash In	Cash Out	Added to Deposit Credit Amount
Y	Y	$\% * (A1 - A2)$
Y	N	$\% * (A1 - A2)$
N	Y	$\% * A1$
N	N	$\% * A1$
N	X	$\% * A1$

Computation Examples

The following examples illustrate the available funds and amount on hold computations that result from the five possible combinations of CASH IN INDICATOR and CASH OUT INDICATOR settings. Each table gives the A1, A2, and A3 variable amounts. All examples use a percentage (%) value of 0.25.

Variables	Indicator		Added to Available Balance	Added to Amount on Hold	Added to Deposit Credit Amount
	Cash In	Cash Out			
A1 = 400	Y	Y	100	300	100
A2 = 0	Y	N	100	300	100
A3 = 0	N	Y	100	300	100
% = 0.25	N	N	100	300	100
	N	X	100	300	100

Variables	Indicator		Added to Available Balance	Added to Amount on Hold	Added to Deposit Credit Amount
	Cash In	Cash Out			
A1 = 400	Y	Y	400	0	0
A2 = 400	Y	N	400	0	0
A3 = 0	N	Y	100	300	100
% = 0.25	N	N	100	300	100
	N	X	100	300	100

Variables	Indicator		Added to Available Balance	Added to Amount on Hold	Added to Deposit Credit Amount
	Cash In	Cash Out			
A1 = 400	Y	Y	325	75	25
A2 = 300	Y	N	325	75	25
A3 = 0	N	Y	100	300	100
% = 0.25	N	N	100	300	100
	N	X	100	300	100

Variables	Indicator		Added to Available Balance	Added to Amount on Hold	Added to Deposit Credit Amount
	Cash In	Cash Out			
A1 = 400	Y	Y	350	50	0
A2 = 400	Y	N	400	0	0
A3 = 50	N	Y	50	350	100
% = 0.25	N	N	100	300	100
	N	X	100	300	100

Variables	Indicator		Added to Available Balance	Added to Amount on Hold	Added to Deposit Credit Amount
	Cash In	Cash Out			
A1 = 400	Y	Y	275	125	25
A2 = 300	Y	N	325	75	25
A3 = 50	N	Y	50	350	100
% = 0.25	N	N	100	300	100
	N	X	100	300	100

Variables	Indicator		Added to Available Balance	Added to Amount on Hold	Added to Deposit Credit Amount
	Cash In	Cash Out			
A1 = 400	Y	Y	50	350	100
A2 = 0	Y	N	100	300	100
A3 = 50	N	Y	50	350	100
% = 0.25	N	N	100	300	100
	N	X	50	350	100

Section 6

BASE24-teller Transaction Impacts on the Database

This section describes the impact of BASE24-teller transactions on the BASE24-teller database.

This section contains BASE24-teller financial transactions plus BASE24-teller file update transactions that affect balances in the Positive Balance File (PBF). It also lists how the transaction amounts carried in the internal message for each of these transactions affects balances in the PBF. The transactions in this section are a subset of the transactions in section 1 and appendix C.

Card-initiated transactions affect the bad PIN tries accumulator in the Cardholder Authorization File (CAF) in addition to the PBF balances. This accumulator is not in this section because its use depends on whether a PIN has been entered instead of the transaction code that has been selected. Refer to the **BASE24 Base Files Maintenance Manual** for descriptions of PIN verification configuration fields located on Institution Definition File (IDF) screen 2 and Card Prefix File (CPF) screen 2.

Positive Balance File (PBF) Impacts

The table in this section shows the impacts to the transaction accumulator fields in the PBF as a result of approved transactions. Each PBF field is identified by its name on the PBF file maintenance screens. PBF screen 1 includes the following accumulator fields:

- CHECKING/SAVINGS AVAILABLE BALANCE or CREDIT CARD AVAILABLE CREDIT
- CHECKING/SAVINGS LEDGER BALANCE or CREDIT CARD CREDIT LIMIT
- CHECKING/SAVINGS AMOUNT ON HOLD or CREDIT CARD CREDIT BALANCE
- CASH IN
- CASH OUT
- LAST DEPOSIT AMOUNT
- LAST WITHDRAWAL AMOUNT

PBF screen 10 includes the following accumulator fields:

- TOTAL DEPOSIT AMOUNT
- AMOUNT DEPOSIT CREDIT
- NUMBER OF DEPOSITS

Amounts in the following internal message fields can impact the PBF. These fields are identified in this section as follows:

- A = **AMT in the WHFF Update token** — The amount of the hold or float being added to or deleted from the Warning/Hold/Float File (WHFF).
- A1 = **AMT-1 in the Financial token** — The first account deposit amount in a split deposit transaction. Otherwise, the original transaction amount.
- A2A = **AMT-2 in the Financial token** — The amount of cash received.

- A2B = **AMT-2 in the Financial token** — In a split deposit transaction, the amount of cash received for the first and second accounts depends on the amount of cash received (A2B) compared to the amount being deposited in the first account (A1).
- If the amount of cash received is less than or equal to the amount being deposited in the first account, the entire cash received amount is added to the first account's CASH IN balance.
 - If the amount of cash received is greater than the amount being deposited in the first account, the amount being deposited in the first account is added to the first account's CASH IN balance and the balance of the cash received is added to the second account's CASH IN balance.
- A3 = **AMT-3 in the Financial token** — The amount of cash disbursed.
- A4 = **AMT-4 in the Financial token** — The second account deposit amount in a split deposit transaction or the fee amount in a purchase transaction.
- A14 = **AMT-1 in the Financial token plus AMT-4 in the Financial token** — AMT-1 is the face amount of the items purchased and AMT-4 is the fee assessed for the purchase. Purchase transactions adjust the PBF balance by the sum of AMT-1 and AMT-4.
- D1 = **Deposit credit portion of AMT-1 in the Financial token** — The deposit amount (A1) minus the portion that is placed on hold (%1).
- D4 = **Deposit credit portion of AMT-4 in the Financial token** — The second deposit amount (A4) minus the portion of that deposit that is placed on hold (%4).

%1 = **AMT-1 in the Financial token, subject to values in fields on IDF screens 24 and 25** — The values in the following fields, in combination with the cash in (A2A or A2B) and cash out (A3) amounts, determine the portion of a deposit that is placed on hold:

- CASH IN INDICATOR on IDF screen 25
- CASH OUT INDICATOR on IDF screen 25
- PERCENT OF DEPOSIT on IDF screen 24
- MAXIMUM DEPOSIT CREDIT on IDF screen 24
- MAXIMUM NUMBER DEPOSITS on IDF screen 24

Refer to section 5 for the results of the various funds availability options.

%4 = **AMT-4 in the Financial token, subject to values in fields on IDF screens 24 and 25** — The values in the following fields, in combination with the cash in (A2A or A2B) and cash out (A3) amounts, determine the portion of a deposit that is placed on hold:

- CASH IN INDICATOR on IDF screen 25
- CASH OUT INDICATOR on IDF screen 25
- PERCENT OF DEPOSIT on IDF screen 24
- MAXIMUM DEPOSIT CREDIT on IDF screen 24
- MAXIMUM NUMBER DEPOSITS on IDF screen 24

Refer to section 5 for the results of the various funds availability options.

A + or – indicates whether the transaction amount is added to or subtracted from the PBF field balance.

The transaction amount without a + or – means that the transaction amount replaces the current PBF field balance.

Tran Code	Transaction Description	Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
10 01 00	Withdrawal from DDA	-A1	-A1			+A3		A1			
10 11 00	Withdrawal from savings	-A1	-A1			+A3		A1			
10 21 00	Withdrawal from NOW	-A1	-A1			+A3		A1			
10 31 00	Withdrawal from credit card	-A1		+A1		+A3		A1			
10 32 00	Withdrawal from credit line	-A1		+A1		+A3		A1			
11 00 00	Cash transit check (log only)										
11 01 00	Cash transit check (DDA)	-A1	-A1			+A3		A1			
11 21 00	Cash transit check (NOW)	-A1	-A1			+A3		A1			
12 00 00	Cash official check (log only)										
12 01 00	Cash official check (DDA)	-A1	-A1			+A3		A1			
12 21 00	Cash official check (NOW)	-A1	-A1			+A3		A1			
13 00 00	Cash CD (log only)										
14 00 00	Cash bond (log only)										

Tran Code	Transaction Description	Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
15 00 00	Cash coupon (log only)										
16 00 00	Miscellaneous debit (log only)										
16 01 00	Miscellaneous debit from DDA	-A1	-A1			+A3		A1			
16 11 00	Miscellaneous debit from savings	-A1	-A1			+A3		A1			
16 12 00	Miscellaneous debit from IRA	-A1	-A1			+A3		A1			
16 21 00	Miscellaneous debit from NOW	-A1	-A1			+A3		A1			
16 31 00	Miscellaneous debit from credit card	-A1		+A1		+A3		A1			
16 32 00	Miscellaneous debit from credit line	-A1		+A1		+A3		A1			
16 41 00	Miscellaneous debit from installment loan							A1			
16 42 00	Miscellaneous debit from mortgage loan							A1			
16 43 00	Miscellaneous debit from commercial loan							A1			

Tran Code	Transaction Description	Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
17 00 00	Cash affiliate/correspondent check (log only)										
17 01 00	Cash affiliate/correspondent check (DDA)	-A1	-A1			+A3		A1			
17 21 00	Cash affiliate/correspondent check (NOW)	-A1	-A1			+A3		A1			
18 01 00	Debit memo post from DDA	-A1	-A1								
18 11 00	Debit memo post from savings	-A1	-A1								
18 12 00	Debit memo post from IRA	-A1	-A1								
18 21 00	Debit memo post from NOW	-A1	-A1								
18 31 00	Debit memo post from credit card	-A1		+A1							
18 32 00	Debit memo post from credit line	-A1		+A1							
18 41 00	Debit memo post from installment loan										
18 42 00	Debit memo post from mortgage loan										

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
18 43 00	Debit memo post from commercial loan											
20 00 01	Deposit to DDA		+D1	+A1	+%1	+A2A	+A3	A1		+A1	+D1	+1
20 00 11	Deposit to savings		+D1	+A1	+%1	+A2A	+A3	A1		+A1	+D1	+1
20 00 12	Deposit to IRA		+D1	+A1	+%1	+A2A	+A3	A1		+A1	+D1	+1
20 00 13	Deposit to CD		+D1	+A1	+%1	+A2A	+A3	A1		+A1	+D1	+1
20 00 21	Deposit to NOW		+D1	+A1	+%1	+A2A	+A3	A1		+A1	+D1	+1
21 01 01	Split deposit	to DDA 1	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to DDA 2	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 01 11	Split deposit	to DDA	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to savings	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 01 12	Split deposit	to DDA	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to IRA	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
21 01 13	Split deposit	to DDA	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to CD	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 01 21	Split deposit	to DDA	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to NOW	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 11 01	Split deposit	to savings	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to DDA	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 11 11	Split deposit	to savings 1	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to savings 2	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 11 12	Split deposit	to savings	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to IRA	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 11 13	Split deposit	to savings	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to CD	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 11 21	Split deposit	to savings	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to NOW	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
21 12 01	Split deposit	to IRA	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to DDA	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 12 11	Split deposit	to IRA	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to savings	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 12 12	Split deposit	to IRA 1	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to IRA 2	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 12 13	Split deposit	to IRA	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to CD	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 12 21	Split deposit	to IRA	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to NOW	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 13 01	Split deposit	to CD	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to DDA	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 13 11	Split deposit	to CD	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to savings	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
21 13 12	Split deposit	to CD	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to IRA	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 13 13	Split deposit	to CD 1	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to CD 2	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 13 21	Split deposit	to CD	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to NOW	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 21 01	Split deposit	to NOW	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to DDA	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 21 11	Split deposit	to NOW	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to savings	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 21 12	Split deposit	to NOW	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to IRA	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
21 21 13	Split deposit	to NOW	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to CD	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
21 21 21	Split deposit	to NOW 1	+D1	+A1	+%1	+A2B	+A3	A1		+A1	+D1	+1
		to NOW 2	+D4	+A4	+%4	+A2B		A4		+A4	+D4	+1
22 00 00	Miscellaneous credit (log only)											
22 00 01	Miscellaneous credit to DDA		+A1	+A1		+A2A		A1				
22 00 11	Miscellaneous credit to savings		+A1	+A1		+A2A		A1				
22 00 12	Miscellaneous credit to IRA		+A1	+A1		+A2A		A1				
22 00 21	Miscellaneous credit to NOW		+A1	+A1		+A2A		A1				
22 00 31	Miscellaneous credit to credit card		+A1		-A1	+A2A		A1				
22 00 32	Miscellaneous credit to credit line		+A1		-A1	+A2A		A1				
22 00 41	Miscellaneous credit to installment loan							A1				
22 00 42	Miscellaneous credit to mortgage loan							A1				
22 00 43	Miscellaneous credit to commercial loan							A1				

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
23 00 01	Credit memo post to DDA		+A1	+A1								
23 00 11	Credit memo post to savings		+A1	+A1								
23 00 12	Credit memo post to IRA		+A1	+A1								
23 00 21	Credit memo post to NOW		+A1	+A1								
23 00 31	Credit memo post to credit card		+A1		−A1							
23 00 32	Credit memo post to credit line		+A1		−A1							
23 00 41	Credit memo post to installment loan											
23 00 42	Credit memo post to mortgage loan											
23 00 43	Credit memo post to commercial loan											
40 01 01	Transfer	from DDA 1	−A1	−A1					A1			
		to DDA 2	+A1	+A1				A1				
40 01 11	Transfer	from DDA	−A1	−A1					A1			
		to savings	+A1	+A1				A1				

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
40 01 12	Transfer	from DDA	–A1	–A1					A1			
		to IRA	+A1	+A1				A1				
40 01 13	Transfer	from DDA	–A1	–A1					A1			
		to CD	+A1	+A1				A1				
40 01 21	Transfer	from DDA	–A1	–A1					A1			
		to NOW	+A1	+A1				A1				
40 11 01	Transfer	from savings	–A1	–A1					A1			
		to DDA	+A1	+A1				A1				
40 11 11	Transfer	from savings 1	–A1	–A1					A1			
		to savings 2	+A1	+A1				A1				
40 11 12	Transfer	from savings	–A1	–A1					A1			
		to IRA	+A1	+A1				A1				
40 11 13	Transfer	from savings	–A1	–A1					A1			
		to CD	+A1	+A1				A1				

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
40 11 21	Transfer	from savings	–A1	–A1					A1			
		to NOW	+A1	+A1				A1				
40 21 01	Transfer	from NOW	–A1	–A1					A1			
		to DDA	+A1	+A1				A1				
40 21 11	Transfer	from NOW	–A1	–A1					A1			
		to savings	+A1	+A1				A1				
40 21 12	Transfer	from NOW	–A1	–A1					A1			
		to IRA	+A1	+A1				A1				
40 21 13	Transfer	from NOW	–A1	–A1					A1			
		to CD	+A1	+A1				A1				
40 21 21	Transfer	from NOW 1	–A1	–A1					A1			
		to NOW 2	+A1	+A1				A1				
50 00 00	Payment (log only)											
50 00 31	Payment to credit card		+A1		–A1	+A2A		A1				

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
50 00 32	Payment to credit line		+A1		−A1	+A2A		A1				
50 00 41	Payment to installment loan							A1				
50 00 42	Payment to mortgage loan							A1				
50 00 43	Payment to commercial loan							A1				
50 00 50	Payment to utility		+A1	+A1		+A2A		A1				
50 00 51	Payment to utility 1		+A1	+A1		+A2A		A1				
50 00 52	Payment to utility 2		+A1	+A1		+A2A		A1				
50 00 53	Payment to utility 3		+A1	+A1		+A2A		A1				
50 00 54	Payment to utility 4		+A1	+A1		+A2A		A1				
50 00 55	Payment to utility 5		+A1	+A1		+A2A		A1				
50 01 00	Payment from DDA		−A1	−A1			+A3		A1			
50 01 31	Payment	from DDA	−A1	−A1					A1			
		to credit card	+A1		−A1			A1				

Number of Deposits	Amt Deposit Credit	Tot Deposit Amt	Last Withdraw Amt	Last Deposit Amt	Cash Out	Cash In	Amount on Hold or Credit Balance	Ledger Balance or Credit Limit	Available Balance or Available Credit	Transaction Description	Tran Code
			A1					-A1	-A1	from DDA	50 01 32
				A1			-A1		+A1	to credit line	
			A1					-A1	-A1	from DDA	50 01 41
				A1						to installment loan	
			A1					-A1	-A1	from DDA	50 01 42
				A1						to mortgage loan	
			A1					-A1	-A1	from DDA	50 01 43
				A1						to commercial loan	
			A1					-A1	-A1	from DDA	50 01 50
				A1				+A1	+A1	to utility	
			A1					-A1	-A1	from DDA	50 01 51
				A1				+A1	+A1	to utility 1	
			A1					-A1	-A1	from DDA	50 01 52
				A1				+A1	+A1	to utility 2	

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
50 01 53	Payment	from DDA	–A1	–A1					A1			
		to utility 3	+A1	+A1				A1				
50 01 54	Payment	from DDA	–A1	–A1					A1			
		to utility 4	+A1	+A1				A1				
50 01 55	Payment	from DDA	–A1	–A1					A1			
		to utility 5	+A1	+A1				A1				
50 11 00	Payment from savings		–A1	–A1			+A3		A1			
50 11 31	Payment	from savings	–A1	–A1					A1			
		to credit card	+A1		–A1			A1				
50 11 32	Payment	from savings	–A1	–A1					A1			
		to credit line	+A1		–A1			A1				
50 11 41	Payment	from savings	–A1	–A1					A1			
		to installment loan						A1				

Number of Deposits	Amt Deposit Credit	Tot Deposit Amt	Last Withdraw Amt	Last Deposit Amt	Cash Out	Cash In	Amount on Hold or Credit Balance	Ledger Balance or Credit Limit	Available Balance or Available Credit	Transaction Description	Tran Code
			A1					-A1	-A1	from savings	50 11 42
				A1						to mortgage loan	
			A1					-A1	-A1	from savings	50 11 43
				A1						to commercial loan	
			A1					-A1	-A1	from savings	50 11 50
				A1				+A1	+A1	to utility	
			A1					-A1	-A1	from savings	50 11 51
				A1				+A1	+A1	to utility 1	
			A1					-A1	-A1	from savings	50 11 52
				A1				+A1	+A1	to utility 2	
			A1					-A1	-A1	from savings	50 11 53
				A1				+A1	+A1	to utility 3	
			A1					-A1	-A1	from savings	50 11 54
				A1				+A1	+A1	to utility 4	

Tran Code	Transaction Description		Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
50 11 55	Payment	from savings	–A1	–A1					A1			
		to utility 5	+A1	+A1				A1				
50 21 00	Payment from NOW		–A1	–A1			+A3		A1			
50 21 31	Payment	from NOW	–A1	–A1					A1			
		to credit card	+A1		–A1			A1				
50 21 32	Payment	from NOW	–A1	–A1					A1			
		to credit line	+A1		–A1			A1				
50 21 41	Payment	from NOW	–A1	–A1					A1			
		to installment loan						A1				
50 21 42	Payment	from NOW	–A1	–A1					A1			
		to mortgage loan						A1				
50 21 43	Payment	from NOW	–A1	–A1					A1			
		to commercial loan						A1				

Number of Deposits	Amt Deposit Credit	Tot Deposit Amt	Last Withdraw Amt	Last Deposit Amt	Cash Out	Cash In	Amount on Hold or Credit Balance	Ledger Balance or Credit Limit	Available Balance or Available Credit	Transaction Description	Tran Code
			A1					-A1	-A1	from NOW	50 21 50
				A1				+A1	+A1	to utility	
			A1					-A1	-A1	from NOW	50 21 51
				A1				+A1	+A1	to utility 1	
			A1					-A1	-A1	from NOW	50 21 52
				A1				+A1	+A1	to utility 2	
			A1					-A1	-A1	from NOW	50 21 53
				A1				+A1	+A1	to utility 3	
			A1					-A1	-A1	from NOW	50 21 54
				A1				+A1	+A1	to utility 4	
			A1					-A1	-A1	from NOW	50 21 55
				A1				+A1	+A1	to utility 5	
										Purchase money order (log only)	60 00 00
			A14			+A3		-A14	-A14	Purchase money order using DDA	60 01 00

Tran Code	Transaction Description	Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
60 11 00	Purchase money order using savings	–A14	–A14			+A3		A14			
60 21 00	Purchase money order using NOW	–A14	–A14			+A3		A14			
60 31 00	Purchase money order using credit card	–A14		+A14		+A3		A14			
61 00 00	Purchase cashier's check (log only)										
61 01 00	Purchase cashier's check using DDA	–A14	–A14			+A3		A14			
61 11 00	Purchase cashier's check using savings	–A14	–A14			+A3		A14			
61 21 00	Purchase cashier's check using NOW	–A14	–A14			+A3		A14			
61 31 00	Purchase cashier's check using credit card	–A14		+A14		+A3		A14			
62 00 00	Purchase traveler's checks (log only)										

Tran Code	Transaction Description	Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
62 01 00	Purchase traveler's checks using DDA	-A14	-A14			+A3		A14			
62 11 00	Purchase traveler's checks using savings	-A14	-A14			+A3		A14			
62 21 00	Purchase traveler's checks using NOW	-A14	-A14			+A3		A14			
62 31 00	Purchase traveler's checks using credit card	-A14		+A14		+A3		A14			
63 00 00	Purchase bank draft (log only)										
63 01 00	Purchase bank draft using DDA	-A14	-A14			+A3		A14			
63 11 00	Purchase bank draft using savings	-A14	-A14			+A3		A14			
63 21 00	Purchase bank draft using NOW	-A14	-A14			+A3		A14			
63 31 00	Purchase bank draft using credit card	-A14		+A14		+A3		A14			
64 00 00	Purchase certified check (log only)										

Tran Code	Transaction Description	Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
64 01 00	Purchase certified check using DDA	-A14	-A14			+A3		A14			
64 11 00	Purchase certified check using savings	-A14	-A14			+A3		A14			
64 21 00	Purchase certified check using NOW	-A14	-A14			+A3		A14			
64 31 00	Purchase certified check using credit card	-A14		+A14		+A3		A14			
65 00 00	Purchase bond (log only)										
65 01 00	Purchase bond using DDA	-A14	-A14			+A3		A14			
65 11 00	Purchase bond using savings	-A14	-A14			+A3		A14			
65 21 00	Purchase bond using NOW	-A14	-A14			+A3		A14			
65 31 00	Purchase bond using credit card	-A14		+A14		+A3		A14			
66 00 00	Purchase miscellaneous (log only)										
66 01 00	Purchase miscellaneous using DDA	-A14	-A14			+A3		A14			

Number of Deposits	Amt Deposit Credit	Tot Deposit Amt	Last Withdraw Amt	Last Deposit Amt	Cash Out	Cash In	Amount on Hold or Credit Balance	Ledger Balance or Credit Limit	Available Balance or Available Credit	Transaction Description	Tran Code
			A14		+A3			-A14	-A14	Purchase miscellaneous using savings	66 11 00
			A14		+A3			-A14	-A14	Purchase miscellaneous using NOW	66 21 00
			A14		+A3		+A14		-A14	Purchase miscellaneous using credit card	66 31 00
										Verify PIN	75 00 00
							+A		-A	Add hold to DDA	85 00 01
							+A		-A	Add hold to savings	85 00 11
							+A		-A	Add hold to IRA	85 00 12
							+A		-A	Add hold to NOW	85 00 21
							+A		-A	Add hold to credit card	85 00 31
							+A		-A	Add hold to credit line	85 00 32
							+A		-A	Add hold to installment loan	85 00 41
							+A		-A	Add hold to mortgage loan	85 00 42

Tran Code	Transaction Description	Available Balance or Available Credit	Ledger Balance or Credit Limit	Amount on Hold or Credit Balance	Cash In	Cash Out	Last Deposit Amt	Last Withdraw Amt	Tot Deposit Amt	Amt Deposit Credit	Number of Deposits
85 00 43	Add hold to commercial loan	−A		+A							
86 00 01	Delete hold from DDA	+A		−A							
86 00 11	Delete hold from savings	+A		−A							
86 00 12	Delete hold from IRA	+A		−A							
86 00 21	Delete hold from NOW	+A		−A							
86 00 31	Delete hold from credit card	+A		−A							
86 00 32	Delete hold from credit line	+A		−A							
86 00 41	Delete hold from installment loan	+A		−A							
86 00 42	Delete hold from mortgage loan	+A		−A							
86 00 43	Delete hold from commercial loan	+A		−A							
87 00 01	Add float to DDA	+A									
87 00 11	Add float to savings	+A									
87 00 12	Add float to IRA	+A									
87 00 21	Add float to NOW	+A									

Number of Deposits	Amt Deposit Credit	Tot Deposit Amt	Last Withdraw Amt	Last Deposit Amt	Cash Out	Cash In	Amount on Hold or Credit Balance	Ledger Balance or Credit Limit	Available Balance or Available Credit	Transaction Description	Tran Code
									+A	Add float to credit card	87 00 31
									+A	Add float to credit line	87 00 32
									+A	Add float to installment loan	87 00 41
									+A	Add float to mortgage loan	87 00 42
									+A	Add float to commercial loan	87 00 43
									−A	Delete float from DDA	88 00 01
									−A	Delete float from savings	88 00 11
									−A	Delete float from IRA	88 00 12
									−A	Delete float from NOW	88 00 21
									−A	Delete float from credit card	88 00 31
									−A	Delete float from credit line	88 00 32
									−A	Delete float from installment loan	88 00 41
									−A	Delete float from mortgage loan	88 00 42
									−A	Delete float from commercial loan	88 00 43

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Appendix A

BASE24-teller Internal Message Header

BASE24-teller internal messages consist of a series of core fields—known as the BASE24-teller Standard Internal Message Header (TSTMH)—followed by a variety of message tokens that contain the information needed for processing.

This appendix describes the fields in the TSTMH. The tokens used by the standard BASE24-teller product are summarized in section 2 and described in greater detail in the ***BASE24 Tokens Manual***.

Teller Standard Internal Message Header (TSTMH)

The following pages describe the Teller Standard Internal Message Header (TSTMH) used by BASE24-teller. The TSTMH is 448 bytes in length.

Position	Level	Field Name and Description	Data Type
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1–92	02	SYS	
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1–4	04	MSG-TYP	PIC 9(4)
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The message type. Valid values are as follows:

- 0200 = Financial Transaction Request
- 0210 = Financial Transaction Response
- 0220 = Financial Transaction Advice
- 0230 = Financial Transaction Advice Response
- 0420 = Financial Transaction Reversal Advice
- 0430 = Financial Transaction Reversal Response
- 0300 = File Inquiry/Update Request
- 0310 = File Inquiry/Update Response
- 0320 = File Inquiry/Update Advice
- 0330 = File Inquiry/Update Advice Response
- 0600 = Administrative Request
- 0610 = Administrative Response
- 0620 = Administrative Advice
- 0630 = Administrative Advice Response

This field is set by the Device Handler, Authorization, or Host Interface process.

5–6	04	PROD-ID	PIC X(2)
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Indicates the BASE24 product originating this transaction. For BASE24-teller transactions, the value in this field is set to 03.

This field is set by the Device Handler.

Position	Level	Field Name and Description	Data Type
7–8	04	DPC-NUM The DPC number identifying the DPC to which the transaction is to be sent. Valid values are 0 through 9999. This field is set by the Authorization process using the value from the RTE-TBL.DPC-NUM field in the BASE24-teller segment of the IDF. The Host Interface process uses the value in this field to identify the DPC to receive the message.	TYPE BINARY
9	04	ORIGINATOR Indicates where the transaction request, reversal, or advice originated. This field applies to 0200, 0300, 0600, 0220, 0320, 0420, and 0620 messages. Valid values are as follows: 1 = Device controlled by BASE24 2 = Device Handler 3 = Authorization process 4 = Host Interface process 5 = Host This field is set by the Authorization, Device Handler, or Host Interface process.	PIC X(1)
10	04	RESPONDER Indicates where the response message (0210, 0310, 0610, 0230, 0330, 0430, or 0630) was generated. Values are as follows: 0 = Response has not been generated (default) 1 = Device controlled by BASE24 2 = Device Handler 3 = Authorization process 4 = Host Interface process 5 = Host This field is set by the Authorization, Device Handler, or Host Interface process.	PIC X(1)

Position	Level	Field Name and Description	Data Type
11–74	04	ORIG The following fields contain transaction origination data.	
11–16	06	TRAN-DAT The date (YYMMDD) the transaction was initiated. This field is set by the Device Handler.	PIC X(6)
17–24	06	TRAN-TIM The time (HHMMSSSTT) the transaction was initiated. This field is set by the Device Handler using the current time and the device's time offset from the TTDF.	PIC X(8)
25–40	06	STA-NAME Identifies the originating terminal. This field is set by the Device Handler using the value from the TERM-ID field in the TTDF.	PIC X(16)
41–42	06	RTE-STAT Indicates the status of a message at the system level. Valid values are as follows: 00 = No error 01 = Processing error 02 = Security device failure 11 = Time out after exit from BASE24 12 = Line down If this field contains 00, the TSTMH response code (in the RESP-HDR.RESP field) is used to determine the status of the message. Nonzero values indicate a failed message. This field is set by the Host Interface process and used by the Authorization process.	PIC 9(2)

Position	Level	Field Name and Description	Data Type
43–58	06	ORIG-PRO-NAME The symbolic name of the BASE24-teller process that originated the transaction. It is used by the Authorization process for return routing. This field is initially set by the Device Handler using its own symbolic name. If a request cannot be handled by the Authorization process and must be routed elsewhere for authorization, the value in this field is moved to the SYS.ORIG.HLD-ORIG-PRO-NAME field, and the symbolic name of the Authorization process routing the transaction is placed in this field. This field can then be used for return routing of the response.	PIC X(16)
59–74	06	HLD-ORIG-PRO-NAME The symbolic name of the BASE24-teller process that originated the transaction. It is used by the Authorization process for return routing. This field is initially blank-filled. If the Authorization process must route a transaction, it moves the value of the SYS.ORIG.ORIG-PRO-NAME field to this field and places its own symbolic name in the SYS.ORIG.ORIG-PRO-NAME field. The value from the SYS.ORIG.ORIG-PRO-NAME field is used for return routing to the Authorization process, and the value from the SYS.ORIG.HLD-ORIG-PRO-NAME field is used by the Authorization process for return routing to the originating process. This field is not used in Release 5.0.	PIC X(16)
75–80	04	TRAN-SEQ-NUM The sequence number associated with this transaction. The value in this field is generated by the Device Handler.	PIC X(6)
81–86	04	DEV-SEQ-NUM The sequence number supplied by the device and printed on the receipt.	PIC 9(6)

Position	Level	Field Name and Description	Data Type
87–92	04	TRACE-NUM The trace number assigned to the transaction. This field is set and used by the Host Interface process to match transactions to their correct response timers when responses are received.	PIC 9(6)
93–244	02	BNK The following fields contain the institution and teller terminal identification data.	
93–96	04	TERM-LN Identifies the logical network of the originating terminal. This field is set by the Device Handler using the value from the TERM-OWNER.LN field in the TTDF.	PIC X(4)
97–100	04	TERM-FIID Identifies the originating terminal's owner. This field is set by the Device Handler using the value from the FIID field in the TTDF.	PIC X(4)
101–104	04	CRD-FIID Identifies the account-owning or card-issuing institution. Whether a card is used to initiate the transaction and whether interbank routing is supported determine the value the Authorization process uses to set this field. If interbank routing is not supported, this field contains the same value as the BNK.TERM-FIID field because the same institution must own the terminal and the account. For card-initiated transactions, the Authorization process sets this field using the value from the FIID field in the Base segment of the CPF. For non-card-initiated transactions, the Authorization process sets this field using the value in the BNK.TERM-FIID field.	PIC X(4)

Position	Level	Field Name and Description	Data Type
		If interbank routing is supported, this field contains the FIID from the Account Routing File (ARF). For card-initiated transactions, the Authorization process selects the ARF record with an FIID that matches the value from the FIID field in the Base segment of the CPF and an account number that matches the characteristics of the account number entered by the teller. For non-card-initiated transactions, the Authorization process selects the ARF record with an account number that matches the characteristics of the account number entered by the teller.	
105–108	04	CRD-LN	PIC X(4)
		The account owner's or card issuer's logical network. This field is set by the Authorization process using the value of the LOGICAL-NET param in the LCONF.	
109–112	04	REGN-ID	PIC X(4)
		Identifies the region of the originating terminal. This field is set by the Device Handler using the value from the TERM-OWNER.REGN-ID field in the TTDF.	
113–116	04	BRCH-ID	PIC X(4)
		Identifies the branch of the originating terminal. This field is set by the Device Handler using the value from the TERM-OWNER.BRCH-ID field in the TTDF.	
117–118	04	DEV-TYP	PIC X(2)
		Indicates the type of terminal from which the transaction originated. For transactions originating at a BASE24 terminal, the Device Handler sets this field using the value from the TERM-TYP field in the TTDF. Values in the range of 50 to 67 are reserved for existing Device Handlers. Values in the range of A0 to A9 are reserved for future development.	

Position	Level	Field Name and Description	Data Type
119–126	04	TLR-ID The teller identification number for the teller who initiated the transaction. The Device Handler fills this field using the value from the TLR-ID field in the TTDF.	PIC X(8)
127–151	04	TERM-LOC The location of the originating terminal. For transactions originating at BASE24-teller terminals, this field is set by the Device Handler using the value from the TERM-LOC field in the TTDF.	PIC X(25)
152–164	04	TERM-CITY The city of the originating terminal. For transactions originating at BASE24-teller terminals, this field is set by the Device Handler using the value from the TERM-CITY field in the TTDF.	PIC X(13)
165–167	04	TERM-ST Identifies the state, province, or political boundary of the originating terminal. In the U.S., this is the two-digit state code. Codes for other countries must be agreed upon by all sharing institutions. For transactions originating at BASE24-teller terminals, this field is set by the Device Handler using the value from the TERM-ST field in the TTDF.	PIC X(3)
168–169	04	TERM-CNTRY Identifies the country of the originating terminal. For transactions originating at BASE24-teller terminals, this field is set by the Device Handler using the value from the TERM-CNTRY field in the TTDF.	PIC X(2)

Position	Level	Field Name and Description	Data Type
170–180	04	ACQ-INST-ID-NUM The terminal owner's routing or issuer identification number. For transactions originating at a BASE24 terminal, the Device Handler sets this field using the value from the ID-NUM field in the TTDF. Data is right-justified and zero-filled on the left.	PIC 9(11)
181–191	04	RCV-INST-ID-NUM The account owner's or card issuer's routing/identification number. For BASE24-teller card issuers, this field is set by the Authorization process using the value from the INST-ID-NUM field in the Base segment of the IDF. Data is right-justified and zero-filled on the left.	PIC 9(11)
192	04	DDA-CUR-FLG A flag specifying whether the PBF for demand deposit accounts is current. Valid values are as follows: Y = Yes, the file is current. N = No, the file is not current. The Authorization process sets this field using the value from the DDA-CUR field in the BASE24-teller segment of the IDF.	PIC X(1)
193	04	SAV-CUR-FLG A flag specifying whether the PBF for savings accounts is current. Valid values are as follows: Y = Yes, the file is current. N = No, the file is not current. The Authorization process sets this field using the value from the SAV-CUR field in the BASE24-teller segment of the IDF.	PIC X(1)

Position	Level	Field Name and Description	Data Type
194	04	CCD-CUR-FLG A flag specifying whether the PBF for credit accounts is current. Valid values are as follows: Y = Yes, the file is current. N = No, the file is not current. The Authorization process sets this field using the value from the CCD-CUR field in the BASE24-teller segment of the IDF.	PIC X(1)
195	04	SPF-CUR-FLG A code specifying whether the SPF and PBF are current. Valid values are as follows: Y = Yes, the files are current. N = No, the files are not current. The Authorization process sets this field using the value from the SPF-CUR field in the BASE24-teller segment of the IDF.	PIC X(1)
196	04	NBF-CUR-FLG A code specifying whether the NBF and PBF are current. Valid values are as follows: Y = Yes, the files are current. N = No, the files are not current. The Authorization process sets this field using the value from the NBF-CUR field in the BASE24-teller segment of the IDF.	PIC X(1)
197	04	WHFF-CUR-FLG A code specifying whether the WHFF and PBF are current. Valid values are as follows: Y = Yes, the files are current. N = No, the files are not current. The Authorization process sets this field using the value from the WHFF-CUR field in the BASE24-teller segment of the IDF.	PIC X(1)

Position	Level	Field Name and Description	Data Type
198–208	04	RTE-GRP The routing group associated with the terminal. This field is set by the Device Handler using the value from the RTE-GRP field in the TTDF. This field is reserved for future use.	PIC X(11)
209	04	INTERBNK-RTG A flag indicating whether the terminal-owning institution supports interbank routing. Valid values are as follows: 0 = Interbank routing is not supported. 1 = Interbank routing is supported. This field is set by the Device Handler using the value from the INTERBNK-RTG field in the BASE24-teller segment of the IDF.	PIC X(1)
210–220	04	BNK-RTG-CDE The bank routing code that can be entered by the teller in order to identify the institution that owns the account. If entered, this routing code is used to locate the appropriate information from the Account Routing File (ARF). The Device Handler fills this field with the information from the teller. This field is right-justified and zero-filled.	PIC X(11)
221–244	04	BNK-RELNSHP The banking relationships defined for the terminal-owning institution. This field is set by the Device Handler using the values from the BNK-RELNSHP fields in the TTDF.	PIC X(1) OCCURS 24 TIMES
245–247	04	CRNCY-CDE The type of currency used during the transaction. This field is set by the Device Handler using the value from the CRNCY-CDE field in the TTDF.	PIC 9(3)
248	04	USER-FLD1 This field is not used.	PIC X(1)

Position	Level	Field Name and Description	Data Type
249–378	02	RQST The following fields contain transaction-specific processing data.	
249–254	04	DEV-TRAN-CDE The device transaction code from the device native message. The content of this field varies for each device type and transaction code.	PIC X(6)
255–262	04	CUST-PASSBOOK-BAL The customer's passbook balance. This value can be entered by the teller. It is used for passbook printing.	TYPE BINARY 64 SIGNED
263	04	AUTH-LVL Indicates the authorizer of the transaction from the perspective of the account owner. Valid values are as follows: 0 = Transaction not supported. 1 = Online authorization, host authorized. 2 = Offline authorization, BASE24 authorized. 3 = Reserved for future use. 4 = Log only; BASE24-teller Authorization process logs transaction to the TTLF. The Positive Balance File (PBF) is updated only for transactions with an AUTH-LVL value of 2 (offline authorization). The Authorization process sets this field using the value from the AUTH-LVL field in the TTF.	PIC X(1)

Position	Level	Field Name and Description	Data Type
264–303	04	TRACK2 The Track 2 data as received from the teller device. The data can be entered electronically or manually. The data is in the following general format: 1. Start sentinel — (;) if data is entered electronically or (M) if data is entered manually 2. PAN, left-justified 3. Field separator (=) 4. Expiration date (MMYY or YYMM) 5. Discretionary data 6. End sentinel (?)	PIC X(40)
304–306	04	MBR-NUM The member number. This field is right-justified and zero-filled.	PIC 9(3)
307–325	04	FROM-ACCT The <i>from</i> account number, or the account number associated with the first account type. If the <i>from</i> account number is not applicable or not available, this field contains blanks.	PIC X(19)
326–344	04	TO-ACCT The <i>to</i> account number, or the account number associated with the second account type. If the <i>to</i> account number is not applicable or not available, this field contains blanks.	PIC X(19)
345–350	04	TRAN The values in the following fields identify the type of transaction and <i>from</i> and <i>to</i> account types. Refer to appendix C for valid transaction code, <i>from</i> and <i>to</i> account type combinations. Note: The actual transactions supported depend on the teller device.	

Position	Level	Field Name and Description	Data Type
345–346	06	CDE The transaction code.	PIC X(2)
347–348	06	FROM-ACCT-TYP The <i>from</i> account type of the transaction.	PIC 9(2)
349–350	06	TO-ACCT-TYP The <i>to</i> account type of the transaction.	PIC 9(2)
351	04	INTL-OVERRD-LVL A code indicating the level of teller initiating the transaction. Each response code that can be overridden has a minimum override level associated with it. The minimum override levels are set in the Override Response File (ORF). The value in the INTL-OVERRD-LVL field identifies which, if any, of the response codes encountered while processing the transaction can be overridden automatically without additional teller action. Valid values are as follows: 0 = No override level — The transaction is being processed without an override level. 1 = Teller override level — The transaction is being processed with an override level of a teller. 2 = Supervisor override level — The transaction is being processed with an override level of a supervisor. 3 = Manager override level — The transaction is being processed with an override level of a manager. This field is set by the Device Handler.	PIC X(1)

Position	Level	Field Name and Description	Data Type
352	04	MAX-TERM-OVRRD-LVL A code indicating the maximum override level that a particular terminal can handle. For example, when the terminal has the capability of processing teller overrides (level 1) and supervisor overrides (level 2), the value in this field should be set to a 2. Valid values are as follows: 1 = Teller level — This terminal has the capability of processing teller overrides. 2 = Supervisor level — This terminal has the capability of processing teller and supervisor overrides. 3 = Manager level — This terminal has the capability of processing teller, supervisor, and manager overrides. This field is set by the Device Handler using the value from the MAX-TERM-OVRRD-LVL field in the TTDF.	PIC X(1)
353	04	PAPERLESS-TRAN A code indicating whether the transaction is paperless or should be processed using a paper source document. The value in this field is not used by BASE24-teller, but is provided to assist the host software. Valid values are as follows: Y = Yes, the transaction is paperless and should be posted to host files from the TTLF. N = No, the transaction is not paperless. It should be posted to host files from a paper source document instead of the TTLF. This field is set by the Authorization process using the value from the PAPERLESS-TRAN field in the TTF.	PIC X(1)

Position	Level	Field Name and Description	Data Type
354	04	COMPL-REQ A code specifying whether BASE24 should send completion messages to the host for transactions. Valid values are as follows: N = No, do not send completion messages to the host. A = Send completion messages to the host for approved transactions only. D = Send completion messages to the host for denied transactions only. B = Send completion messages to the host for both approved and denied transactions. This field is set by the Authorization process using the value from the COMPL-REQ field in the TTF.	PIC X(1)
355	04	ADVC-RESP-REQ A code specifying whether a response is required for a teller-initiated advice or reversal advice message. Valid values are as follows: 0 = No, a response is not required. 1 = Yes, a response is required. This field is set by the Device Handler.	PIC X(1)
356	04	CRD-PRESENT A code indicating whether the transaction was initiated by a card at the teller terminal. Valid values are as follows: Y = Yes, the transaction was initiated by a card. N = No, the transaction was not initiated by a card. This field is set by the Device Handler.	PIC X(1)

Position	Level	Field Name and Description	Data Type
357–359	04	ENTRY-MDE	PIC X(3)
		The entry mode of the teller transaction. It identifies both how the card information was entered (positions 1 and 2) and the device PIN entry capability (position 3). Valid values for positions 1 and 2 are as follows: 00 = Unspecified 01 = Manual 02 = Magnetic stripe 03 = Bar code 04 = Optical character recognition (OCR) 05 = Integrated circuit card 06–60 = Reserved for ISO use 61–80 = Reserved for national use 81–99 = Reserved for private use Valid values for position 3 are as follows: 0 = No PIN entry capability 1 = PIN entry capability 2–5 = Reserved for ISO use 6–7 = Reserved for national use 8–9 = Reserved for private use This field is set by the Device Handler.	
360–361	04	RSN-CDE	PIC X(2)
		An informational field which can be used to distinguish between different types of the same transaction. This field is frequently used for miscellaneous debit/credit transactions or debit/credit memo post transactions to indicate the specific reason for the transaction. The values associated with this field are user defined and can vary for each BASE24-teller customer. This field is set by the Device Handler.	

Position	Level	Field Name and Description	Data Type
362	04	NOT-ON-US-FLG Indicates whether the transaction is on-us. Valid values are as follows: 0 = Terminal owner and account owner are the same (on-us). 1 = Terminal owner and account owner are not the same (not-on-us). This field is set by the Authorization process.	PIC 9(1)
363	04	MULT-ACCT Indicates whether the terminal is capable of handling multiple account data. Valid values are as follows: 0 = Terminal does not support multiple account selection; default to primary account. 1 = Terminal supports multiple account selection; return account selection information. This field is set by the Device Handler using the value from the MULT-ACCT field in the TTDF.	PIC 9(1)
364–367	04	SAVE-ACCT The actual <i>from</i> and <i>to</i> account types. These fields are set by the Authorization process using the value from the ACCT. TYP field in the Accounts segment of the CAF. Valid values are as follows: 00 = No account specified 01–09 = Checking account type 11, 14–19 = Savings account type 12 = IRA 13 = Certificate of Deposit 21 = NOW 31, 33–39 = Credit account type 32 = Credit line 41 = Installment loan 42 = Mortgage loan 43 = Commercial loan 50 = Utility payment	

Position	Level	Field Name and Description	Data Type
		51 = Utility 1 payment	
		52 = Utility 2 payment	
		53 = Utility 3 payment	
		54 = Utility 4 payment	
		55 = Utility 5 payment	
364–365	06	FROM-ACCT-TYP	PIC X(2)
366–367	06	TO-ACCT-TYP	PIC X(2)
368	04	MULT-ACCT-RQST-FLG	PIC 9(1)
		Indicates whether this is a multiple account selection request. Valid values are as follows:	
		0 = Normal request (default). This is not a multiple account selection request.	
		1 = Multiple account selection request.	
		The Device Handler sets this field to 1 when account information has been selected or when additional account information is requested from the CAF.	
369	04	COMPLETE-TRACK2-DATA	PIC X(1)
		Indicates the ability of the terminal or acquirer to capture and transmit complete Track 2 data on card swipe transactions. Valid values are as follows:	
		N = No, this terminal or acquirer does not capture and transmit Track 2 data on card swipe transactions.	
		Y or 0 = Yes, this terminal or acquirer can capture and transmit Track 2 data on card swipe transactions.	
		The Device Handler sets the value in this field.	

Position	Level	Field Name and Description	Data Type
370–377	04	OVRRD-TLR-ID The identification of the teller, supervisor, or manager that is overriding the transaction. The Device Handler sets the value in this field.	PIC X(8)
378	04	USER-FLD2 This field is not used.	PIC X(1)
379–380	02	TIM-OFST The time difference between the transaction-initiating terminal and the Tandem processor location. It is the signed number of minutes to be added to the Tandem system time in order to obtain the terminal time. This field is set by the Device Handler using the value from the TIM-OFST field in the TTDF.	TYPE BINARY
381–424	02	RESP-HDR The following fields contain transaction response data.	
381	04	CRD-ACTION The return/retain card action set by the transaction authorizer. This is the card action for the original response, prior to any override. Valid values are as follows: 0 = Return card. 1 = Retain card.	PIC X(1)
382–384	04	RESP-CDE The response code associated with the transaction. Approved response codes begin with the letter A, fatal response codes begin with the letter F, and response codes that can be overridden begin with the letter O. Refer to appendix B for valid response codes.	PIC X(3)

Position	Level	Field Name and Description	Data Type
385–390	04	POST-DAT The date (YYMMDD) the transaction will be posted by the financial institution. This field is set by the Device Handler using the value from the POST-DAT field in the TTDF.	PIC X(6)
391–396	04	AUTH-ID-RESP The authorization ID response. The value in this field is not generated by BASE24-teller, but can optionally be generated by the authorizing host.	PIC X(6)
397	04	ERR-FLG A code used to provide additional information regarding the disposition of the transaction. Valid values are as follows: C = Card verification failed K = KMAC synchronization error M = MAC failure S = Sanity check error T = Token error 0 or $\emptyset$ = Not applicable (default) ($\emptyset$ indicates a blank character)	PIC X(1)
398	04	MIN-OVRRD-LVL A code indicating the minimum level of override necessary to authorize the transaction in situations when the transaction is normally declined. This level is defined in the Override Response File (ORF). Valid values are as follows: 0 = No override required — The transaction is approved and does not require an authorization override. 1 = Teller authorizes override — The transaction can be approved by a teller.	PIC X(1)

Position	Level	Field Name and Description	Data Type
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- | | | | |
|---|---|--|--|
| 2 | = | Supervisor authorizes override — The transaction can be approved by a supervisor. | |
| 3 | = | Manager authorizes override — The transaction can be approved by a manager. | |
| 4 | = | Override restricted — The transaction is declined and cannot be approved with an authorization override. | |

399	04	ACCT-IND	PIC X(1)
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A code indicating which account is associated with the declined transaction. This field is used when a transaction is declined and the response code is associated with a specific account.

A transaction can be declined with a fatal response code (a response code beginning with the letter F). A transaction can also be declined with an override-possible response code (a response code beginning with the letter O) that cannot be overridden with this transaction because its override level is higher than the maximum override level of the terminal where the transaction originated. Response codes that can be associated with a specific account include the following:

- | | | |
|--------------------|---|--------------------------------------|
| F32, F33, F34, F3C | = | Fatal account errors on the CAF |
| F40 | = | Fatal PBF error |
| F50, F51 | = | Fatal SPF errors |
| F60, F61 | = | Fatal WHFF errors |
| F70–F73 | = | Fatal NBF errors |
| O40–O49 | = | PBF errors that cannot be overridden |
| O50–O52 | = | SPF errors that cannot be overridden |
| O60 | = | WHFF error that cannot be overridden |
| O70, O71 | = | NBF errors that cannot be overridden |

Valid values are as follows:

- | | | |
|--------|---|--|
| 0 or ␣ | = | Not applicable (␣ indicates a blank character) |
| F | = | <i>From</i> account |
| T | = | <i>To</i> account |
| C | = | Credit line/backup account |

Position	Level	Field Name and Description	Data Type
		This field is set by the Device Handler, Authorization, and Host Interface processes.	
400	04	CRD-VRFY-FLG	PIC X(1)
		A flag indicating whether the card has been verified. Valid values are as follows:	
		Y = Card verification was performed and the card verification digits are valid. Card verification should not be performed again.	
		C = Card verification was performed, the card verification digits are invalid, and transaction authorization is being continued. Card verification should not be attempted again.	
		D = Card verification was performed, the card verification digits are invalid, and the transaction is being declined with a value of C (card verification failure) in the RESP-HDR.ERR-FLG field.	
		N or 0 = Card verification was not attempted or a security device error occurred.	
401–414	04	USER-FLD4	PIC X(14)
		This field is not used.	
415–424	04	USER-FLD5	PIC X(10)
		This field is not used.	
425–432	02	ENTRY-TIM	TYPE BINARY 64 SIGNED
		The time, in Greenwich Mean Time (GMT), that the transaction entered the BASE24 system for the first time.	
		The Device Handler sets this field when the TSTMH is initialized.	

433–440	02	EXIT-TIM	TYPE BINARY 64 SIGNED
The time, in GMT, that the transaction left the BASE24 system for authorization through the Host Interface process.			
441–448	02	RE-ENTRY-TIM	TYPE BINARY 64 SIGNED
The time, in GMT, when the response to the transaction reentered the BASE24 system.			

Appendix B

BASE24-teller Authorization Response Codes

This appendix includes a description of each response code and how it is generated by the Authorization process. It also contains each response code's default override level in the Override Response File (ORF).

Authorization Response Codes

Transaction authorizers use authorization response codes to indicate how they want transactions to be handled. When a transaction is authorized, the authorizer returns these codes to inform the party requesting the authorization of the actions to take. BASE24-teller carries the response code in the RESP-HDR.RESP-CDE field of the Teller Standard Internal Message Header (TSTMH).

Authorization response codes are three-position codes indicating how the transaction authorizer wants the transaction handled. The first position indicates the action to be taken on the transaction. Valid values are as follows:

- A = Approved. The transaction authorizer approved the transaction, and the teller can complete the transaction as is.
- F = Declined, fatal error. The transaction authorizer encountered a fatal error while processing the transaction. The transaction cannot be completed.
- O = Declined, nonfatal error (i.e., error that can be overridden). The transaction authorizer declined the transaction because of a nonfatal error. The transactions can subsequently be approved by a teller, supervisor, or manager with the appropriate override level. Override levels are established in the Override Response File (ORF).

The second and third positions of the RESP-HDR.RESP-CDE field of the TSTMH contain a code indicating the reason the transaction was approved or declined.

To keep explanations brief, the following file name abbreviations have been used throughout this appendix:

- CAF = Cardholder Authorization File
- CPF = Card Prefix File
- IDF = Institution Definition File
- KEYA = Key Authorization File
- NBF = No Book File
- PBF = Positive Balance File
- SPF = Stop Payment File
- TTDF = Teller Terminal Data File
- WHFF = Warning/Hold/Float File

Default Override Levels

BASE24-teller uses levels to determine whether an operator can post a transaction or override a response code at a teller terminal, as well as the maximum level of override that can be performed from a given teller terminal. The following codes identify these levels in BASE24:

- 0 = No override required. The transaction is approved and does not require an authorization override.
- 1 = The teller authorizes the override. The transaction can be approved if entered by a teller.
- 2 = The supervisor authorizes the override. The transaction can be approved if entered by a supervisor.
- 3 = The manager authorizes the override. The transaction can be approved if entered by a manager.
- 4 = Override restricted. The transaction is declined and cannot be approved with an authorization override.

The Override Response File (ORF) contains one record for each response code supported in the BASE24-teller system for each ORF profile. An ORF profile is an identifier used to group institutions together for override processing. The ORF profile enables institutions in the same logical network to share ORF records instead of duplicating the ORF records for each institution.

At the time BASE24-teller is installed, ACI installs a complete set of ORF records under a default profile. This default ORF profile contains all of the response codes BASE24-teller can support and must be copied for each profile in the system. The default override levels in this appendix are taken from this default profile.

Note: While the override levels in the ORF can be changed for all response codes, changing the override level of approved (Axx) or fatal (Fxx) response codes does *not* affect the processing performed by BASE24-teller. BASE24-teller uses the override levels in the ORF only for the response codes that can be overridden (Oxx).

Response Code Descriptions

The following pages provide a description of each response code supported by BASE24-teller. The ORF default override level is provided for each response code.

Each description includes the response code text, a general description of the response code, and the circumstances under which the BASE24-teller Authorization process generates the code. BASE24-teller supports some response codes that the BASE24-teller Authorization process does not generate. These response codes can be generated by the host, the BASE24-teller Host Interface process, or some BASE24-teller Device Handlers.

Resp Code	Description	ORF Default
A00	APPROVED TRANSACTION The transaction was unconditionally approved. The transaction was verified against all criteria available. The BASE24-teller Authorization process returns this response code when the transaction was approved and account balances were impacted, if required. The BASE24-teller Authorization process can return this response code for all message types except administrative messages.	0
A01	APPROVED WITH NO BALANCES The transaction was unconditionally approved, but the balances cannot be returned. The BASE24-teller Authorization process returns this response code in the following cases: • Account types are not specified in the financial transaction request or advice transaction codes (e.g., 120000, 130000, 140000). • Transactions are specified as log-only transactions in the TTF record.	0
A02	APPROVED WITH WARNINGS The transaction was approved, but there are currently warnings on file. The BASE24-teller Authorization process does not generate this response code.	0

Resp Code	Description	ORF Default
A03	APPROVED WITH OVERRIDE The transaction was approved, but it had to be overridden by the teller, supervisor, or manager. The BASE24-teller Authorization process returns this response code when it encounters an error that can be overridden and both of the following conditions exist: • The transaction is a financial request or advice or a file inquiry/update request or advice. • The level of the teller is sufficient to automatically override the error condition.	0
A04	APPROVED BY DEVICE HANDLER The Device Handler approved the transaction. The BASE24-teller Authorization process does not generate this response code.	0
A05	APPROVED WITH OVERDRAFT The transaction was approved by taking the account balance negative up to the overdraft limit specified in the PBF because the account did not contain sufficient funds to cover the debit portion of the financial transaction. The BASE24-teller Authorization process can return this response code when processing the following transactions: • Deposit with cash back • Split deposit with cash back • Cash check • Payment from • Purchase • Transfer • Withdrawal	0

Resp Code	Description	ORF Default
A06	APPROVED WITH PASSBOOK CURRENT The transaction was approved for a passbook account, but there are no NBF records to be sent to the device for passbook printing. The BASE24-teller Authorization process can return this response code when processing passbook print requests.	0
A07	APPROVED WITH BACKUP ACCOUNT The transaction was approved by using the credit line/backup account. The BASE24-teller Authorization process returns this response code when the account has a backup account configured and funds were transferred from the backup account to the <i>from</i> account in order to cover the transaction. This can take place on the following transactions: • Cash check • Payment from • Purchase • Transfer • Withdrawal	0
A08	APPROVED SAF BY DEVICE HANDLER The Store-and-Forward (SAF) transaction was sent to the Device Handler and it was not processing the SAF date. The BASE24-teller Authorization process does not generate this response code.	0
A90	APPROVED ADMINISTRATIVE REQUEST The transaction was an approved administrative transaction (i.e., signon, signoff, logon, logoff). The BASE24-teller Authorization process returns this response code when the transaction is an approved administrative transaction request or advice.	0

Resp Code	Description	ORF Default
A91– A99	APPROVED ADMINISTRATIVE REQUEST The transaction was an approved administrative transaction (i.e., signon, signoff, logon, logoff). Response codes A90 through A99 have the same meaning; however, the BASE24-teller Authorization process generates response code A90 only.	0
F00	INVALID MESSAGE LENGTH The transaction was declined because the transaction authorizer detected a message length error. The BASE24-teller Authorization process does not generate this response code.	4
F01	MESSAGE EDIT ERROR The transaction was declined because the message failed field editing. The BASE24-teller Authorization process returns this response code in any of the following situations: • It cannot find a required token. • It cannot update a token. • It encounters an invalid length when accessing a variable-length token (e.g., the CAF Inquiry, Account, Customer Name, NBF, SPF Inquiry, or WHFF Inquiry tokens). • The teller requested a CAF inquiry or update and the CRD-PRESENT flag in the TSTMH is not set to Y.	4
F02	FIID NOT PROCESSING TELLER The transaction was declined because either the terminal owner or the account-owning institution does not support BASE24-teller. The BASE24-teller Authorization process returns this response code when it cannot find an entry for the institution in the IDF extended memory table.	4

Resp Code	Description	ORF Default
F03	TELLER NOT SIGNED ON The transaction was declined because the teller has not signed on to the BASE24-teller system. The BASE24-teller Authorization process does not generate this response code.	4
F04	BAD BUSINESS DATE The transaction was declined because the business date in the transaction was not available to BASE24. The BASE24-teller Authorization process returns this response code when the posting date from the request message does not match either the current or next business date found on the terminal-owning institution's IDF table entry.	4
F05	NEXT BUSINESS DATE NOT AVAILABLE The transaction was declined because the teller attempted to log on for the next business day before the time allowed by the institution. The BASE24-teller Authorization process does not generate this response code.	4
F06	AUTHORIZATION NOT RESPONDING The transaction was declined because a transaction timer expired. The BASE24-teller Authorization process does not generate this response code.	4
F07	HOST IS DOWN The transaction was declined because the host station was down. The BASE24-teller Authorization process returns this response code when the value in the SYS.RTE-STAT field in the TSTMH indicates that a line is down or a message timed out after leaving BASE24.	4

Resp Code	Description	ORF Default
F08	DATABASE PROBLEM The transaction was declined because of a system problem. The BASE24-teller Authorization process returns this response code in any of the following situations: • It is unable to write to the CAF, NBF, PBF, SPF, or WHFF. • It is unable to position or read the CAF, NBF, PBF, SPF, or WHFF. • It cannot derive valid card/PIN verification parameters for the transaction from the CPF. • It cannot derive valid expiration date check parameters for the transaction from the CPF. • It finds duplicate account types and numbers on the CAF record. • It encounters errors with the passbook print or automatic passbook print portion of the transaction. • It cannot find the KEYA record for PIN verification. • The card-owning institution does not have a valid filename configured for the CAF, NBF, SPF, or WHFF. • An invalid or unknown card status is specified in the CAF. • The information given to modify an account number for interbank routing would create an invalid account number. • The PBF record does not have a BASE24-teller segment defined. • An invalid parameter is passed to a procedure (indicates program error; refer to log messages for more information). • The IDF extended memory table entry is marked as having a fatal error. • Interbank routing is supported by the terminal-owning institution, but the ARF has not been loaded into the extended memory table. • The same institution does not own the <i>to</i> account and the <i>from</i> account on a two-sided transaction. • The <i>to</i> account and the backup account are the same on a two-sided transaction. • The currency code used in the transaction does not match the currency code of the card-owning institution.	4

Resp Code	Description	ORF Default
F09	ENTER LESSER AMOUNT The transaction was declined because the requested amount would cause the customer's daily withdrawal limit to be exceeded. In this case, a portion of the daily withdrawal amount remains, but it is less than the requested amount. The BASE24-teller Authorization process does not generate this response code.	4
F0A	DAILY WITHDRAWAL LIMIT REACHED The transaction was declined because the customer previously reached the personal daily withdrawal limit for the business day. The BASE24-teller Authorization process does not generate this response code.	4
F0B	AMOUNT GREATER THAN SYSTEM LIMIT The transaction was declined because the teller requested an amount that exceeds the maximum supported by BASE24. The BASE24-teller Authorization process returns this response code when the amount being added to a PBF amount field would cause the PBF field balance to exceed the maximum amount that the field can hold. The following transactions can increase amounts in PBF balance fields: • Transferring money from the backup account to the <i>from</i> account (available balance, ledger balance). • Reversing backup account information for any transaction using a backup account (ledger balance, available balance, credit balance). • Adding a deposit float record to the WHFF (available balance). • Adding or deleting the WHFF hold record (amount on hold, available balance). • Posting a credit transaction (available balance, ledger balance, cash in today). • Posting a debit transaction (cash out today, credit balance).	4

Resp Code	Description	ORF Default
F0B	AMOUNT GREATER THAN SYSTEM LIMIT <i>continued</i> • Posting a deposit transaction (cash in today, cash out today, available balance, ledger balance, total deposit amount, amount on hold, amount of deposit credit). • Posting a payment transaction (cash in today, cash out today, available balance, credit balance, ledger balance). • Posting a purchase transaction (cash out today, calculation of transaction amount plus fee amount, credit balance). • Posting a transfer transaction (cash in today, cash out today, available balance, credit balance, ledger balance). • Posting a withdrawal transaction (cash out today, credit balance). • Reversing a credit transaction (credit balance). • Reversing a debit transaction (ledger balance, available balance). • Reversing a deposit transaction (deposit credit amount). • Reversing a payment transaction (ledger balance, available balance, credit balance). • Reversing a purchase transaction (ledger balance, calculation of transaction amount plus fee amount, available balance). • Reversing a transfer transaction (ledger balance, available balance, credit balance). • Reversing a withdrawal transaction (ledger balance, available balance).	4
F0C	RETRY TRANSACTION The transaction was declined because of a temporary system problem. The transaction should be retried. The BASE24-teller Authorization process returns this response code when either the PBF or the NBF for this transaction is not current, the transaction could possibly involve passbook printing, and the passbook print flag in the IDF is set to decline transactions when the PBF or NBF is not current.	4

Resp Code	Description	ORF Default
F0D	INVALID MESSAGE DESTINATION The transaction was declined because an invalid destination was supplied. The BASE24-teller Authorization process returns this response code when the transaction is configured to be authorized by the host but there is no host destination in the IDF routing table.	4
F0E	INELIGIBLE AUTOMATIC PASSBOOK TRANSACTION The transaction was declined because automatic passbook printing is not allowed for this financial transaction. The BASE24-teller Authorization process returns this response code when automatic passbook printing is required for the transaction, but automatic passbook printing is not allowed for this transaction due to any of the following conditions: • The transaction is a financial advice message. • The transaction is a two-sided transaction. • The transaction is not included in the Automatic Passbook Print Transaction Table (i.e., cannot be configured for automatic passbook printing).	4
F0F	ACCOUNT OWNER NOT IN LOGICAL NETWORK The transaction was declined because BASE24-teller cannot determine the owner of the account. The BASE24-teller Authorization process returns this response code when the account owner cannot be found using the information loaded in the ARF extended memory table. The account owner cannot be determined as a result of encountering any of the following conditions: • A valid ARF extended memory table entry cannot be found. • The FIIDs of the ARF extended memory table entry and the CPF do not match on a card-initiated transaction. (Applies to both account routing and bank routing.) • The BASE24-teller Authorization process cannot find a matching account number in the ARF data extended memory table when performing account routing.	4

Resp Code	Description	ORF Default
F0G	CUSTOMER NOT SUPPORTED AT INSTITUTION The transaction was declined because the terminal owner and account owner have not established a valid banking relationship. The BASE24-teller Authorization process returns this response code when the issuer institution and the acquirer institution do not have a banking relationship for interbank routing.	4
F0H	TRANSACTION NOT SUPPORTED AT NOT-ON-US TERMINAL The transaction was declined because it was a not-on-us transaction that is not supported by the account owner (i.e., the OUR CUST/NOT-ON-US TERM field in the TTF is set to N). The BASE24-teller Authorization process returns this response code when the terminal-owning institution does not allow this transaction for customers other than their own.	4
F0I	SYSTEM ERROR The transaction was declined because the Host Interface process cannot translate the PIN or authenticate the message due to a problem with the security device. The BASE24-teller Authorization process returns this response code when any of the following errors occur: • A security device error occurs during card verification. • A security device error occurs during PIN verification. • A sanity check error occurs during PIN verification.	4
F0J	RECORD ALREADY EXISTS The transaction was declined because the Authorization process attempted to add a record that already existed. The BASE24-teller Authorization process returns this response code when the record that is being added already exists. This response code can be returned when adding SPF or WHFF records.	4

Resp Code	Description	ORF Default
F0Z	OVERRIDE NEEDED The transaction was declined because the Authorization process determined the transaction requires an override. The BASE24-teller Authorization process returns this response code during transaction processing when it encounters one or more response codes that could be overridden manually but not automatically. Unlike other Fxx response codes, this response code indicates the transaction can be processed successfully with a higher override level.	4
F10	INELIGIBLE TRANSACTION The transaction was declined because the teller attempted a transaction that is not supported by BASE24-teller. The BASE24-teller Authorization process returns this response code when any of the following situations occurs: • It does not recognize the transaction code for the financial transaction request or advice. • It does not recognize the transaction code for the file maintenance transaction request or advice. • It does not recognize the transaction code for the financial reversal transaction request. • A passbook print or reprint transaction is requested with an advice message type of 0320. • A transaction other than a deposit was attempted using a deposit-only or business deposit card.	4

Resp Code	Description	ORF Default
F11	INVALID TELLER LEVEL The transaction was declined because the teller level was invalid or insufficient for this transaction. The BASE24-teller Authorization process returns this response code when any of the following situations occurs: • The teller requesting the transaction does not have the minimum teller level sufficient to perform the transaction. • The initial teller override level received in the message header is not one of the valid values. • The maximum terminal override level is not one of the valid values. • The initial teller override level is greater than the maximum terminal override level.	4
F12	TRANSACTION NOT SUPPORTED BY ISSUER The transaction was declined because a banking relationship exists between the terminal owner and the account issuer, but the account issuer does not support or allow this transaction. The BASE24-teller Authorization process returns this response code when either of the following situations occurs: • The TTF extended memory table entry for the transaction cannot be found for the account-owning institution. • The TTF extended memory table entry for the institution acquiring the transaction indicates that the transaction is not supported or indicates a reserved authorization level.	4
F13	TRANSACTION NOT SUPPORTED BY TERMINAL OWNER The transaction was declined because the transaction is not supported by the terminal owner (i.e., the TERM TRAN ALLOWED field in the TTF is set to N). The BASE24-teller Authorization process does not generate this response code.	4

Resp Code	Description	ORF Default
F20	CPF RECORD NOT FOUND The transaction was declined because no CPF record exists for the prefix from the card. The BASE24-teller Authorization process returns this response code when either of the following situations occurs: • A card-initiated transaction is requested but the CPF extended memory segment was not built at initialization (i.e., the TLR-PROCESS-CRD-PIN param in the Logical Network Configuration File (LCONF) is set to N). • The specific CPF extended memory table entry cannot be located.	4
F21	INVALID TRACK 2 DATA The transaction was declined because a Track 2 offset was being used and invalid data was encountered in the Track 2 field. The BASE24-teller Authorization process returns this response code when pertinent information needed from the Track 2 data cannot be retrieved for any of the following reasons: • The format of the Track 2 data area is invalid. • The offsets specified on the CPF extended memory table entry are incorrect. • The attempt to verify the card fails.	4
F22	AMOUNT LESS THAN MINIMUM CREDIT CARD ADVANCE The transaction was declined because the customer attempted to obtain an amount of cash that was below the minimum established by that institution or the customer attempted to obtain an amount of cash that was below the amount increment established by the institution. The BASE24-teller Authorization process does not generate this response code.	4
F23	CARD OWNER NOT SUPPORTED The transaction was declined because the customer attempted to use a card that was not supported. The BASE24-teller Authorization process does not generate this response code.	4

Resp Code	Description	ORF Default
F30	CAF RECORD NOT FOUND The transaction was declined because no CAF record exists for the cardholder account number. The BASE24-teller Authorization process returns this response code when the attempt to read the CAF record fails with record not found.	4
F31	BAD CARD STATUS The transaction was declined because the card status in the CAF was other than open or restricted. This response code could also be the result of a restricted status in the CAF when the requested transaction would result in a withdrawal of funds. The BASE24-teller Authorization process returns this response code when the CAF contains card status code 6 (reserved for BASE24-pos only) because BASE24-teller cannot use cards with this status.	4
F32	ACCOUNT TYPE NOT FOUND IN CAF The transaction was declined because a specific account type and/or account number was entered for a transaction and the account type was not found in the CAF record. The BASE24-teller Authorization process searches the CAF's account table in either of the following situations: • It is verifying that the requested account type and number specified for the transaction exist on the CAF record. • It is performing account selection for transactions which specify account types but not account numbers. The BASE24-teller Authorization process returns this response code when either of the following situations occurs: • The CAF Inquiry token provides an invalid index into the account table of the CAF record. • No accounts or no more accounts can be returned to the requesting process from the CAF record account table.	4

Resp Code	Description	ORF Default
F33	MULTIPLE CAF ACCOUNT TYPES The transaction was declined because a specific account type was entered for a transaction and the entered account type has more than one account associated with it. The user should inquire to the CAF to determine which account is the desired account and perform the transaction manually. The BASE24-teller Authorization process returns this response code in any of the following situations: • The account select indicator in the IDF is set to 0 (institution does not support multiple account selection). • The account select indicator in the IDF is set to 2 (multiple account selection performed at teller device), but the account select indicator in the TTDF is set to N (device making the request does not support multiple account selection). • The message is an advice (message types 0220, 0320, or 0420) and multiple account selection is not allowed on advice messages.	4
F34	PBF ACCOUNT NOT LINKED TO CAF The transaction was declined because the PBF account was not linked to the account specified in the CAF. The BASE24-teller Authorization process returns this response code for a card-initiated transaction when it cannot locate a PBF record for the account specified in the CAF record account table.	4
F35	CARD MUST BE SWIPED The transaction was declined because the PIN offset data has been configured to be on Track 2 (requiring that the data be read electronically) and the Track 2 information has been entered manually.	4
F37	LOST CARD STATUS The transaction was declined because the card status found in the CAF indicates a lost card. The BASE24-teller Authorization process can return this response code with card status code 2 (lost).	4

Resp Code	Description	ORF Default
F38	STOLEN CARD STATUS The transaction was declined because the card status found in the CAF indicates a stolen card. The BASE24-teller Authorization process can return this response code with card status code 3 (stolen).	4
F39	CLOSED CARD STATUS The transaction was declined because the card status found in the CAF indicates a closed card. The BASE24-teller Authorization process can return this response code with card status code 9 (closed).	4
F3A	DENIED CARD STATUS The transaction was declined because the card status found in the CAF indicates a denied card. The BASE24-teller Authorization process can return this response code with card status code C (denial).	4
F3B	EXPIRED CARD The transaction was declined because the card status found in the CAF indicates an expired card or because the expiration date check failed. The BASE24-teller Authorization process can return this response code with card status code F (expired card).	4
F3C	INELIGIBLE ACCOUNT STATUS This response code is reserved for future use.	4
F3Z	MULTIPLE ACCOUNT SELECTION The BASE24-teller Authorization process returns this response code when multiple accounts have been found on the CAF record and multiple account selection is supported by both the device and the institution. This is not a fatal error. It indicates that account selection must be performed and an account must be selected to complete the transaction.	4

Resp Code	Description	ORF Default
F40	PBF RECORD NOT FOUND The transaction was declined because the Authorization process checked the PBF for this account and could not find it. The BASE24-teller Authorization process returns this response code for a transaction initiated without a card when a record not found error is returned when reading the PBF.	4
F50	SPF RECORD NOT FOUND The transaction was declined because the SPF record requested for an inquiry or delete transaction was not in the file.	4
F51	INVALID STOP PAY/WARN STATUS The transaction was declined because a stop pay/warning status was entered by the teller and the value was invalid. The BASE24-teller Authorization process does not generate this response code.	4
F60	WHFF RECORD NOT FOUND The transaction was declined because the WHFF record requested for an inquiry or delete transaction was not in the file.	4
F61	MULTIPLE WHFF RECORDS The transaction was declined because the Authorization process attempted to delete a WHFF record for the partial key specified in the transaction and the WHFF has more than one record associated with this key.	4
F70	NBF RECORD NOT FOUND The transaction was declined because the NBF record requested for an inquiry was not in the file.	4

Resp Code	Description	ORF Default
F71	PBF ACCOUNT NOT LINKED TO NBF The transaction was declined because the PBF account was not a passbook account. The BASE24-teller Authorization process can return this response code with the following transactions: • Financial transaction with automatic passbook printing • NBF inquiry • Passbook print • Passbook reprint	4
F72	NBF RECORD NOT FOUND FOR REVERSAL The transaction was declined because a teller-initiated reversal was performed for a passbook account and BASE24 could not locate the NBF record for the original financial transaction.	4
F73	NO NBF RECORDS TO REPRINT The transaction was declined because a passbook reprint transaction was performed and all of the printed NBF records have already been reprinted.	4
O31	PIN INCORRECT The transaction was declined because the customer's Personal Identification Number (PIN) was entered incorrectly. The BASE24-teller Authorization process returns this response code when the PIN is entered incorrectly or does not match the PIN associated with the card used in the transaction.	2
O32	PIN TRIES EXCEEDED The transaction was declined because the customer exceeded the maximum number of PIN tries for the current usage period.	2

Resp Code	Description	ORF Default
O33	UNABLE TO VERIFY PIN The transaction was declined because the PIN could not be verified due to a failure within the system (configuration or device). The BASE24-teller Authorization process returns this response code under either of the following circumstances: • The PIN offset is configured to be located on the CAF record but the transaction is to be authorized by the host. (The PIN offset must be part of the Track 2 data when PIN verification is part of the transaction screening.) • The Security Utilities return an invalid PIN data or invalid PIN parameter error when attempting PIN verification.	4
O34	PIN METHOD — NO PIN PRESENT The transaction was declined because a PIN verification method has been configured but a PIN was not present in the message.	2
O35	PIN PRESENT — NO PIN METHOD The transaction was declined because a PIN was present in the message but no PIN verification method has been configured.	2
O36	CARD NOT ACTIVATED The transaction was declined because the card status found in the CAF indicated that the card has not been activated. The BASE24-teller Authorization process can return this response code with card status code 0 (issued but not active).	2
O37	CALL FOR APPROVAL The transaction was declined because the card status found in the CAF indicated to call for approval. The BASE24-teller Authorization process can return this response code with card status code A (referral).	3

Resp Code	Description	ORF Default
O38	USER DEFINED The transaction was declined because the card status found in the CAF indicated a user-defined status. The BASE24-teller Authorization process can return this response code with card status code B (maybe).	1
O39	SIGNATURE REQUIRED The transaction was declined because the card status found in the CAF indicated that a signature is required for approval. The BASE24-teller Authorization process can return this response code with card status code D (signature required).	2
O40	BAD PBF ACCOUNT STATUS The transaction was declined because the status of the account, stored on the PBF record, indicates that the account is restricted from transactions of this type. The BASE24-teller Authorization process can return this response code with the following account status codes: • 0, A, B, or C (inactive account) • 9, V, W, X, Y, or Z (closed account)	2
O41	INELIGIBLE ACCOUNT The transaction was declined because the status of the account, stored on the PBF record, indicates that no transactions are allowed on this account. The BASE24-teller Authorization process can return this response code with the following account status codes: • 2, J, K, or L (restricted to deposits) • 4, S, T, or U (restricted primary account)	2

Resp Code	Description	ORF Default
O42	TODAY'S CASH OUT EXCEEDED The transaction was declined because the customer exceeded his or her maximum withdrawal limit for the day. The BASE24-teller Authorization process declines a transaction with this response code when the customer has already exceeded the institution's limit for the maximum amount of cash which can be withdrawn during the business day. This limit is defined in the IDF's customer class table. The cash out amount is checked for the following financial transactions: • Miscellaneous debit • Deposit with cash back • Split deposit with cash back • Purchase • Cash check • Payment from • Withdrawal	3
O43	WILL EXCEED TODAY'S CASH OUT The transaction was declined because the customer will exceed his or her maximum withdrawal limit for the day if the transaction completed as entered. The BASE24-teller Authorization process returns this response code with a transaction that would cause the customer to exceed the institution's limit for the maximum amount of cash which can be withdrawn during the business day. This limit is defined in the IDF's customer class table. The cash out amount is checked for the following financial transactions: • Miscellaneous debit • Deposit with cash back • Split deposit with cash back • Purchase • Cash check • Payment from • Withdrawal	3

Resp Code	Description	ORF Default
O44	INSUFFICIENT FUNDS The transaction was declined because the transaction would cause funds to be removed from an account, but the available balance and the ledger balance in the account are not sufficient to cover the amount requested. The BASE24-teller Authorization process returns this response code with a transaction that requests funds from a primary or backup account in which the available funds are less than or equal to zero. The transaction amount is checked for the following financial transactions: • Deposit with cash back • Split deposit with cash back • Payment from • Purchase • Cash check • Transfer • Withdrawal • Transactions accessing a backup account	2

Resp Code	Description	ORF Default
O45	INSUFFICIENT AVAILABLE FUNDS The transaction was declined because the transaction would cause funds to be removed from an account, but the available balance in the account was not sufficient to cover the amount requested. However, the ledger balance in the account is sufficient to cover the amount requested. The BASE24-teller Authorization process returns this response code with a transaction that requests funds from a primary or backup account in which the available funds are greater than zero, but less than the requested amount. The transaction amount is checked for the following transactions: • Deposit with cash back • Split deposit with cash back • Payment from • Purchase • Cash check • Transfer • Withdrawal • Transactions accessing a backup account • Transactions adding a hold record to the WHFF	2

Resp Code	Description	ORF Default
O46	ACCOUNT EXCEEDS OVERDRAFT AMOUNT The transaction was declined because the account will exceed the overdraft amount. The BASE24-teller Authorization process declines a withdrawal transaction with this response code when there are not enough funds in the account and the requested amount exceeds the overdraft amount for the account. The overdraft limit is checked on the following financial transactions: • Deposit with cash back • Split deposit with cash back • Payment from • Purchase • Cash check • Transfer • Withdrawal	3
O47	EXCEEDED CASH IN LIMIT The transaction was declined because the cash in limit defined in the PBF was exceeded. The BASE24-teller Authorization process returns this response code with a transaction that exceeds the maximum amount of cash which can be deposited during the current business day. The cash in amount is checked for the following financial transactions: • Payment to • Miscellaneous credit • Deposit • Split deposit	2

Resp Code	Description	ORF Default
O48	EXCEEDED CASH OUT LIMIT The transaction was declined because the cash out limit defined in the PBF was exceeded. The BASE24-teller Authorization process returns this response code with a transaction that exceeds the maximum amount of cash which can be withdrawn during the current business day. The cash out amount is checked for the following financial transactions: • Miscellaneous debit • Deposit • Split deposit • Payment from • Purchase • Transfer • Cash check • Withdrawal	2
O49	PASSBOOK BALANCE DIFFERENCE The transaction was declined because the customer's passbook balance does not equal BASE24's passbook balance. The BASE24-teller Authorization process returns this response code when the passbook balance found on the PBF record and the balance in the request message do not match. Balances are checked the following transactions: • Financial transactions with automatic passbook printing • Passbook print transactions	1
O50	MATCH ON CHECK NUMBER/AMOUNT The cash check transaction was declined because the authorizer detected a match between the check number and amount entered and a check number and amount in the database. The BASE24-teller Authorization process returns this response code when it finds an SPF record that matches on the check number and on the amount from the request message.	1

Resp Code	Description	ORF Default
O51	MATCH ON CHECK NUMBER The cash check transaction was declined because the authorizer detected a match between the check number entered and a check number in the database. The BASE24-teller Authorization process returns this response code when it finds an SPF record that matches the check number from the request message.	1
O52	MATCH ON CHECK RANGE The cash check transaction was declined because the authorizer detected a match between the check number entered and a check range in the database. The BASE24-teller Authorization process returns this response code when it finds an SPF record that defines a range of check numbers which contains the check number from the request message.	1
O60	WARNING RECORD(S) FOUND The transaction was declined because the authorizer detected one or more warnings in the database. The BASE24-teller Authorization process returns this response code when a warning record for the account is found in the WHFF. The WHFF is searched if it is specified on the TTF for this transaction and the account's PBF record indicates that warning records exist. The TTF and PBF are checked for all PBFs (up to 3) involved in the transaction for all financial requests and advices.	1
O70	TOO MANY NBF RECORDS A passbook print or automatic passbook print transaction was declined because the number of NBF records for the account exceeds the maximum allowed. This condition can cause a slow response time. The BASE24-teller Authorization process compares the number of NBF records for the account to a number specified by the TLR-MAX-NBF-CNT parameter in the LCONF when performing teller-initiated or automatic passbook printing transactions.	2

Resp Code	Description	ORF Default
O71	PBF/NBF BALANCE DIFFERENCE A passbook print transaction was declined because BASE24's passbook balance does not equal BASE24's ledger balance. The BASE24-teller Authorization process compares these balances for the following transactions: • Financial transactions with automatic passbook printing • Passbook print transactions	3

Appendix C

Supported BASE24-teller Transactions

BASE24-teller supports over 300 transactions. This appendix provides the six-digit transaction code and a brief description for each of the BASE24-teller transactions. Similar transactions are grouped together.

Transaction Identifiers

Each BASE24-teller transaction has a unique identifier, known as a transaction code, that is recognized by BASE24-teller. This transaction code consists of three two-digit parts: transaction code, *from* account type, and *to* account type. The two-digit transaction code identifies the type of processing being performed. The *from* account type and *to* account type identify the type of account receiving funds from the transaction or providing the funds for the transaction. BASE24-teller uses the account type codes defined in section 1 of this manual to identify the *to* and *from* accounts. For example, transaction code 401101, when divided into its three two-digit parts, identifies a transfer (40) from savings (11) to checking or DDA (01).

The term transaction code identifies the six-digit identifier or the first two-digit part of the six-digit identifier. References to a two-digit transaction code typically include all of the six-digit transaction codes that begin with the two-digit transaction code. For example, a reference to transaction code 11 (cash transit check) includes three six-digit transaction codes:

110000 = Cash transit check (log only)
110100 = Cash transit check (DDA)
112100 = Cash transit check (NOW)

Terminology

The following definitions are used in the transaction descriptions.

PBF Balances, Accumulators, and Statuses

BASE24 maintains accountholder balance, accumulator, and status fields in each accountholder's PBF record. Each BASE24-teller transaction code affects these balances, accumulators, and statuses in a different way. Some of the transaction descriptions in the remainder of this appendix summarize these effects because the change in the value of a PBF field is sometimes the only factor that distinguishes one group of transactions from another. Refer to section 6 of this manual for the specific impacts from each transaction code on the balance, accumulator, and status fields summarized here.

Balances

BASE24-teller transactions affect the following PBF account balances:

Available balance	The current balance of funds available for withdrawal from a checking, savings, CD, IRA, NOW, or utility payment account; or the funds that can still be advanced to the customer from a credit account.
Credit balance	The current balance of a credit card, line of credit, installment loan, mortgage loan, or commercial loan account.
Hold amount	The current balance of funds in a noncredit account that are currently unavailable for withdrawal. Funds are placed on hold to protect a financial institution from losses in case checks that it has accepted for deposit are not paid by the issuing financial institution. The ledger balance minus the hold amount equals the available balance.
Ledger balance	The current balance of a checking, savings, CD, IRA, NOW, or utility payment account.

Accumulators

BASE24-teller transactions affect the following PBF accumulators:

Cash in	The total amount of cash received during the current processing day. This accumulator is reset to zero when the PBF record is refreshed or when the first transaction of the day is received.
Cash out	The total amount of cash disbursed during the current processing day. This accumulator is reset to zero when the PBF record is refreshed or when the first transaction of the day is received.
Deposit credit amount	The total amount added to the available balance as a result of deposits made to an account via BASE24-teller terminals during the current BASE24 processing day. This accumulator is reset to zero when the account is refreshed or when the first transaction of the day is received.
Number of deposits	The total number of deposits made to an account via BASE24-teller terminals during the current BASE24 processing day. This accumulator is reset to zero when the account is refreshed or when the first transaction of the day is received.
Total deposit amount	The total amount deposited to an account via BASE24-teller terminals during the current BASE24 processing day. This accumulator is reset to zero when the account is refreshed or when the first transaction of the day is received.

Statuses

BASE24-teller transactions affect the following PBF statuses:

Last deposit/ payment amount	The amount of the most recent deposit to a noncredit account or payment to a credit account. This status is updated only for transactions authorized by BASE24.
Last deposit/ payment date	The date of the most recent deposit to a noncredit account or payment to a credit account. This status is updated only for transactions authorized by BASE24.
Last withdrawal/ charge amount	The amount of the most recent withdrawal from a noncredit account or charge to a credit account. This status is updated only for transactions authorized by BASE24.
Last withdrawal/ charge date	The date of the most recent withdrawal from a noncredit account or charge to a credit account. This status is updated only for transactions authorized by BASE24.

Insufficient Funds

BASE24-teller uses overdraft limits and automatic overdraft protection to authorize checks and withdrawals for an accountholder even though he or she does not have sufficient funds in the account. A financial institution may prefer to overdraw a customer's account or transfer funds between a customer's accounts to cover the shortage of funds instead of returning checks or withdrawals unpaid, either as a service to the customer or as a way to reduce the number of items that the financial institution must handle manually.

An overdraft limit is the amount that a customer's primary account can be overdrawn and still have checks and withdrawals authorized. BASE24-teller can approve a transaction without an override even if the transaction amount exceeds the available balance, as long as the resulting negative balance does not exceed the overdraft limit amount. The value in the OVERDRAFT LIMIT field in the accountholder's PBF record controls this limit.

Automatic overdraft protection is the transfer of funds to an accountholder's primary account so that checks and withdrawals can be authorized without overdrawing the account. BASE24-teller can transfer funds from the accountholder's line of credit/backup account as part of the processing it performs

for a financial transaction. Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with automatic overdraft protection.

Processing performed by the BASE24-teller Authorization process for a subset of the BASE24-teller transaction set includes overdraft limits, automatic overdraft protection, or both. Transactions that include processing for overdraft limits are identified by a ✓ in the OL column of the tables presented in the remainder of this appendix. Transactions that include processing for automatic overdraft protection are identified by a ✓ in the OP column of the tables presented in the remainder of this appendix.

Preauthorized Holds

A merchant using BASE24-pos can place preauthorization holds on a customer's account before the customer makes a purchase to ensure that funds will be available when the purchase is actually made. As an example, a hotel clerk could place a preauthorization hold on a customer's account when he or she checks in to ensure that funds will be available in the account to pay the bill when the customer is ready to check out. When processing transactions that would reduce the balance of an account, BASE24-teller can check for preauthorized holds when computing account balances. Refer to section 4 of this manual for the available options.

Transactions that can include checking for preauthorized holds are identified by a ✓ in the PH column of the tables presented in the remainder of this appendix.

Passbooks

Each savings, IRA, certificate of deposit (CD), NOW, installment loan, mortgage loan, and commercial loan account maintained by BASE24-teller can be handled as a passbook or nonpassbook account. With a passbook account, a financial institution provides the customer with a passbook as evidence of the account and records each transaction in the passbook. With a nonpassbook account, a financial institution typically uses cards or coupons as evidence of the account and usually provides the customer with a receipt as record of each transaction. A flag set in the PBF when an account is established determines whether BASE24-teller handles the account as a passbook or nonpassbook account.

When processing a financial transaction involving a passbook account, BASE24-teller processes the transaction as if the passbook is not available, writing the transaction to a No Book File (NBF) for later posting to the customer's passbook. Without the automatic passbook printing feature (described later), the

teller must first post the financial transaction and then use another transaction to print the entries in the customer's passbook. Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with passbook processing.

Processing performed by the BASE24-teller Authorization process for a subset of the BASE24-teller transaction set can include writing the transaction to the NBF for later posting to the customer's passbook. These transactions are identified by a ✓ in the PB column of the tables presented in the remainder of this appendix.

Automatic Passbook Printing

While processing certain transactions, BASE24-teller can automatically print the transaction currently being processed plus any transactions performed since the last time the passbook was updated. With the automatic passbook printing feature, BASE24-teller still adds the transaction to the NBF, but then automatically starts the passbook print transaction without any extra action by the teller. Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with automatic passbook printing.

Processing performed by the BASE24-teller Authorization process for a subset of the BASE24-teller transaction set can include automatically starting a passbook print transaction. These transactions are identified by a ✓ in the AP column of the tables presented in the remainder of this appendix.

Stop Payments

As part of the processing performed for some of the check cashing transactions, BASE24-teller searches the Stop Payment File (SPF) for stop payment information before authorizing the transaction. A stop payment order can be placed on a check written by one of a financial institution's own customers to prevent the check from being cashed by anyone. Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with stop payments.

Stop payment processing is not identified by a separate column in the tables presented in the remainder of this appendix because all transactions affected by this processing appear in a single transaction group. The following two-digit transaction codes identify transactions that include stop payment processing: 11, 12, and 17.

Warning, Hold, and Deposit Float Records

As part of the processing performed for financial transactions other than log-only transactions, BASE24-teller can search the Warning/Hold/Float File (WHFF) for warning records. Warnings alert tellers about possible problems involving a particular account.

The WHFF also contains hold and float information. Holds decrease the available funds balance and increase the amount on hold for a noncredit account or decrease the available credit and increase the credit balance for a credit account. Deposit float increases the available funds balance for a noncredit account or increase the available credit for a credit account. BASE24-teller does not search the WHFF for hold and float information because balances in the Positive Balance File (PBF) already reflect all hold and float amounts.

Refer to section 4 of this manual for the options and section 5 of this manual for the processing involved with warning, hold, and deposit float records.

Warning record processing is not identified by a separate column in the tables presented in the remainder of this appendix. Financial transactions other than log-only transactions can be affected by this processing. The following two-digit transaction codes identify financial transactions: 10 through 23, 40 through 66, and 75.

Withdrawal Transactions

BASE24-teller supports the withdrawal transactions shown in the table below.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
10 01 00	✓	✓			✓	Withdrawal from DDA — The disbursement of funds from a customer's checking account.
10 11 00	✓	✓	✓	✓	✓	Withdrawal from savings — The disbursement of funds from a customer's savings account.
10 21 00	✓	✓	✓	✓		Withdrawal from NOW — The disbursement of funds from a customer's NOW account.
10 31 00	✓				✓	Withdrawal from credit card — The advance of funds from a customer's credit card account.
10 32 00	✓					Withdrawal from credit line — The advance of funds from a customer's line of credit account.

Support Key:

OL = Overdraft limit

PB = Passbooks

PH = Preauthorized hold checks

OP = Overdraft protection

AP = Automatic passbook printing

Check Cashing Transactions

BASE24-teller supports the check cashing transactions shown in the table below.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
11 00 00						Cash transit check (log only) — The disbursement of funds to a customer in exchange for a check they have presented to our institution for payment. This transaction is logged by BASE24-teller and is not verified against or used to update any BASE24 records.
11 01 00	✓	✓			✓	Cash transit check (DDA) — The disbursement of funds to a customer in exchange for a check they have presented to our institution for payment. BASE24-teller uses the check writer's account number to check the PBF for sufficient funds, search the SPF for stop payment orders, and search the WHFF for warnings before approving this transaction.
11 21 00	✓	✓	✓	✓		Cash transit check (NOW) — The disbursement of funds to a customer in exchange for a check they have presented to our institution for payment. BASE24-teller uses the check writer's account number to check the PBF for sufficient funds, search the SPF for stop payment orders, and search the WHFF for warnings before approving this transaction.
12 00 00						Cash official check (log only) — The disbursement of funds to a customer in exchange for an official check they have presented to our institution for payment. This transaction is logged by BASE24-teller and is not verified against or used to update any BASE24 records.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
12 01 00	✓	✓			✓	Cash official check (DDA) — The disbursement of funds to a customer in exchange for an official check they have presented to our institution for payment. BASE24-teller uses the check writer's account number to check the PBF for sufficient funds, search the SPF for stop payment orders, and search the WHFF for warnings before approving this transaction.
12 21 00	✓	✓	✓	✓		Cash official check (NOW) — The disbursement of funds to a customer in exchange for an official check they have presented to our institution for payment. BASE24-teller uses the check writer's account number to check the PBF for sufficient funds, search the SPF for stop payment orders, and search the WHFF for warnings before approving this transaction.
13 00 00						Cash CD (log only) — The disbursement of funds to a customer in exchange for a certificate of deposit (CD) they have presented to our institution for payment. This transaction is logged by BASE24-teller and is not verified against or used to update any BASE24 records.
14 00 00						Cash bond (log only) — The disbursement of funds to a customer in exchange for a bond they have presented to our institution for payment. This transaction is logged by BASE24-teller and is not verified against or used to update any BASE24 records.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
15 00 00						Cash coupon (log only) — The disbursement of funds to a customer in exchange for a coupon they have presented to our institution for payment. This transaction is logged by BASE24-teller and is not verified against or used to update any BASE24 records.
17 00 00						Cash affiliate/correspondent check (log only) — The disbursement of funds to a customer in exchange for a check drawn on an affiliate or correspondent institution that they have presented to our institution for payment. This transaction is logged by BASE24-teller and is not verified against or used to update any BASE24 records.
17 01 00	✓	✓			✓	Cash affiliate/correspondent check (DDA) — The disbursement of funds to a customer in exchange for a check drawn on an affiliate or correspondent institution that they have presented to our institution for payment. BASE24-teller uses the check writer's account number to check the PBF for sufficient funds, search the SPF for stop payment orders, and search the WHFF for warnings before approving this transaction.
17 21 00	✓	✓	✓	✓		Cash affiliate/correspondent check (NOW) — The disbursement of funds to a customer in exchange for a check drawn on an affiliate or correspondent institution that they have presented to our institution for payment. BASE24-teller uses the check writer's account number to check the PBF for sufficient funds, search the SPF for stop payment orders, and search the WHFF for warnings before approving this transaction.

Support Key:

OL = Overdraft limit

PB = Passbooks

PH = Preauthorized hold checks

OP = Overdraft protection

AP = Automatic passbook printing

Miscellaneous Debit Transactions

BASE24-teller supports the miscellaneous debit transactions shown in the table below.

The customer's presence typically distinguishes between miscellaneous debit and debit memo post transactions. Miscellaneous debit transactions are treated as customer-initiated transactions and update the PBF withdrawal date and amount.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
16 00 00						Miscellaneous debit (log only) — A debit transaction that is logged by BASE24 and does not update any BASE24 records.
16 01 00					✓	Miscellaneous debit from DDA — A disbursement of funds that can be made from a checking account for a variety of reasons. This transaction is processed without checking the PBF overdraft limit amount.
16 11 00			✓	✓	✓	Miscellaneous debit from savings — A disbursement of funds that can be made from a savings account for a variety of reasons. This transaction is processed without checking the PBF overdraft limit amount.
16 12 00			✓	✓		Miscellaneous debit from IRA — A disbursement of funds that can be made from an IRA account for a variety of reasons. This transaction is processed without checking the PBF overdraft limit amount.
16 21 00			✓	✓		Miscellaneous debit from NOW — A disbursement of funds that can be made from a NOW account for a variety of reasons. This transaction is processed without checking the PBF overdraft limit amount.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
16 31 00					✓	Miscellaneous debit from credit card — A disbursement of funds that can be posted to a credit card account for a variety of reasons. This transaction is processed without checking the PBF overdraft limit amount.
16 32 00						Miscellaneous debit from credit line — A disbursement of funds that can be posted to a line of credit account for a variety of reasons. This transaction is processed without checking the PBF overdraft limit amount.
16 41 00			✓	✓		Miscellaneous debit from installment loan — A debit transaction that can be posted to an installment loan account for a variety of reasons. This transaction does not update PBF installment loan account balances.
16 42 00			✓	✓		Miscellaneous debit from mortgage loan — A debit transaction that can be posted to a mortgage loan account for a variety of reasons. This transaction does not update PBF mortgage loan account balances.
16 43 00			✓	✓		Miscellaneous debit from commercial loan — A debit transaction that can be posted to a commercial loan account for a variety of reasons. This transaction does not update PBF commercial loan account balances.

Support Key:

OL = Overdraft limit

PB = Passbooks

PH = Preauthorized hold checks

OP = Overdraft protection

AP = Automatic passbook printing

Debit Memo Post Transactions

BASE24-teller supports the debit memo post transactions shown in the table below.

The customer's presence typically distinguishes between miscellaneous debit and debit memo post transactions. Debit memo post transactions are treated as being performed without the customer present, so no PBF accumulators or statuses are updated for any of these transactions.

Note: Cash affiliate/correspondent check transactions (transaction code 17) are included with check cashing transactions earlier in this appendix.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
18 01 00					✓	Debit memo post from DDA — A noncash transaction that decreases the balance of a checking account. This transaction is processed without checking the PBF overdraft limit amount.
18 11 00			✓	✓	✓	Debit memo post from savings — A noncash transaction that decreases the balance of a savings account. This transaction is processed without checking the PBF overdraft limit amount.
18 12 00			✓	✓		Debit memo post from IRA — A noncash transaction that decreases the balance of an IRA account. This transaction is processed without checking the PBF overdraft limit amount.
18 21 00			✓	✓		Debit memo post from NOW — A noncash transaction that decreases the balance of a NOW account. This transaction is processed without checking the PBF overdraft limit amount.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
18 31 00					✓	Debit memo post from credit card — A noncash transaction that decreases the available credit balance and increases the credit balance of a credit card account. This transaction is processed without checking the PBF overdraft limit amount.
18 32 00						Debit memo post from credit line — A noncash transaction that decreases the available credit balance and increases the credit balance of a line of credit account. This transaction is processed without checking the PBF overdraft limit amount.
18 41 00			✓	✓		Debit memo post from installment loan — A noncash debit transaction involving an installment loan account. This transaction does not update PBF installment loan account balances and is processed without checking the PBF overdraft limit amount.
18 42 00			✓	✓		Debit memo post from mortgage loan — A noncash debit transaction involving a mortgage loan account. This transaction does not update PBF mortgage loan accounts and is processed without checking the PBF overdraft limit amount.
18 43 00			✓	✓		Debit memo post from commercial loan — A noncash debit transaction involving a commercial loan account. This transaction does not update PBF commercial loan accounts and is processed without checking the PBF overdraft limit amount.

Support Key:

OL = Overdraft limit

OP = Overdraft protection

PB = Passbooks

AP = Automatic passbook printing

PH = Preauthorized hold checks

Regular Deposit Transactions

BASE24-teller supports the regular deposit transactions shown in the table below.

The PBF overdraft limit amount is checked using the difference between the cash out and cash in amounts specified in the transaction message when the cash out amount is greater than the cash in amount.

All regular deposit transactions can be performed with a deposit-only (BD) card.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
20 00 01	✓					Deposit to DDA — A deposit of funds to a customer's checking account.
20 00 11	✓		✓	✓		Deposit to savings — A deposit of funds to a customer's savings account.
20 00 12	✓		✓	✓		Deposit to IRA — A deposit of funds to a customer's IRA account.
20 00 13	✓		✓	✓		Deposit to CD — A deposit of funds to a customer's certificate of deposit (CD) account.
20 00 21	✓		✓	✓		Deposit to NOW — A deposit of funds to a customer's NOW account.

Support Key:

OL = Overdraft limit

PB = Passbooks

PH = Preauthorized hold checks

OP = Overdraft protection

AP = Automatic passbook printing

Split Deposit Transactions

BASE24-teller supports the split deposit transactions shown in the table below.

Any cash in amount specified in the message is applied to the first account unless the cash in amount exceeds the first account's deposit amount. The second account receives the cash in excess of the amount being deposited in the first account. The cash out amount specified in the message is applied to the first account.

The PBF overdraft limit amount for the first account is checked using the difference between the cash out and cash in amounts for that account when the cash out amount is greater than the cash in amount. The PBF overdraft limit for the second account is not checked because its cash out amount is always zero. Refer to sections 5 and 6 of this manual for additional information.

All split deposit transactions can be performed with a deposit-only (BD) card as long as the cash out amount is zero.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
21 01 01	✓					Split deposit to DDA and DDA — A deposit of funds to two checking accounts.
21 01 11	✓		✓			Split deposit to DDA and savings — A deposit of funds to a checking account and a savings account. Only the savings account can have a passbook.
21 01 12	✓		✓			Split deposit to DDA and IRA — A deposit of funds to a checking account and an IRA account. Only the IRA account can have a passbook.
21 01 13	✓		✓			Split deposit to DDA and CD — A deposit of funds to a checking account and a certificate of deposit (CD) account. Only the CD account can have a passbook.
21 01 21	✓		✓			Split deposit to DDA and NOW — A deposit of funds to a checking account and a NOW account. Only the NOW account can have a passbook.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
21 11 01	✓		✓			Split deposit to savings and DDA — A deposit of funds to a savings account and a checking account. Only the savings account can have a passbook.
21 11 11	✓		✓			Split deposit to savings and savings — A deposit of funds to two savings accounts.
21 11 12	✓		✓			Split deposit to savings and IRA — A deposit of funds to a savings account and an IRA account.
21 11 13	✓		✓			Split deposit to savings and CD — A deposit of funds to a savings account and a certificate of deposit (CD) account.
21 11 21	✓		✓			Split deposit to savings and NOW — A deposit of funds to a savings account and a NOW account.
21 12 01	✓		✓			Split deposit to IRA and DDA — A deposit of funds to an IRA account and a checking account. Only the IRA account can have a passbook.
21 12 11	✓		✓			Split deposit to IRA and savings — A deposit of funds to an IRA account and a savings account.
21 12 12	✓		✓			Split deposit to IRA and IRA — A deposit of funds to two IRA accounts.
21 12 13	✓		✓			Split deposit to IRA and CD — A deposit of funds to an IRA account and a certificate of deposit (CD) account.
21 12 21	✓		✓			Split deposit to IRA and NOW — A deposit of funds to an IRA account and a NOW account.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
21 13 01	✓		✓			Split deposit to CD and DDA — A deposit of funds to a certificate of deposit (CD) account and a checking account. Only the CD account can have a passbook.
21 13 11	✓		✓			Split deposit to CD and savings — A deposit of funds to a certificate of deposit (CD) account and a savings account.
21 13 12	✓		✓			Split deposit to CD and IRA — A deposit of funds to a certificate of deposit (CD) account and an IRA account.
21 13 13	✓		✓			Split deposit to CD and CD — A deposit of funds to two certificate of deposit (CD) accounts.
21 13 21	✓		✓			Split deposit to CD and NOW — A deposit of funds to a certificate of deposit (CD) account and a NOW account.
21 21 01	✓		✓			Split deposit to NOW and DDA — A deposit of funds to a NOW account and a checking account. Only the NOW account can have a passbook.
21 21 11	✓		✓			Split deposit to NOW and savings — A deposit of funds to a NOW account and a savings account.
21 21 12	✓		✓			Split deposit to NOW and IRA — A deposit of funds to a NOW account and an IRA account.
21 21 13	✓		✓			Split deposit to NOW and CD — A deposit of funds to a NOW account and a certificate of deposit (CD) account.
21 21 21	✓		✓			Split deposit to NOW and NOW — A deposit of funds to two NOW accounts.

Support Key:

OL = Overdraft limit

OP = Overdraft protection

PB = Passbooks

AP = Automatic passbook printing

PH = Preauthorized hold checks

Miscellaneous Credit Transactions

BASE24-teller supports the miscellaneous credit transactions shown in the table below.

The customer's presence typically distinguishes between miscellaneous credit and credit memo post transactions. Miscellaneous credit transactions are treated as customer-initiated transactions and update the PBF deposit date and amount.

All miscellaneous credit transactions can be performed with a deposit-only (BD) card.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
22 00 00						Miscellaneous credit (log only) — A miscellaneous credit transaction that is logged by BASE24 and does not update any BASE24 records.
22 00 01						Miscellaneous credit to DDA — A cash receipt that can be applied to a checking account for a variety of reasons.
22 00 11			✓	✓		Miscellaneous credit to savings — A cash receipt that can be applied to a savings account for a variety of reasons.
22 00 12			✓	✓		Miscellaneous credit to IRA — A cash receipt that can be applied to an IRA account for a variety of reasons.
22 00 21			✓	✓		Miscellaneous credit to NOW — A cash receipt that can be applied to a NOW account for a variety of reasons.
22 00 31						Miscellaneous credit to credit card — A cash receipt that can be applied to a credit card account for a variety of reasons.
22 00 32						Miscellaneous credit to credit line — A cash receipt that can be applied to a line of credit account for a variety of reasons.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
22 00 41			✓	✓		Miscellaneous credit to installment loan — A cash receipt that can be applied to an installment loan account for a variety of reasons. This transaction does not update PBF installment loan account balances.
22 00 42			✓	✓		Miscellaneous credit to mortgage loan — A cash receipt that can be applied to a mortgage loan account for a variety of reasons. This transaction does not update PBF mortgage loan account balances.
22 00 43			✓	✓		Miscellaneous credit to commercial loan — A cash receipt that can be applied to a commercial loan account for a variety of reasons. This transaction does not update PBF commercial loan account balances.

Support Key:

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PH = Preauthorized hold checks

Credit Memo Post Transactions

BASE24-teller supports the credit memo post transactions shown in the table below.

The customer's presence typically distinguishes between miscellaneous credit and credit memo post transactions. Credit memo post transactions are treated as being performed without the customer present, so no PBF accumulators or statuses are updated for any of these transactions.

All credit memo transactions can be performed with a deposit-only (BD) card.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
23 00 01						Credit memo post to DDA — A noncash transaction that increases the balance of a checking account.
23 00 11			✓	✓		Credit memo post to savings — A noncash transaction that increases the balance of a savings account.
23 00 12			✓	✓		Credit memo post to IRA — A noncash transaction that increases the balance of an IRA account.
23 00 21			✓	✓		Credit memo post to NOW — A noncash transaction that increases the balance of a NOW account.
23 00 31						Credit memo post to credit card — A noncash transaction that increases the available credit balance and decreases the credit balance of a credit card account.
23 00 32						Credit memo post to credit line — A noncash transaction that increases the available credit balance and decreases the credit balance of a line of credit account.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
23 00 41			✓	✓		Credit memo post to installment loan — A noncash transaction involving an installment loan account. This transaction does not update PBF installment loan account balances.
23 00 42			✓	✓		Credit memo post to mortgage loan — A noncash transaction involving a mortgage loan account. This transaction does not update PBF mortgage loan account balances.
23 00 43			✓	✓		Credit memo post to commercial loan — A noncash transaction involving a commercial loan account. This transaction does not update PBF commercial loan account balances.

Support Key:

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Positive Balance File Inquiry Transactions

BASE24-teller supports the Positive Balance File (PBF) inquiry transactions shown in the table below. Inquiries do not affect PBF balances, accumulators, or statuses.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
30 00 01					✓	PBF inquiry on DDA — A request for information contained on the PBF record of a checking account.
30 00 11					✓	PBF inquiry on savings — A request for information contained on the PBF record of a savings account.
30 00 12						PBF inquiry on IRA — A request for information contained on the PBF record of an IRA account.
30 00 13						PBF inquiry on CD — A request for information contained on the PBF record of a certificate of deposit (CD) account.
30 00 21						PBF inquiry on NOW — A request for information contained on the PBF record of a NOW account.
30 00 31					✓	PBF inquiry on credit card — A request for information contained on the PBF record of a credit card account.
30 00 32						PBF inquiry on credit line — A request for information contained on the PBF record of a line of credit account.
30 00 41						PBF inquiry on installment loan — A request for information contained on the PBF record of an installment loan account.
30 00 42						PBF inquiry on mortgage loan — A request for information contained on the PBF record of a mortgage loan account.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
30 00 43						PBF inquiry on commercial loan — A request for information contained on the PBF record of a commercial loan account.
31 00 01					✓	PBF short inquiry on DDA — A request for the available and ledger balances from the PBF record of a checking account.
31 00 11					✓	PBF short inquiry on savings — A request for the available and ledger balances from the PBF record of a savings account.
31 00 12						PBF short inquiry on IRA — A request for the available and ledger balances from the PBF record of an IRA account.
31 00 13						PBF short inquiry on CD — A request for the available and ledger balances from the PBF record of a certificate of deposit (CD) account.
31 00 21						PBF short inquiry on NOW — A request for the available and ledger balances from the PBF record of a NOW account.
31 00 31					✓	PBF short inquiry on credit card — A request for the available and ledger balances from the PBF record of a credit card account.
31 00 32						PBF short inquiry on credit line — A request for the available and ledger balances from the PBF record of a line of credit account.
31 00 41						PBF short inquiry on installment loan — A request for the available and ledger balances from the PBF record of an installment loan account.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
31 00 42						PBF short inquiry on mortgage loan — A request for the available and ledger balances from the PBF record of a mortgage loan account.
31 00 43						PBF short inquiry on commercial loan — A request for the available and ledger balances from the PBF record of a commercial loan account.

Support Key:

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Stop Payment File Inquiry Transactions

BASE24-teller supports the following Stop Payment File (SPF) inquiry transactions.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
32 00 01						SPF inquiry on DDA — A request for stop payment information about a checking account.
32 00 21						SPF inquiry on NOW — A request for stop payment information about a NOW account.

Support Key:

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Cardholder Authorization File Inquiry Transactions

BASE24-teller supports the following Cardholder Authorization File (CAF) inquiry transactions. These transactions must be initiated with information read from the magnetic stripe on a card. BASE24-teller does not perform the expiration date or card status checks on CAF inquiry and update transactions; permitting a teller to review the information and make corrections, if necessary.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
33 00 00						CAF inquiry on all accounts — A request for account information contained on a CAF record regardless of account type.
33 00 01						CAF inquiry on DDA — A request for information about checking accounts contained on a CAF record.
33 00 11						CAF inquiry on savings — A request for information about savings accounts contained on a CAF record.
33 00 12						CAF inquiry on IRA — A request for information about IRA accounts contained on a CAF record.
33 00 13						CAF inquiry on CD — A request for information about certificate of deposit (CD) accounts contained on a CAF record.
33 00 21						CAF inquiry on NOW — A request for information about NOW accounts contained on a CAF record.
33 00 31						CAF inquiry on credit card — A request for information about credit card accounts contained on a CAF record.
33 00 32						CAF inquiry on credit line — A request for information about line of credit accounts contained on a CAF record.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
33 00 41						CAF inquiry on installment loan — A request for information about installment loan accounts contained on a CAF record.
33 00 42						CAF inquiry on mortgage loan — A request for information about mortgage loan accounts contained on a CAF record.
33 00 43						CAF inquiry on commercial loan — A request for information about commercial loan accounts contained on a CAF record.

Support Key:

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AP = Automatic passbook printing

No Book File Inquiry Transactions

BASE24-teller supports the No Book File (NBF) inquiry transactions shown in the table below.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
34 00 11			✓			NBF inquiry on savings — A request for entries currently stored in the NBF for a savings account.
34 00 12			✓			NBF inquiry on IRA — A request for entries currently stored in the NBF for an IRA account.
34 00 13			✓			NBF inquiry on CD — A request for entries currently stored in the NBF for a certificate of deposit (CD) account.
34 00 21			✓			NBF inquiry on NOW — A request for entries currently stored in the NBF for a NOW account.
34 00 41			✓			NBF inquiry on installment loan — A request for entries currently stored in the NBF for an installment loan account.
34 00 42			✓			NBF inquiry on mortgage loan — A request for entries currently stored in the NBF for a mortgage loan account.
34 00 43			✓			NBF inquiry on commercial loan — A request for entries currently stored in the NBF for a commercial loan account.

Support Key:

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PB = Passbooks

PH = Preauthorized hold checks

OP = Overdraft protection

AP = Automatic passbook printing

Passbook Print Transactions

BASE24-teller supports the passbook print transactions shown in the table below.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
35 00 11			✓			Passbook print for savings — A request for unprinted entries currently stored in the No Book File (NBF) for a savings account.
35 00 12			✓			Passbook print for IRA — A request for unprinted entries currently stored in the NBF for an IRA account.
35 00 13			✓			Passbook print for CD — A request for unprinted entries currently stored in the NBF for a certificate of deposit (CD) account.
35 00 21			✓			Passbook print for NOW — A request for unprinted entries currently stored in the NBF for a NOW account.
35 00 41			✓			Passbook print for installment loan — A request for unprinted entries currently stored in the NBF for an installment loan account.
35 00 42			✓			Passbook print for mortgage loan — A request for unprinted entries currently stored in the NBF for a mortgage loan account.
35 00 43			✓			Passbook print for commercial loan — A request for unprinted entries currently stored in the NBF for a commercial loan account.

Support Key:

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AP = Automatic passbook printing

Passbook Reprint Transactions

BASE24-teller supports the passbook reprint transactions shown in the table below.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
36 00 11			✓			Passbook reprint for savings — A request for previously printed entries currently stored in the NBF for a savings account.
36 00 12			✓			Passbook reprint for IRA — A request for previously printed entries currently stored in the NBF for an IRA account.
36 00 13			✓			Passbook reprint for CD — A request for previously printed entries currently stored in the NBF for a certificate of deposit (CD) account.
36 00 21			✓			Passbook reprint for NOW — A request for previously printed entries currently stored in the NBF for a NOW account.
36 00 41			✓			Passbook reprint for installment loan — A request for previously printed entries currently stored in the NBF for an installment loan account.
36 00 42			✓			Passbook reprint for mortgage loan — A request for previously printed entries currently stored in the NBF for a mortgage loan account.
36 00 43			✓			Passbook reprint for commercial loan — A request for previously printed entries currently stored in the NBF for a commercial loan account.

Support Key:

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Warning/Hold/Float File Inquiry Transactions

BASE24-teller supports the Warning/Hold/Float File (WHFF) inquiry transactions shown in the table below.

The WHFF contains warning, hold, and deposit float information. WHFF warning entries force the BASE24-teller Authorization process to decline transactions that might otherwise be approved. WHFF hold entries decrease the PBF available funds balance and increase the PBF amount on hold balance. WHFF deposit float entries increase the PBF available funds balance but do not affect the PBF amount on hold balance.

Each WHFF inquiry transaction permits the teller to specify the type of record (i.e., warning, hold, deposit float, or all three) for which he or she is searching. If no type is specified, all records on the WHFF can be returned for the account in question.

The BASE24-teller Authorization process does not check the PBF overdraft limit when adding hold entries to the WHFF or removing deposit float entries from the WHFF.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
37 00 01						WHFF inquiry on DDA — A request for warning, hold, and deposit float information for a checking account.
37 00 11						WHFF inquiry on savings — A request for warning, hold, and deposit float information for a savings account.
37 00 12						WHFF inquiry on IRA — A request for warning, hold, and deposit float information for an IRA account.
37 00 13						WHFF inquiry on CD — A request for warning, hold, and deposit float information for a certificate of deposit (CD) account.
37 00 21						WHFF inquiry on NOW — A request for warning, hold, and deposit float information for a NOW account.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
37 00 31						WHFF inquiry on credit card — A request for warning, hold, and deposit float information for a credit card account.
37 00 32						WHFF inquiry on credit line — A request for warning, hold, and deposit float information for a line of credit account.
37 00 41						WHFF inquiry on installment loan — A request for warning, hold, and deposit float information for an installment loan account.
37 00 42						WHFF inquiry on mortgage loan — A request for warning, hold, and deposit float information for a mortgage loan account.
37 00 43						WHFF inquiry on commercial loan — A request for warning, hold, and deposit float information for a commercial loan account.

Support Key:

OL = Overdraft limit

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Transfer Transactions

BASE24-teller supports the transfer transactions shown in the table below.

Overdraft limits, overdraft protection, and preauthorized holds apply to the *from* account, and are checked using the transaction amount. Overdraft limits, overdraft protection, and preauthorized holds are not checked for the *to* account because funds are not being disbursed from that account.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
40 01 01	✓	✓			✓	Transfer from DDA to DDA — The withdrawal of funds from one checking account and credit of funds to another checking account.
40 01 11	✓	✓	✓		✓	Transfer from DDA to savings — The withdrawal of funds from a checking account and credit of funds to a savings account. Only the savings account can have a passbook.
40 01 12	✓	✓	✓		✓	Transfer from DDA to IRA — The withdrawal of funds from a checking account and credit of funds to an IRA account. Only the IRA account can have a passbook.
40 01 13	✓	✓	✓		✓	Transfer from DDA to CD — The withdrawal of funds from a checking account and credit of funds to a certificate of deposit (CD) account. Only the CD account can have a passbook.
40 01 21	✓	✓	✓		✓	Transfer from DDA to NOW — The withdrawal of funds from a checking account and credit of funds to a NOW account. Only the NOW account can have a passbook.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
40 11 01	✓	✓	✓		✓	Transfer from savings to DDA — The withdrawal of funds from a savings account and credit of funds to a checking account. Only the savings account can have a passbook.
40 11 11	✓	✓	✓		✓	Transfer from savings to savings — The withdrawal of funds from one savings account and credit of funds to another savings account.
40 11 12	✓	✓	✓		✓	Transfer from savings to IRA — The withdrawal of funds from a savings account and credit of funds to an IRA account.
40 11 13	✓	✓	✓		✓	Transfer from savings to CD — The withdrawal of funds from a savings account and credit of funds to a certificate of deposit (CD) account.
40 11 21	✓	✓	✓		✓	Transfer from savings to NOW — The withdrawal of funds from a savings account and credit of funds to a NOW account.
40 21 01	✓	✓	✓			Transfer from NOW to DDA — The withdrawal of funds from a NOW account and credit of funds to a checking account. Only the NOW account can have a passbook.
40 21 11	✓	✓	✓			Transfer from NOW to savings — The withdrawal of funds from a NOW account and credit of funds to a savings account.
40 21 12	✓	✓	✓			Transfer from NOW to IRA — The withdrawal of funds from a NOW account and credit of funds to an IRA account.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
40 21 13	✓	✓	✓			Transfer from NOW to CD — The withdrawal of funds from a NOW account and credit of funds to a certificate of deposit (CD) account.
40 21 21	✓	✓	✓			Transfer from NOW to NOW — The withdrawal of funds from one NOW account and credit of funds to another NOW account.

Support Key:

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Payment Transactions

BASE24-teller supports the payment transactions shown in the table below.

When the transaction includes a *from* account, overdraft limits, overdraft protection, and preauthorized holds apply to the *from* account and are checked using the transaction amount. Even though all but four payment transactions (500000, 500100, 501100, and 502100) involve a *to* account, overdraft limits, overdraft protection, and preauthorized holds are not checked for the *to* account because funds are not being disbursed from that account.

Only a subset of these payment transactions can be performed with a deposit-only (BD) card. Payment transactions involving a withdrawal of any kind (i.e., the *from* account type is a value other than 00) cannot be performed with a deposit-only card even though part of the transaction is a deposit. Transactions that are valid with a deposit-only card are identified within the descriptions.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
50 00 00						Payment (log only) — A payment transaction that is logged by BASE24 and does not update any BASE24 records.
50 00 31						Payment to credit card — A receipt of funds that increases the available credit balance and decreases the credit balance of a credit card account. This transaction is valid with a deposit-only card.
50 00 32						Payment to credit line — A receipt of funds that increases the available credit balance and decreases the credit balance of a line of credit account. This transaction is valid with a deposit-only card.
50 00 41			✓	✓		Payment to installment loan — A payment transaction involving an installment loan account. This transaction is valid with a deposit-only card, but does not update PBF installment loan account balances.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
50 00 42			✓	✓		Payment to mortgage loan — A payment transaction involving a mortgage loan account. This transaction is valid with a deposit-only card, but does not update PBF mortgage loan account balances.
50 00 43			✓	✓		Payment to commercial loan — A payment transaction involving a commercial loan account. This transaction is valid with a deposit-only card, but does not update PBF commercial loan account balances.
50 00 50						Payment to utility — A receipt of funds that increases the balance of a utility account. This transaction is valid with a deposit-only card.
50 00 51						Payment to utility 1 — A receipt of funds that increases the balance of a utility 1 account. This transaction is valid with a deposit-only card.
50 00 52						Payment to utility 2 — A receipt of funds that increases the balance of a utility 2 account. This transaction is valid with a deposit-only card.
50 00 53						Payment to utility 3 — A receipt of funds that increases the balance of a utility 3 account. This transaction is valid with a deposit-only card.
50 00 54						Payment to utility 4 — A receipt of funds that increases the balance of a utility 4 account. This transaction is valid with a deposit-only card.
50 00 55						Payment to utility 5 — A receipt of funds that increases the balance of a utility 5 account. This transaction is valid with a deposit-only card.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
50 01 00	✓	✓			✓	Payment from DDA — A withdrawal of funds from a customer's checking account.
50 01 31	✓	✓			✓	Payment from DDA to credit card — The withdrawal of funds from a checking account and payment of funds to a credit card account. The credit card account's available credit balance is increased and its credit balance is decreased.
50 01 32	✓	✓			✓	Payment from DDA to credit line — The withdrawal of funds from a checking account and credit of funds to a line of credit account. The line of credit account's available credit balance is increased and its credit balance is decreased.
50 01 41	✓	✓	✓		✓	Payment from DDA to installment loan — The withdrawal of funds from a checking account and payment of funds to an installment loan account. Only the installment loan account can have a passbook, but this transaction updates only the checking account balances in the PBF.
50 01 42	✓	✓	✓		✓	Payment from DDA to mortgage loan — The withdrawal of funds from a checking account and payment of funds to a mortgage loan account. Only the mortgage loan account can have a passbook, but this transaction updates only the checking account balances in the PBF.
50 01 43	✓	✓	✓		✓	Payment from DDA to commercial loan — The withdrawal of funds from a checking account and payment of funds to a commercial loan account. Only the commercial loan account can have a passbook, but this transaction updates only the checking account balances in the PBF.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
50 01 50	✓	✓			✓	Payment from DDA to utility — The withdrawal of funds from a checking account and payment of funds to a utility account.
50 01 51	✓	✓			✓	Payment from DDA to utility 1 — The withdrawal of funds from a checking account and payment of funds to a utility 1 account.
50 01 52	✓	✓			✓	Payment from DDA to utility 2 — The withdrawal of funds from a checking account and payment of funds to a utility 2 account.
50 01 53	✓	✓			✓	Payment from DDA to utility 3 — The withdrawal of funds from a checking account and payment of funds to a utility 3 account.
50 01 54	✓	✓			✓	Payment from DDA to utility 4 — The withdrawal of funds from a checking account and payment of funds to a utility 4 account.
50 01 55	✓	✓			✓	Payment from DDA to utility 5 — The withdrawal of funds from a checking account and payment of funds to a utility 5 account.
50 11 00	✓	✓	✓	✓	✓	Payment from savings — A cash withdrawal from a customer's savings account.
50 11 31	✓	✓	✓		✓	Payment from savings to credit card — The withdrawal of funds from a savings account and payment of funds to a credit card account. The credit card account's available credit balance is increased and its credit balance is decreased. Only the savings account can have a passbook.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
50 11 32	✓	✓	✓		✓	Payment from savings to credit line — The withdrawal of funds from a savings account and payment of funds to a line of credit account. The line of credit account's available credit balance is increased and its credit balance is decreased. Only the savings account can have a passbook.
50 11 41	✓	✓	✓		✓	Payment from savings to installment loan — The withdrawal of funds from a savings account and payment of funds to an installment loan account. This transaction does not update PBF installment loan account balances.
50 11 42	✓	✓	✓		✓	Payment from savings to mortgage loan — The withdrawal of funds from a savings account and payment of funds to a mortgage loan account. This transaction does not update PBF mortgage loan account balances.
50 11 43	✓	✓	✓		✓	Payment from savings to commercial loan — The withdrawal of funds from a savings account and payment of funds to a commercial loan account. This transaction does not update PBF commercial loan account balances.
50 11 50	✓	✓	✓		✓	Payment from savings to utility — The withdrawal of funds from a savings account and payment of funds to a utility account. Only the savings account can have a passbook.
50 11 51	✓	✓	✓		✓	Payment from savings to utility 1 — The withdrawal of funds from a savings account and payment of funds to a utility 1 account. Only the savings account can have a passbook.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
50 11 52	✓	✓	✓		✓	Payment from savings to utility 2 — The withdrawal of funds from a savings account and payment of funds to a utility 2 account. Only the savings account can have a passbook.
50 11 53	✓	✓	✓		✓	Payment from savings to utility 3 — The withdrawal of funds from a savings account and payment of funds to a utility 3 account. Only the savings account can have a passbook.
50 11 54	✓	✓	✓		✓	Payment from savings to utility 4 — The withdrawal of funds from a savings account and payment of funds to a utility 4 account. Only the savings account can have a passbook.
50 11 55	✓	✓	✓		✓	Payment from savings to utility 5 — The withdrawal of funds from a savings account and payment of funds to a utility 5 account. Only the savings account can have a passbook.
50 21 00	✓	✓	✓	✓		Payment from NOW — A cash withdrawal from a customer's NOW account.
50 21 31	✓	✓	✓			Payment from NOW to credit card — The withdrawal of funds from a NOW account and payment of funds to a credit card account. The credit card account's available credit balance is increased and its credit balance is decreased. Only the NOW account can have a passbook.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
50 21 32	✓	✓	✓			Payment from NOW to credit line — The withdrawal of funds from a NOW account and credit of funds to a line of credit account. The line of credit account's available credit balance is increased and its credit balance is decreased. Only the NOW account can have a passbook.
50 21 41	✓	✓	✓			Payment from NOW to installment loan — The withdrawal of funds from a NOW account and payment of funds to an installment loan account. This transaction does not update PBF installment loan account balances.
50 21 42	✓	✓	✓			Payment from NOW to mortgage loan — The withdrawal of funds from a NOW account and payment of funds to a mortgage loan account. This transaction does not update PBF mortgage loan account balances.
50 21 43	✓	✓	✓			Payment from NOW to commercial loan — The withdrawal of funds from a NOW account and payment of funds to a commercial loan account. This transaction does not update PBF commercial loan account balances.
50 21 50	✓	✓	✓			Payment from NOW to utility — The withdrawal of funds from a NOW account and payment of funds to a utility account. Only the NOW account can have a passbook.
50 21 51	✓	✓	✓			Payment from NOW to utility 1 — The withdrawal of funds from a NOW account and payment of funds to a utility 1 account. Only the NOW account can have a passbook.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
50 21 52	✓	✓	✓			Payment from NOW to utility 2 — The withdrawal of funds from a NOW account and payment of funds to a utility 2 account. Only the NOW account can have a passbook.
50 21 53	✓	✓	✓			Payment from NOW to utility 3 — The withdrawal of funds from a NOW account and payment of funds to a utility 3 account. Only the NOW account can have a passbook.
50 21 54	✓	✓	✓			Payment from NOW to utility 4 — The withdrawal of funds from a NOW account and payment of funds to a utility 4 account. Only the NOW account can have a passbook.
50 21 55	✓	✓	✓			Payment from NOW to utility 5 — The withdrawal of funds from a NOW account and payment of funds to a utility 5 account. Only the NOW account can have a passbook.

Support Key:

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PH = Preauthorized hold checks

Purchase Transactions

BASE24-teller supports the purchase transactions shown in the table below.

When the purchase transaction includes a *from* account, the PBF overdraft limit amount in the *from* account is checked using the sum of the item being purchased plus the fee assessed with the transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
60 00 00						Purchase money order (log only) — A money order purchase transaction that is logged by BASE24 and does not update any BASE24 records.
60 01 00	✓	✓			✓	Purchase money order using DDA — The withdrawal of funds from a checking account to purchase a money order.
60 11 00	✓	✓	✓	✓	✓	Purchase money order using savings — The withdrawal of funds from a savings account to purchase a money order.
60 21 00	✓	✓	✓	✓		Purchase money order using NOW — The withdrawal of funds from a NOW account to purchase a money order.
60 31 00	✓				✓	Purchase money order using credit card — A cash advance from a credit card account to purchase a money order.
61 00 00						Purchase cashier's check (log only) — A cashier's check purchase transaction that is logged by BASE24 and does not update any BASE24 records.
61 01 00	✓	✓			✓	Purchase cashier's check using DDA — The withdrawal of funds from a checking account to purchase a cashier's check.
61 11 00	✓	✓	✓	✓	✓	Purchase cashier's check using savings — The withdrawal of funds from a savings account to purchase a cashier's check.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
61 21 00	✓	✓	✓	✓		Purchase cashier's check using NOW — The withdrawal of funds from a NOW account to purchase a cashier's check.
61 31 00	✓				✓	Purchase cashier's check using credit card — A cash advance from a credit card account to purchase a cashier's check.
62 00 00						Purchase traveler's checks (log only) — A traveler's check purchase transaction that is logged by BASE24 and does not update any BASE24 records.
62 01 00	✓	✓			✓	Purchase traveler's checks using DDA — The withdrawal of funds from a checking account to purchase traveler's checks.
62 11 00	✓	✓	✓	✓	✓	Purchase traveler's checks using savings — The withdrawal of funds from a savings account to purchase traveler's checks.
62 21 00	✓	✓	✓	✓		Purchase traveler's checks using NOW — The withdrawal of funds from a NOW account to purchase traveler's checks.
62 31 00	✓				✓	Purchase traveler's checks using credit card — A cash advance from a credit card account to purchase traveler's checks.
63 00 00						Purchase bank draft (log only) — A bank draft purchase transaction that is logged by BASE24 and does not update any BASE24 records.
63 01 00	✓	✓			✓	Purchase bank draft using DDA — The withdrawal of funds from a checking account to purchase a bank draft.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
63 11 00	✓	✓	✓	✓	✓	Purchase bank draft using savings — The withdrawal of funds from a savings account to purchase a bank draft.
63 21 00	✓	✓	✓	✓		Purchase bank draft using NOW — The withdrawal of funds from a NOW account to purchase a bank draft.
63 31 00	✓				✓	Purchase bank draft using credit card — A cash advance from a credit card account to purchase a bank draft.
64 00 00						Purchase certified check (log only) — A certified check purchase transaction that is logged by BASE24 and does not update any BASE24 records.
64 01 00	✓	✓			✓	Purchase certified check using DDA — The withdrawal of funds from a checking account to purchase a certified check.
64 11 00	✓	✓	✓	✓	✓	Purchase certified check using savings — The withdrawal of funds from a savings account to purchase a certified check.
64 21 00	✓	✓	✓	✓		Purchase certified check using NOW — The withdrawal of funds from a NOW account to purchase a certified check.
64 31 00	✓				✓	Purchase certified check using credit card — A cash advance from a credit card account to purchase a certified check.
65 00 00						Purchase bond (log only) — A bond purchase transaction that is logged by BASE24 and does not update any BASE24 records.
65 01 00	✓	✓			✓	Purchase bond using DDA — The withdrawal of funds from a checking account to purchase a bond.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
65 11 00	✓	✓	✓	✓	✓	Purchase bond using savings — The withdrawal of funds from a savings account to purchase a bond.
65 21 00	✓	✓	✓	✓		Purchase bond using NOW — The withdrawal of funds from a NOW account to purchase a bond.
65 31 00	✓				✓	Purchase bond using credit card — A cash advance from a credit card account to purchase a bond.
66 00 00						Purchase miscellaneous (log only) — A miscellaneous purchase transaction that is logged by BASE24 and does not update any BASE24 records.
66 01 00	✓	✓			✓	Purchase miscellaneous using DDA — The withdrawal of funds from a checking account for a miscellaneous purchase.
66 11 00	✓	✓	✓	✓	✓	Purchase miscellaneous using savings — The withdrawal of funds from a savings account for a miscellaneous purchase.
66 21 00	✓	✓	✓	✓		Purchase miscellaneous using NOW — The withdrawal of funds from a NOW account for a miscellaneous purchase.
66 31 00	✓				✓	Purchase miscellaneous using credit card — A cash advance from a credit card account for a miscellaneous purchase.

Support Key:

OL = Overdraft limit

PB = Passbooks

PH = Preauthorized hold checks

OP = Overdraft protection

AP = Automatic passbook printing

Cardholder Authorization File Status Update Transactions

BASE24-teller supports the CAF status update transactions shown in the table below. These transactions must be initiated with information read from the magnetic stripe on a card. BASE24-teller does not perform the expiration date or card status checks on CAF inquiry and update transactions; permitting a teller to review the information and make corrections, if necessary.

The card status transaction replaces the value in the CARD STATUS field in the CAF. The account status transactions replace the account status values in the CAF and PBF. In the following descriptions, account status refers to the STATUS (account status) field in the CAF and the ACCOUNT STATUS field in the PBF.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
73 00 00						Change CAF card status — A request to replace the value in the Card Status field of the CAF with the card status value contained in this transaction.
74 00 01						Change CAF/PBF DDA status — A request to replace the value in the account status fields of the CAF and PBF for a checking account with the account status value contained in this transaction.
74 00 11						Change CAF/PBF savings status — A request to replace the value in the account status fields of the CAF and PBF for a savings account with the account status value contained in this transaction.
74 00 12						Change CAF/PBF IRA status — A request to replace the value in the account status fields of the CAF and PBF for an IRA account with the account status value contained in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
74 00 13						Change CAF/PBF CD status — A request to replace the value in the account status fields of the CAF and PBF for a certificate of deposit (CD) account with the account status value contained in this transaction.
74 00 21						Change CAF/PBF NOW status — A request to replace the value in the account status fields of the CAF and PBF for a NOW account with the account status value contained in this transaction.
74 00 31						Change CAF/PBF credit card status — A request to replace the value in the account status fields of the CAF and PBF for a credit card account with the account status value contained in this transaction.
74 00 32						Change CAF/PBF credit line status — A request to replace the value in the account status fields of the CAF and PBF for a line of credit account with the account status value contained in this transaction.
74 00 41						Change CAF/PBF installment loan status — A request to replace the value in the account status fields of the CAF and PBF for an installment loan account with the account status value contained in this transaction.
74 00 42						Change CAF/PBF mortgage loan status — A request to replace the value in the account status fields of the CAF and PBF for a mortgage loan account with the account status value contained in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
74 00 43						Change CAF/PBF commercial loan status — A request to replace the value in the account status fields of the CAF and PBF for a commercial loan account with the account status value contained in this transaction.

Support Key:

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PIN Verification Transaction

BASE24-teller supports the following PIN verification transaction. This transaction must be initiated with information read from the magnetic stripe on a card.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
75 00 00						Verify PIN — A request to verify the accuracy of the PIN contained in this transaction.

Support Key:

OL = Overdraft limit

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OP = Overdraft protection

AP = Automatic passbook printing

Stop Payment File Update Transactions

BASE24-teller supports the following Stop Payment File (SPF) update transactions.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
80 00 01						Add stop payment to DDA — A request to add stop payment information for a checking account to the SPF.
80 00 21						Add stop payment to NOW — A request to add stop payment information for a NOW account to the SPF.
81 00 01						Delete stop payment from DDA — A request to delete stop payment information for a checking account from the SPF.
81 00 21						Delete stop payment from NOW — A request to delete stop payment information for a checking account from the SPF.

Support Key:

OL = Overdraft limit

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AP = Automatic passbook printing

Positive Balance File Account Status Update Transactions

BASE24-teller supports the following PBF account status update transactions.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
82 00 01						Change PBF account status for DDA — A request to replace the value in the ACCOUNT STATUS field of the PBF for a checking account with the account status value contained in this transaction.
82 00 11						Change PBF account status for savings — A request to replace the value in the ACCOUNT STATUS field of the PBF for a savings account with the account status value contained in this transaction.
82 00 12						Change PBF account status for IRA — A request to replace the value in the ACCOUNT STATUS field of the PBF for an IRA account with the account status value contained in this transaction.
82 00 13						Change PBF account status for CD — A request to replace the value in the ACCOUNT STATUS field of the PBF for a certificate of deposit (CD) account with the account status value contained in this transaction.
82 00 21						Change PBF account status for NOW — A request to replace the value in the ACCOUNT STATUS field of the PBF for a NOW account with the account status value contained in this transaction.
82 00 31						Change PBF account status for credit card — A request to replace the value in the ACCOUNT STATUS field of the PBF for a credit card account with the account status value contained in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
82 00 32						Change PBF account status for credit line — A request to replace the value in the ACCOUNT STATUS field of the PBF for a line of credit account with the account status value contained in this transaction.
82 00 41						Change PBF account status for installment loan — A request to replace the value in the ACCOUNT STATUS field of the PBF for an installment loan account with the account status value contained in this transaction.
82 00 42						Change PBF account status for mortgage loan — A request to replace the value in the ACCOUNT STATUS field of the PBF for a mortgage loan account with the account status value contained in this transaction.
82 00 43						Change PBF account status for commercial loan — A request to replace the value in the ACCOUNT STATUS field of the PBF for a commercial loan account with the account status value contained in this transaction.

Support Key:

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Positive Balance File Stop Pay/Warning Status Update Transactions

BASE24-teller supports the following transactions to update the value in the Stop Pay/Warning Status field of the PBF.

This flag can be manually updated in situations when the SPF or WHFF has been modified without updating this flag. The transaction codes used to add or delete SPF stop pay records or WHFF warning records can also set this flag.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
83 00 01						Change PBF stop pay/warning status for DDA — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for a checking account with the status value contained in this transaction.
83 00 11						Change PBF stop pay/warning status for savings — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for a savings account with the status value contained in this transaction.
83 00 12						Change PBF stop pay/warning status for IRA — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for an IRA account with the status value contained in this transaction.
83 00 13						Change PBF stop pay/warning status for CD — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for a certificate of deposit (CD) account with the status value contained in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
83 00 21						Change PBF stop pay/warning status for NOW — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for a NOW account with the status value contained in this transaction.
83 00 31						Change PBF stop pay/warning status for credit card — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for a credit card account with the status value contained in this transaction.
83 00 32						Change PBF stop pay/warning status for credit line — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for a line of credit account with the status value contained in this transaction.
83 00 41						Change PBF stop pay/warning status for installment loan — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for an installment loan account with the status value contained in this transaction.
83 00 42						Change PBF stop pay/warning status for mortgage loan — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for a mortgage loan account with the status value contained in this transaction.
83 00 43						Change PBF stop pay/warning status for commercial loan — A request to replace the value in the STOP PAY/WARNING STATUS field of the PBF for a commercial loan account with the status value contained in this transaction.

Support Key:

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Warning/Hold/Float File Update Transactions

BASE24-teller supports the Warning/Hold/Float File (WHFF) update transactions shown in the table below.

The WHFF can contain warning, hold, and deposit float information. WHFF warning entries cause the BASE24-teller Authorization process to decline transactions that might otherwise be approved. The warning update transactions automatically add and delete WHFF warning records.

Hold update transactions decrease the PBF available funds balance and increase the PBF amount on hold balance. The BASE24-teller Authorization process does not consider the PBF overdraft limit when determining whether a hold can be added. Hold update transactions automatically adjust the PBF available funds and amount on hold balances. The value of the TLR-WRITE-HLDS-DEP-FLOATS param in the Logical Network Configuration File (LCONF) controls whether hold update transactions also add and delete WHFF hold records.

Deposit float update transactions increase the PBF available funds balance without affecting the PBF amount on hold balance. The BASE24-teller Authorization process does not consider the PBF overdraft limit when determining whether a deposit float entry can be deleted. Deposit float update transactions automatically adjust the PBF available funds balance. The value of the TLR-WRITE-HLDS-DEP-FLOATS param in the LCONF controls whether deposit float update transactions also add and delete WHFF deposit float records.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
84 00 01						Add warning to DDA — A request to add a warning record to the WHFF for the checking account specified in this transaction.
84 00 11						Add warning to savings — A request to add a warning record to the WHFF for the savings account specified in this transaction.
84 00 12						Add warning to IRA — A request to add a warning record to the WHFF for the IRA account specified in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
84 00 13						Add warning to CD — A request to add a warning record to the WHFF for the certificate of deposit (CD) account specified in this transaction.
84 00 21						Add warning to NOW — A request to add a warning record to the WHFF for the NOW account specified in this transaction.
84 00 31						Add warning to credit card — A request to add a warning record to the WHFF for the credit card account specified in this transaction.
84 00 32						Add warning to credit line — A request to add a warning record to the WHFF for the line of credit account specified in this transaction.
84 00 41						Add warning to installment loan — A request to add a warning record to the WHFF for the installment loan account specified in this transaction.
84 00 42						Add warning to mortgage loan — A request to add a warning record to the WHFF for the mortgage loan account specified in this transaction.
84 00 43						Add warning to commercial loan — A request to add a warning record to the WHFF for the commercial loan account specified in this transaction.
85 00 01						Add hold to DDA — A request to add a hold to the checking account specified in this transaction.
85 00 11						Add hold to savings — A request to add a hold to the savings account specified in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
85 00 12						Add hold to IRA — A request to add a hold to the IRA account specified in this transaction.
85 00 21						Add hold to NOW — A request to add a hold to the NOW account specified in this transaction.
85 00 31						Add hold to credit card — A request to add a hold to the credit card account specified in this transaction.
85 00 32						Add hold to credit line — A request to add a hold to the line of credit account specified in this transaction.
85 00 41						Add hold to installment loan — A request to add a hold to the installment loan account specified in this transaction.
85 00 42						Add hold to mortgage loan — A request to add a hold to the mortgage loan account specified in this transaction.
85 00 43						Add hold to commercial loan — A request to add a hold to the commercial loan account specified in this transaction.
86 00 01						Delete hold from DDA — A request to delete a hold from the checking account specified in this transaction.
86 00 11						Delete hold from savings — A request to delete a hold from the savings account specified in this transaction.
86 00 12						Delete hold from IRA — A request to delete a hold from the IRA account specified in this transaction.
86 00 21						Delete hold from NOW — A request to delete a hold from the NOW account specified in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
86 00 31						Delete hold from credit card — A request to delete a hold from the credit card account specified in this transaction.
86 00 32						Delete hold from credit line — A request to delete a hold from the line of credit account specified in this transaction.
86 00 41						Delete hold from installment loan — A request to delete a hold from the installment loan account specified in this transaction.
86 00 42						Delete hold from mortgage loan — A request to delete a hold from the mortgage loan account specified in this transaction.
86 00 43						Delete hold from commercial loan — A request to delete a hold from the commercial loan account specified in this transaction.
87 00 01						Add float to DDA — A request to add deposit float to the checking account specified in this transaction.
87 00 11						Add float to savings — A request to add deposit float to the savings account specified in this transaction.
87 00 12						Add float to IRA — A request to add deposit float to the IRA account specified in this transaction.
87 00 21						Add float to NOW — A request to add deposit float to the NOW account specified in this transaction.
87 00 31						Add float to credit card — A request to add deposit float to the credit card account specified in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
87 00 32						Add float to credit line — A request to add deposit float to the line of credit account specified in this transaction.
87 00 41						Add float to installment loan — A request to add deposit float to the installment loan account specified in this transaction.
87 00 42						Add float to mortgage loan — A request to add deposit float to the mortgage loan account specified in this transaction.
87 00 43						Add float to commercial loan — A request to add deposit float to the commercial loan account specified in this transaction.
88 00 01						Delete float from DDA — A request to delete deposit float from the checking account specified in this transaction.
88 00 11						Delete float from savings — A request to delete deposit float from the savings account specified in this transaction.
88 00 12						Delete float from IRA — A request to delete deposit float from the IRA account specified in this transaction.
88 00 21						Delete float from NOW — A request to delete deposit float from the NOW account specified in this transaction.
88 00 31						Delete float from credit card — A request to delete deposit float from the credit card account specified in this transaction.
88 00 32						Delete float from credit line — A request to delete deposit float from the line of credit account specified in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
88 00 41						Delete float from installment loan — A request to delete deposit float from the installment loan account specified in this transaction.
88 00 42						Delete float from mortgage loan — A request to delete deposit float from the mortgage loan account specified in this transaction.
88 00 43						Delete float from commercial loan — A request to delete deposit float from the commercial loan account specified in this transaction.
89 00 01						Delete warning from DDA — A request to delete a warning record from the WHFF for the checking account specified in this transaction.
89 00 11						Delete warning from savings — A request to delete a warning record from the WHFF for the savings account specified in this transaction.
89 00 12						Delete warning from IRA — A request to delete a warning record from the WHFF for the IRA account specified in this transaction.
89 00 13						Delete warning from CD — A request to delete a warning record from the WHFF for the certificate of deposit (CD) account specified in this transaction.
89 00 21						Delete warning from NOW — A request to delete a warning record from the WHFF for the NOW account specified in this transaction.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
89 00 31						Delete warning from credit card — A request to delete a warning record from the WHFF for the credit card account specified in this transaction.
89 00 32						Delete warning from credit line — A request to delete a warning record from the WHFF for the line of credit account specified in this transaction.
89 00 41						Delete warning from installment loan — A request to delete a warning record from the WHFF for the installment loan account specified in this transaction.
89 00 42						Delete warning from mortgage loan — A request to delete a warning record from the WHFF for the mortgage loan account specified in this transaction.
89 00 43						Delete warning from commercial loan — A request to delete a warning record from the WHFF for the commercial loan account specified in this transaction.

Support Key:

OL = Overdraft limit

OP = Overdraft protection

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PH = Preauthorized hold checks

Administrative Transactions

BASE24-teller supports the following administrative transactions.

Tran Code	O L	O P	P B	A P	P H	Transaction Type and Description
90 00 00						Logon (start of day) — An administrative transaction logged by BASE24 to identify the start of a business day and set the transaction posting date in the BASE24-teller Terminal Data File (TTDF).
91 00 00						Logoff (end of day) — An administrative transaction logged by BASE24 to identify the end of a business day and set the transaction posting date in the TTDF to zeros.
92 00 00						Signon (balancing day) — An administrative transaction logged by BASE24 to identify the start of a work session or balancing period and set the transaction posting date in the TTDF.
93 00 00						Signoff (balancing day) — An administrative transaction logged by BASE24 to identify the end of a work session or balancing period and set the transaction posting date in the TTDF to zeros.

Support Key:

OL	=	Overdraft limit	OP	=	Overdraft protection
PB	=	Passbooks	AP	=	Automatic passbook printing
PH	=	Preauthorized hold checks			

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